How older adults use measures of brain activity in real life: A Design Case Study

Master thesis

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Abstract

Every year, the number of elderly people suffering from MCI considerably increased. Consequently, their subjective well-being and life quality are negatively affected. Artifacts from the digital world could play an important role in preventing or even reversing MCI. A smartphone application (Bo app) that uses neurofeedback for cognitive interventions was developed using participatory design methods in this study. As a result, it has been learnt that designing minimalist apps will keep the users focused, and various feedback such as Real-time neurofeedback, Daily feedback, Monthly feedback and Competitive feedback enthusiast them to conduct cognitive training sessions. Moreover, in-app-app step-by-step video guidance guides them in using new devices when they face a new device for the first time. Game party functionality can satisfy their social needs. The PD allows users to benefit from the smartphone applications based on their preferences, expectations, and needs. Co-design encourages elderly participants to involve more productive and creative. Thanks to our participatory design approach, the Bo app is considered to be a step forward compared to existing solutions and the results and insights are encouraging.

List of contents

List of figures ..... **iv**

List of tables ..... **vi**

Index of Abbreviations ..... **viii**

1 Introduction **1**

2 Related work and background **4**

2.1 Technology appropriation for older adults **4**

2.2 Benefits and Challenges of Participatory Design Research with Older Adults **6**

2.2.1 Accessibility and Acceptability 7

2.2.2 Feedback 8

2.2.3 Motivation 8

2.2.4 Considering heterogeneous expectations and needs 8

2.2.5 Sociality 9

2.2.6 Gamification 10

2.3 fNIRS and MCI prevention **11**

2.3.1 Meditation and Mindfulness role in preventing and delaying MCI 14

2.3.2 Exergames, music, and Cognitive Stimulation game's role for preventing and delaying MCI 18

2.4 Research Question **19**

3 Methodology **21**

3.1 Design Case Study **21**

3.2 EVITA research project **24**

3.3 Participants **27**

3.4 Materials and methods **28**

3.5 Pre-study methodology **31**

3.5.1 Interviews 31

3.5.2 Observation 32

3.6 Design **32**

3.6.1 Basic Consideration of prototyping stage 32

3.6.2 Instant collaborative instant co-design with the interviewee 35
3.7 Appropriation and Mid-appropriation phase 36
3.7.1 Usability test - Thinking aloud 36
3.8 Final appropriation 37
3.8.1 Expert review 37
3.8.2 Self-report 39
4 Concept and ideation 40
5 Description of the Bo System 42
5.1 ABC App and Cloud-Based System (STRAPI) 42
5.2 BrainHQ 43
5.3 NeU sensor and fNIRS 45
5.4 Describe design concept and development of Bo app Error! Bookmark not defined.
5.4.1 Development of the Bo-App 48
6 Results of interviews, observation and usability tests ..... Error! Bookmark not defined.
6.1 Results of pre-study based on view of Sessions 57
6.2 Results of appropriation phase based on view of Sessions ..... Error! Bookmark not defined.
6.3 Results of Mid-appropriation phase ..... Error! Bookmark not defined.
6.4 Results of Final-appropriation phase ..... Error! Bookmark not defined.
6.5 Results of the expert interviews 70
7 Discussion and Lessons Learned 74
7.1 Motivational factors and challenges 75
7.2 Design implications and methodological considerations 77
7.3 Limitations and Future Work 87
8 Conclusion and outlook 89
References 92
Appendix 106

List of figures

Figure 1 Cognitive functions (Reynolds et al., 2021) 12
Figure 2 Essential psychological needs (Hassenzahl et al., 2013) 12
Figure 3 Functional near-infrared spectroscopy (Hoshi, 2016) 14
Figure 4 An illustration of the meditation process 17
Figure 5 Design Case Study structure (Wulf et al., 2015) 21
Figure 6 The prefrontal cortex 27
Figure 7 The mind map for analyzing early stages ideas based on the core idea 40
Figure 8 The concept map to find the relations between the project elements 41
Figure 9 Wireframe of the Bo App 41
Figure 10 Different pages of the ABC app 43
Figure 11 fNIRS sensor function 46
Figure 13 Bo-App infrastructure and design goals 48
Figure 14 Features of the Bo-App 49
Figure 15 Iterative improvement of daily feedback. a) V1.0 b) V2.0 c) V3.0 49
Figure 16 Application Tour 51
Figure 17 Inter app step-by-step manual. 53
Figure 18 Diaphragmatic breathing guidance as a cognitive intervention 55
Figure 19 Inviting friends to the Bo app 56
Figure 20 Screenshot of the homepage during P2' interaction with the app 60
Figure 22 Home page of ABC app 61
Figure 23 Screenshot from the game in ABC when P2 was playing 61
Figure 24 Screenshot from P2 playing with BrainHQ 62
Figure 25 Bo app logo 63
Figure 26 Screenshot of activity page in the Bo app 64
Figure 27 Different pages of the ABC app feedback 65
Figure 28 On-boarding 66
Figure 29 Relaxation page shown before every activity 67
Figure 30 Bo app versions 70
Figure 31 Flowchart of Bo app 110
Figure 32 On-boarding pages 110

Figure 33 Bo app functions 110
Figure 34 NeU device step-by-step tutorial 111
Figure 35 Diaphragmatic relaxation and mindfulness guidance 111
Figure 36 Daily feedback 112
Figure 37 (a) Long-time feedback, (b) Competitive feedback 112
Figure 38 Cognitive training games 113
Figure 39 Bluetooth setup 113
Figure 40 Error informing 113
Figure 41 Bo app funcionalities 114
Figure 42 A game sample of BrainHQ with background knowledge 114
Figure 43 Initializing guidance of the game sample 114

List of tables

Table 1 Demographic information of participants 28
Table 2 Prototype Versions of the Bo-App 51

Index of Abbreviations

HCI	Human Computer Interaction
ICT	Information Communication Technology
PD	Participatory Design
WHO	World Health Organization
PwD	People with Dementia
AD	Alzheimer Disease
MCI	Mild Cognitive Impairment
AHA	Active and Healthy Aging
fNIRS	functional Near infrared spectroscopy
CBF	Cerebral Blood Flow
Prefrontal Cortex	PFC
rCBF	Regional Crebral Blood Flow
FAM	Focused Attention Meditation
OMM	Open-Monitoring Meditation
WMC	Working Memory Capacity
DLPFC	Dorsolateral Prefrontal Cortex
HRV	Heart Rate Variability
CT-NF	Cognitive Training with Neurofeedback
vHIE	virtual HIE-based intervention
IVR	immerse virtual reality
HIE	high-intensity intermittent aerobic exercise
HbT	Total-hemoglobin
ABC	Active Brain C
IT	Information Technology
GPs	General Practitioners

1 Introduction

According to WHO (*Ageing and Health*, 2022), one billion people were over the age of 60 in 2019. By 2050, the population will reach 2.1 billion. Growing older is not only associated with losing friends and family, but also with sensory, executive and cognitive functions (e.g., concentration) and short-term memory, and physical deterioration (Chung et al., 2016). Cognitive functions refer to a set of abilities, including memory, concentration, attention, and reasoning which, when used, make it possible to carry out everyday tasks like learning and communicating. The number of neurons is a fundamental, innate factor that cannot be manipulated, but lifestyle factors like nutrition, physical activity, education, and a high intellectual level throughout one's life are all factors that contribute to cognitive ability. Mental or neurological disorders can be caused by any or all of these factors. Neurocognitive disorders (dementia), anxiety, depression, and substance abuse are the main disorders. Dementia (*Dementia*, 2022) is a syndrome characterized by deterioration of cognitive function. As of September 2019, approximately 50 million people worldwide suffered from dementia WHO (World Health Organization) (*Dementia*, 2022). In 2030, there will be 82 million people with dementia (PwD) and 152 million in 2050. Neurodegenerative pathologies such as dementia gradually impair autonomy and competency (Hassenzahl et al., 2013) and increase the workload for caregivers, and institutionalization may be required when the disease is advanced (Wrobel et al., 2014). These diseases result in severe emotional and financial stress for those who suffer from them, their families, and society as a whole (Moussard, 2018). Consequently, seniors' quality of life suffers as a result of these pathologies. In addition, it is difficult for people to receive help due to the stigma associated with mental illnesses like dementia and misdiagnosis. Dementia often begins with mild cognitive impairment (MCI). Therefore, people who suffer from MCI are more likely to develop dementia than those with normal cognitive functioning. If appropriate treatment is administered, even MCI can return to a normal functioning (Shimada et al., 2017). Active interventions may be able to delay the progression of this syndrome and improve cognitive functions (Zhang et al., 2022). Diseases can be avoided by taking steps to promote aging well. These include, for brief mention, activities such as reading, playing a musical instrument, playing games, and

volunteering are cognitive activities (Moussard, 2018) or meditation (Crescentini et al., 2017), among others. Performing these activities throughout the life span is important, even for the elderly. These activities can be carried out individually or in groups, at home or outside through associations. MCI affects negatively the well-being of elderly adults; one solution could be engaging them in a set of positive experiences and activities which eventually affect positively their well-being. Digital tools have the potential to improve the quality of life for seniors (Kolland, 2014). For instance, it has been proved that a portable cognitive training with neurofeedback (CT-NF) system using 2ch NIRS affects positively life quality of the users (Nouchi et al., 2021), or in another study in order to improve both users' health and cognitive performance by using the designed app to learn how to change their physiological activity such as diaphragmatic breathing exercise (Chelidoni et al., 2020). Information and communication technologies (ICTs) designs could be helpful to, directly and indirectly, fulfill essential psychological needs, such as autonomy, competence, relatedness, popularity, stimulation, and safety (Hassenzahl et al., 2013) through their well-designed elements. There are two possibilities: The designs could be used by elderly people supervised and unsupervised and each method has advantages and disadvantages. On the one hand, it may be difficult for the elderly to use measuring technologies independently in their private home. On the other hand they may be unmotivated to attend training sessions on a regular basis. Elderly people require caregivers to carry out daily activities, making them more dependent and incapacitated. In other words, their sense of self-sufficiency, experience, and competence will gradually decline, causing them to feel less confident while also reducing their cognitive abilities and increasing their risk of developing dementia-related diseases. This thesis focuses on unsupervised designs that could probably fulfill the psychological needs of elderly people more efficiently. Researches rarely discuss a design that enables people to measure and improve their level of brain activity in later life while in their private homes unsupervised. In order to increase the elderly's sense of self-confidence and self-sufficiency and make them more independent of caregivers and relatives, as well as reduce the cost of hiring caregivers, a protocol was developed in this thesis so that seniors could hold practice sessions autonomously at home. Brain cognitive training sessions should be conducted by users regularly to improve their brain activity level. Applications with a profound

scientific potential to improve brain function are included in training sessions. So that, while holding brain training sessions, the user can simultaneously perform the relevant activity in the course of his brain activity, which not only helps to get their brain working more, but it will also motivate them to continue exercising on a daily basis, as measured by the new device from the frontal lobe of the brain. Another goal of this study is to give people a sense of control over their daily lives. Another advantage is the ability to detect MCI in its early stages, as one of the first symptoms of dementia and MCI disease is a lack of hemoglobin cells in brain blood flow, particularly in the prefrontal cortex (Wierenga et al., 2014). The thesis investigated an innovative solution aimed at providing needs-based and practice-oriented solutions for unsupervised cognitive sessions through Design case Study approach. As a result of elderly people's competence and autonomy, they may feel capable and respected for their life experiences (Hassenzahl et al., 2013). Following is the structure of the rest of the thesis. Chapter 2 gives an overview of the studies that looked at Design Case Study in living labs about technology usage, then studies that worked on the brain activity measurement device and cognitive interventions are introduced. Chapter 3 addresses the various method used during the study to learn about Design Case Study and Participatory Design and how this led to design practice as a part of e-VITA project (a European and Japanese collaboration to develop a virtual coaching service for older adults that uses ICT technology) (Chapter 3.2). Chapter 4 provides concept and ideation. Chapter 5 describes the Bo system and integrated elements. Chapter 6 presents the results of the pre-study and assessments of how design practices have changed during these phases in and after mid-evaluation. Chapter 7 discusses the results and compares them with studies that have been done before as well as limitations. Finally, this thesis ends with a conclusion in Chapter 8.

2 Related work and background

Brain functionality impairment in the elderly people target group is a common problem and people are always seeking a way to prevent their brain functionality reduction caused by the aging process. One reason is that they would prefer to do their daily activities, particularly cognitive-related activities, independently from other family members or caregivers. As mentioned previously, cognitive ability and doing daily activities are interdependently related to each other. Besides, given the increasing prevalence rate and the limitations of available pharmacological treatments, Alzheimer's Disease (AD), Dementia and MCI are global health issues. Consequently, accessible nonpharmacological interventions are crucial (Fang et al., 2017). To prevent and/or postpone MCI as an early stage of Dementia related diseases, subjective wellbeing needed to be increased. For doing this, fundamental psychological needs (e.g., autonomy, competence, popularity, stimulation, and safety) (Hassenzahl et al., 2013) needed to be fulfilled, eventually subjective wellbeing is likely to be increased (Hassenzahl et al., 2010). In other words, by and for doing daily activities independently from others, cognitive ability and subjective well-being will be increased and possible MCI in later life will be prevented. Considering all these, designing an appropriate technology which covers the elderly fundamental psychological needs is necessary to strengthen cognitive ability.

2.1 Technology appropriation for older adults

Dementia design has become increasingly popular in recent years and on the other hand, as human being's lives become increasingly reliant on digital artifacts, they have to learn about how to work with these tools on a daily basis throughout their entire lives. Some of these artifacts are, however, not well designed for elderly people to learn how to use these tools, and many of them find it challenging to integrate digital tools into their daily lives. Furthermore, parts of this target group are suffering from Mild Cognitive Impairment (MCI), and regardless of their mental and physical impairments, they should be able to use the ICTs easily. Another problem that makes the design process even harder, is that

this target group members are originally selected based solely on their age and it is highly heterogeneous in terms of social class, education, and technical ability (Cerna, Katerina et al., 2020), therefore, they have different learning pace. Different usage of smartphones by older adults may be primarily explained by differences in culture, values, and lifestyle, but some behaviour differences may be largely explained by cognitive decline (Gordon et al., 2019). Considering all these, the design process must take into account the diverse skills of stakeholders. Their tech needs may vary (not just learning needs) and consequently, no single discipline can effectively address learning for life for older adults and the adoption of technology by them is a multidisciplinary issue, bringing together gerontologists, anthropologists, sociologists of later life, special education, social workers, care scientists, and user-oriented IT researchers (Cerna, Katerina et al., 2020). Moreover, older adults are often critical of technology if they do not see the value in using it every day. As long as they see real-world applications for the technology, their attitudes can change and they can become motivated to use it to support their everyday lives (Bengs et al., 2018). Digital tools, also, can be difficult to use due to language barriers caused by technical and/or English terms. The challenge is particularly difficult for people who do not speak English or have little technical knowledge (Müller et al., 2015a). Various concerns and fears impede trial and error learning, such as a lack of trust and a fear of unwanted costs resulting from using digital tools and related support services (Müller et al., 2015a). Moreover, there are barriers that include usability problems and a lack of understanding of their needs and preferences (Czaja & Lee, 2007).

Digital tools have the potential to improve the quality of life for seniors (Kolland, 2014) but researchers and developers are missing promising opportunities to support older adults or offer solutions to insignificant problems because of inadequate understanding of how older adults use smartphones, how their usage varies from younger users and the causes for those differences (Gordon et al., 2019). They concluded that the results presented in the quasi-experiment are sufficiently strong to predict older adult's cognitive function from smartphone usage. Different usage of smartphones by older adults may be primarily explained by differences in culture, values, and lifestyle, but some behaviour differences may be largely explained by cognitive decline (Gordon et al., 2019). The design of ICTs that can be used in a real-world context is dependent on the diversity of stakeholders,

their individual interests and motivations as well as personal skills and learning requirements. Associated strategies and decisions influence both the design of the project and the final ICT product (Müller et al., 2015a). Participatory approaches with older adults are needed for that to create technology that appropriately fits their needs in real-world context.

2.2 Benefits and Challenges of Participatory Design Research with Older Adults

Participatory design holds that designers need to prosper by gaining a better understanding of the usage contexts of older people by involving them in the design of systems they will be using (Muller & Kuhn, 1993). In participatory design (PD) contexts, people can easily understand a technical tool if they feel comfortable using it (Joshi, 2016). In order to support the appropriation of digital tools in learning spaces, technology adoption and ownership of the learning process should be considered, as well as the possibility of influencing the learning setting. A learning process literacy must also be addressed besides digital literacy (Cerna, Katerina et al., 2020). Additionally, Gasteiger et al. did a four-year international project participatory design in 2022, by collaborating with experts, carers, relatives and older adults with mild cognitive impairment (MCI) to address real-life challenges. This study revealed that for robots and cognitive stimulating games to be effective, useful, acceptable and user-friendly, it is crucial to engage end users in the design process (Gasteiger et al., 2022). It is essential to conduct research with older adults in a context that takes their social, political, and other factors into account (Cerna, Katerina et al. ,2020; Evans et al. ,2018; Gerling et al. ,2020); therefore, a "broad" and holistic approach is required. In this sense, "pure" PD is not sufficient, but a Living Lab can be a breakthrough infrastructure that includes all stakeholders in real-life situations. To conduct PD with a broad stakeholder view in real-life environments, the Living Lab approach has been suggested in order to set up a pragmatic research infrastructure. Living labs that use ethnographic methods, such as diary studies, participant observation, or cultural probes, can provide a wealth of qualitative data that researchers can use to understand various practices on a very detailed, individual level,

as well as identify general attitudes, needs, and problems (Crabtree, 1998). So, information and communication technology (ICT) service providers use Living Labs to meet the challenges of innovation faced by their users (Følstad, 2008). These methods supplement more traditional methods of information gathering, such as home ethnography and laboratory research field (Müller et al., 2015b). Based on another PD and a qualitative living lab (Vaziri et al., 2019) study to demonstrate the diversity and complexity of older adults' views on health and technology use, about active and healthy aging (AHA), it was found that firstly, taking into account older adults' heterogeneous expectations and requirements when designing technology is key to ensuring older adults' continued use of it, another research (Callari et al., 2012) found out that being user friendly is correlated directly with motivation to use the digital artifact. Secondly a variety of design factors such as (1) Build trust, (2) Improve comprehensibility, (3) ease of use, (4) Raise awareness appear to be relevant for the successful adoption of such platforms by older adults that should be taken into account in designing a mobile health technology with integrative capabilities for older adults.

2.2.1 Accessibility and Acceptability

A research study (Callari et al., 2012), which was conducted to identify strategies that would improve the quality of life for seniors, examined the accessibility and acceptability of such interaction by senior citizens with technological devices for motor and cognitive training. According to the findings, elderly people may accept technological artifacts if they perceive that they will improve their well-being and health and it was found that seniors preferred technological devices that offered them tailored information and could be incorporated into their daily routine. The study suggested that to gain elderly peoples' trust to technology if interfaces and technological artifacts consider their physical characteristics and their cognitive processes (Callari et al., 2012). Aged adults found the robot and games to be acceptable and useful for providing stimulating activities and reminders to check their health and wellbeing (Gasteiger et al., 2022). In order to ensure accessibility and positive experience in developing a typing tool with PD, another study (Silva et al., 2019) the interface was kept minimalist, which participants found appealing.

2.2.2 Feedback

It was very important for the participants to receive feedback on their performance during training sessions with motor and cognitive devices. In training sessions with motor trainers and cognitive psychologists, participants were more motivated and interested in using devices when feedback was timed and tailored, and tracks of performance trends were shown(Callari et al., 2012). Despite of talking about PD, based on the Rienzo et al. studies on cognitive training games and older adult preferences the most used gamification elements that seek to motivate participants to continue in the game are achievement and feedback. In addition immediate feedback can increase motivation and help redirect behavior towards goal achievement; scoring and design factors are also important aspects of the game's usability for the older user (Rienzo et al., 2021).

2.2.3 Motivation

It was found that a user-friendly device perception correlated directly with motivation to use the device. According to the results, elderly people are more motivated to use technological devices when they perceive them as user-friendly and when they believe that the technology will help them stay healthy and safe(Callari et al., 2012). Considering target group heterogeneous expectations is motivating for them (Vaziri et al., 2019).

Provide intuitive and easy-to-use games that do not exceed the effort made by the elderly in their daily lives, so as not to cause stress and avoid abandonment of the activity(Rienzo et al., 2021)..

2.2.4 Considering heterogeneous expectations and needs

The study (Callari et al., 2012) suggested that elderly people mistrust technology because of interfaces and technological artifacts that still fail to consider their physical characteristics (e.g. physical limitations in sight, hearing, movement) and their cognitive processes (selective memory often driven by practical needs) (Callari et al., 2012).

Voilmy & Duchêne stressed the importance of a real-life setting and user awareness in (Voilmy & Duchêne, 2014). They explained that firstly, the Living Lab methodology can be used to investigate a wide range of user needs (observable, tacit, and latent) and to

integrate design research methodologies that address specific categories of user needs, and secondly, context is leveraged as an essential component of the design process to allow users to interact with new products and services in their daily lives. Apart from talking about PD, different kind of tactile feedback such as vibration feedback, cause improvement in the users' satisfaction (Gao & Zhou, 2020, page 316). It is common for the elderly to experience visual and auditory difficulties, as well as reduced cognitive and motor abilities. The elderly may have difficulty using mobile devices due to these problems. However, older users can cope with these physical changes with the help of current feedback technology in mobile devices (Gao & Zhou, 2020, page 316). Another project in 2021 (Ertl et al., 2021) was a Design Fiction Workshop, with the intention of including stakeholders who are currently excluded from the concept. They are commonly marginalized, such as migrants, refugees, older adults, children, or those who are neurodiverse, such as those suffering from mental health challenges such as autism or dementia. People who are physically challenged. Virtual systems must be capable of identifying individual states (e.g., training progress or motivational states) and adapting the gaming and training experiences accordingly (Kickmeier-Rust et al., 2012). Diversity of stakeholders, their individual interests and motivations as well as personal skills and learning requirements influence both the design of the project and the final ICT product (Müller et al., 2015a). Aside from talking about PD, taking into account the full spectrum of cognitive functions could be beneficial for providing a better support for all older adults (Gordon et al., 2019). Imagery and speech should be customized according to the individual's needs and preferences to optimized interaction (Gasteiger et al., 2022).

2.2.5 Sociality

A multidisciplinary and translational research project that aims to foster a new learning culture for later-life learning was promotes learning to address aging issues. This project promotes digital literacy by providing socially-grounded learning opportunities for older people with limited technology skills. A paper reported fieldwork (Cerna & Müller, 2021), that was a part of the project, in which researchers collaborated with older adults to create mobile demo-kits as an easily accessible didactic prototype to foster digital literacy in older adults and help them to be more autonomous. In this work, a mobile demo-kit and

a sustainable PD approach were combined. The results of this project were used and connections with participants were kept even after finishing the project. These elderly people are encouraged to discuss and share their experiences about technologies with each other and other stakeholders by developing a mobile demonstration kit comprised of a variety of devices (Cerna, Katerina et al., 2020). It was concluded from Vaziri et. al. studies that functionalities that facilitate and foster social participation and inclusion and answer significantly older adults' expectations (Vaziri et al., 2019). Moreover, in another study, after using the 'exergames' for six months in a living lab environment, the participants found that their primary motivation to use the system was not so much physical activity, but socializing. During workshops or social events, users exchanged experiences with other participants or played the 'exergames' with their grandchildren (Meurer et al., 2018).

As above mentioned, many investigations have been conducted on improving well-being for later life, but they rarely focused on developing a protocol for conducting unsupervised cognitive sessions in the homes of elderly people. Therefore, more research is required on this topic.

2.2.6 Gamification

Cognitive Stimulation games appear to be a promising approach in the fight against the physical and mental decline of elderly people and related harms to life quality (Kickmeier-Rust et al., 2012). Since elderly people have specific needs and expectations, gamification through PD approaches could be very beneficial to them. In 2021 a research carried out, to gather the main aspects that should be considered in the design of a Cognitive Stimulation game for the cognitive training of older adults, in order to guarantee a good development (Rienzo et al., 2021). It is recommended to involve users (older adults) and stakeholders in design evaluation, adapting software to user profiles; consider aspects such as age, familiarity with the use of technologies and educational level. For the activities of the game, it is recommended to adopt the content to situations of daily life, or themes of the real world. To achieve adequate training, it is essential that it be adapted to personal or sociocultural variables, which requires generating different tasks and suggesting, based on prior evaluation, the most appropriate one for the older user

and design different tasks that involve different cognitive domains in order not to bore the user with the same activity (Rienzo et al., 2021). The instructions of the activity must be clear, brief, striking and must be presented for a prolonged period of time, or maintained during the development of the activity, since some older adults mention having difficulty remembering the instructions (Rodríguez-Almendros et al., 2021). However, it is not recommended to use elements such as insignia and competition since in the study by Altmeyer, Lessel et al (Altmeyer et al., 2018) adults mention that their use makes them feel under pressure and they do not find a favorable meaning in it. In conclusion, based on the analyses, the main aspects that should be considered in the design of a digital game for the cognitive training of these adults are presented. As for the gamification elements, immediate feedback can increase motivation and help redirect behavior towards goal achievement; scoring and design factors are also important aspects of the game's usability for the older user and its corresponding performance measurement.

2.3 fNIRS and MCI prevention

This part is addressing measurement method for CBF by fNIRS and how this measurement could lead to MCI prevention will be explained. MCI is distinct from dementia; abnormal cerebral blood flow (an oxidative parameter) has been linked to this condition (Maruya et al., 2021). Using a summary analysis, it has been found that, on average, 4.2% of MCIs are converted to dementias every year (Albert et al., 2011). The annual conversion rate can range from 10 to 15% in other short-term clinical observations (Morris et al., 2001). Petersen et al. (2005) propose criteria for MCI for a person or a family member who complains of cognitive decline and objective impairment of one or more cognitive functions (p. 20-30). However, daily activities are typical and not characteristic of dementia. As a result, it is difficult to detect MCI in its earliest stages because patients with this condition are able to continue with normal daily activities (Maruya et al., 2021).

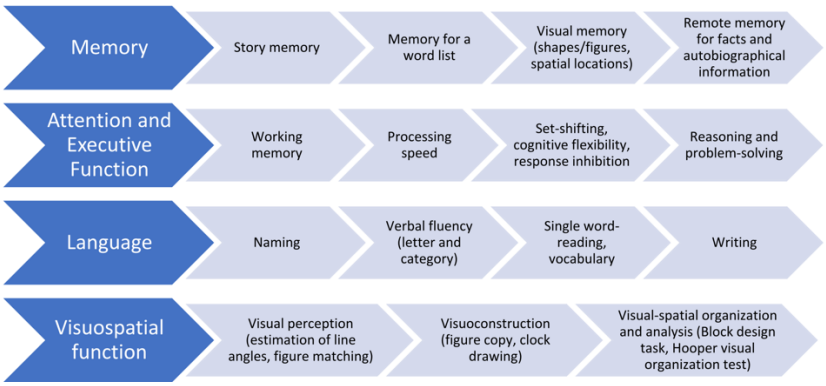


Figure 1 Cognitive functions (Reynolds et al., 2021)

In another word, MCI is regarded as an early stage of dementia. Individuals with MCI are more likely to develop dementia than those with normal cognitive function. However, even MCI can return to normal cognitive function if the appropriate treatment is administered (Shimada et al., 2017). Hassenzahl et al. (2013) interpreted well-being as the result of engaging in positive experiences on a daily basis. This type of experience is positive when it fulfills essential psychological needs, such as autonomy, competence, relatedness, popularity, stimulation, and safety (Hassenzahl et al., 2013) (Figure 2).

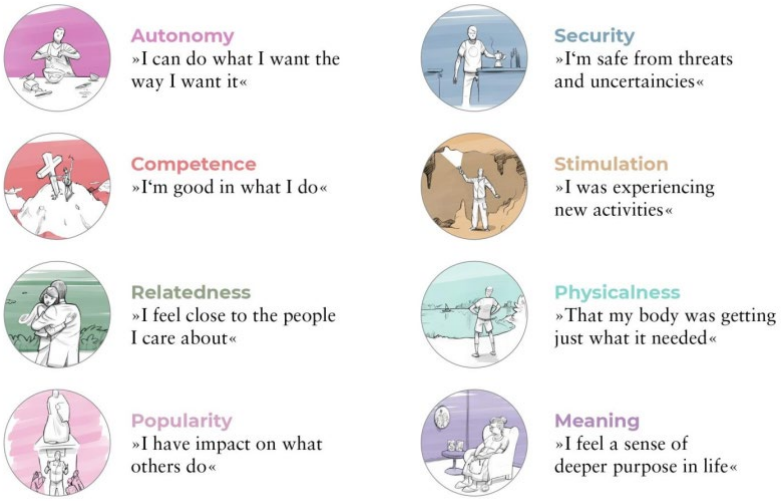


Figure 2 Essential psychological needs (Hassenzahl et al., 2013)

MCI affects negatively the well-being of elderly adults (Wilson et al., 2013); one solution could be engaging them in a set of positive experiences and activities which eventually

affect positively their well-being. Cognitive disorders should also be detected as early as possible so that appropriate support can be provided as soon as possible. A number of prospective observational studies indicate that people who engage in mentally stimulating activities - such as reading, playing games, and learning - at an older age have a lower risk of developing dementia (Fratiglioni & Wang, 2007). In contrast to those who do not engage in cognitively stimulating activities, elderly people who are cognitively active have higher degrees of dementia and MCI symptoms (Middleton & Yaffe, 2009). A change in cerebral blood flow (CBF) may contribute to cognitive decline (Iturria-Medina et al., 2016). The prefrontal cortex needs oxygen and glucose to function properly, and the brain increases blood flow constantly to deliver these nutrients quickly. The brain is in a state of activation when blood flow is active in the it and brain sensors can detect when blood flow increases in the prefrontal cortex (Solutions, 2022). Moreover, cerebral blood flow changes take place before the onset of MCI, supporting the idea that early intervention for MCI and dementia may be possible (Maruya et al., 2021). The findings show that cerebral blood flow changes happen before the development of frailty or MCI, bolding a potential approach for early intervention for MCI and dementia and differences in CBF could be a sign of cognitive decline and can be detected earlier (Maruya et al., 2021). Functional near-infrared spectroscopy (fNIRS) is a portable optical sensor that allows real-time monitoring of the brain hemodynamics (Lin et al., 2009; Son and Yazici, 2006; Son & Yazici, 2006). As a neuroimaging tool that quantifies changes in the concentration of oxygenated hemoglobin (oxyHb) and de oxygenated Hemoglobin (deoxyHb) non-invasively in the human brain, this sensor is especially suitable for investigating the effects of physical activity on cerebral oxygenation and hemodynamics (Herold et al., 2018). fNIRS helps to evaluate the brain cognitive ability level as well as brain training of users during the cognitive training sessions, and eventually could be helpful in MCI prevention and reduce the cognitive brain deterioration.

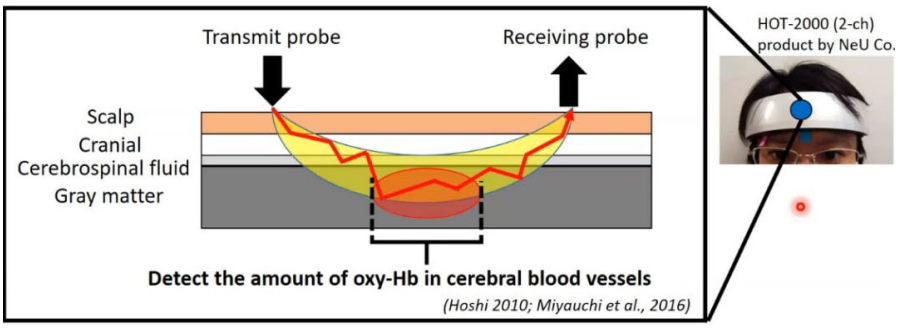


Figure 3 Functional near-infrared spectroscopy (Hoshi, 2016)

Several research investigated the efficacy of cognitive training sessions on brain cognitive ability improvement and MCI prevention. For example, a large randomized controlled trial showed that a 7-month multidomain intervention for mild cognitive impairment, consisting of cognitive training, aerobic exercise, and music therapy, improved global cognition and Para hippocampal cerebral blood flow compared with a control group receiving usual care. According to these results, non-pharmacological, multicomponent interventions can improve brain health and cognitive status in MCI participants (Train the Brain Consortium, 2017, Gates et al, 2011). They reported that the different methods of cognitive training sessions could be generally helpful in MCI patients and prevention. The result is that cognitive exercises can produce moderate to-large beneficial effects on memory-related outcomes (Gates et al., 2011). Additionally, a set of research supports the efficacy of holistic approaches (e.g., combinations of Cognitive Stimulation and mindfulness, neurofeedback, and exercise and dietary modifications) that together may help improve cognition and increase gray matter volume in older adult MCI patients (Fotuhi et al., 2016; Kane et al., 2017; Köbe et al., 2016). Therefore, focus of this thesis is designing a protocol to conduct cognitive training sessions through cognitive stimulations, mindfulness and neurofeedback.

2.3.1 Meditation and Mindfulness role in preventing and delaying MCI

In the absence of disease-modifying pharmacological treatments for mild cognitive impairment and dementia or slow its progression, nonpharmacological interventions for these conditions are developed. Numerous treatment modalities, such as mindfulness,

meditation and cognitive training, aim to prevent or delay cognitive decline, particularly in the early stages of the disease, while also potentially improving mood and quality of life (Reynolds et al., 2021) and improving cognitive functions, such as executive functions and attention, as well as brain structure (Crescentini et al., 2017). Moreover, there has also been research showing the effectiveness of meditation in reducing depression, stress, and anxiety, as well as MCI. Recent research has also suggested that meditation can delay cognitive decline in the elderly (Crescentini et al., 2017) which could also be caused by MCI. Because MCI is associated with motor dysfunction, which is linked to a potential increase in cognitive decline in the elderly (Maruya et al., 2021). A research laboratory in 2015 showed that meditation significantly increased cerebral oxygenation, enhanced performance, and activated the prefrontal cortex (PFC). Also, fNIRS was used to evaluate the relative hemodynamic changes in the PFC during a cognitive task and an increase in prefrontal cerebral blood flow (rCBF) due to meditation was observed (Deepeshwar et al., 2015). There have also been researches showing the effectiveness of meditation in reducing depression, stress, and anxiety, as well as substance abuse problems. Recent research has also suggested that meditation can delay cognitive decline in the elderly (Crescentini et al., 2017). For example, Park et al. (2017), have reviewed how meditation can be used to reduce stress and improve cognition, eventually reducing dementia symptoms. These studies have demonstrated meaningful results in terms of reduced cognitive decline, reduced perception of stress and increased quality of life, as well as increasing functional connectivity, percentage changes in the brain, and cerebral blood flow in the cortex (Park & Han, 2017). In another project Reynolds et al. (2021) also reported that there is potential benefit in cognitive and mood improvements from mindfulness-based and cognitive training interventions for adults with MCI which is a safe approach in elderly people. They suggested mindfulness could benefit aspects of attention, psychomotor function, and memory in MCI, while cognitive training has shown positive benefits on general cognition, memory, working memory, and executive function (Reynolds et al., 2021). Cognitive training is a mean to enhance cognitive functions and improve performance in any activity requiring cognitive abilities (Simons et al., 2016; Tang & Posner, 2014), and mindfulness meditation is a type of cognitive training that enhances attentional control, emotional regulation, and self-awareness by developing brain

networks (Tang et al., 2015; Tang & Posner, 2014). There are two main types of mindfulness meditation: focused attention meditation (FAM) and open-monitoring meditation (OMM)(Lutz et al., 2008; Miyoshi et al., 2020). When FAM is used as an inhibitory control of attention towards irrelevant stimuli within a cognitive task, it increases the top-down attention control (Colzato et al., 2015, 2016; Deepeshwar et al., 2015; Duncan, 2010; Lippelt et al., 2014). In one study (Yamaya et al., 2021) the effect of one-session FAM and control intervention method on Working memory capacity (WMC) were compared. In their study, Su-soku (Chiesa, 2009; Lutz et al., 2008) meditation as a FAM intervention was compared with random thinking intervention as a control. Su-soku meditation is a traditional meditation method in which the participants by focusing on their breathing, counting their breathing cycles, and maintaining their focus on their breath. In their study, the researchers found that top-down attention control during FAM is associated with bilateral dorsolateral prefrontal cortex (DLPFC) activation as well as increasing WMC among meditation novices (Yamaya et al., 2021). It has been shown that mindfulness therapy can be effective in preventing dementia in MCI through a low-cost and self-directed psychosocial program. Studies have also shown that mindfulness therapy can reverse cognitive decline in MCI (Ng et al., 2021). Studies regarding the effects of one-session FAM on novices' WMCs and monitoring their activation of the DLPFC by using fNIRS are rare and Yamaya et al. (2021) are the first one. According to the study results, significant bilateral activation of the DLPFC was observed, which was also correlated with a significant WMC improvement. A one-session FAM leads to a change in consciousness following top-down attention control (including inhibition) or DLPFC activation. Conducting a one-session FAM before a class or job can be much more helpful. Additionally, these findings may assist in establishing the impact of one-session FAM on meditation novices (Yamaya et al., 2021).

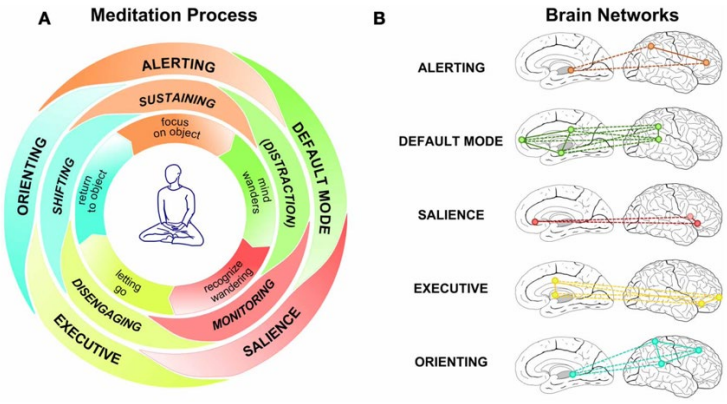


Figure 4 An illustration of the meditation process
is shown in diagram (A). The inner-circle illustrates the phenomenological layer by showing the typical order (clockwise) of meditation activities. This inner circle represents the process of attention that lies beneath, while the outer circle represents the different brain networks that are involved in these functions. There is some overlap between the attentional processes and the brain networks to illustrate that in many cases more than one process/network is involved. diagram(B) illustrates the major brain regions related to each of the five networks. Anatomical details are discussed in the main text (Malinowski, 2013).

Using an app-based breathing intervention, Chelidoni et al. (2020) examined the effectiveness of the intervention in enhancing physiological recovery. Study participants were examined for changes in Heart Rate Variability (HRV), which is a biomarker of cardiovascular health and a measure of the autonomic nervous system. Participants were guided to breathe according to their heart rates using a commercially available app. As part of this study, participants who participated in the BioBase breathing condition engaged in a guided diaphragmatic breathing exercise for five minutes, focusing on breathing, using all of their lung capacity, and decreasing their breathing rate to trigger the vagus nerve. Using the commercially available app to deliver biofeedback breathing interventions can facilitate stress recovery and have multiple positive health effects, the authors concluded. Individuals can use these apps to learn how to change their physiological activity in order to improve both their health and cognitive performance (Chelidoni et al., 2020).

2.3.2 Exergames, music, and Cognitive Stimulation game's role for preventing and delaying MCI

a) ZPLAY

ZPLAY is a web-based game-based user interface system that provides Alzheimer disease (AD) intervention therapy through a game-based interface (Makedon et al., 2010). The purpose of this game is to improve the well-being of caregivers and family members of people diagnosed with AD or MCI as well as to assist General Physicians (GPs) in diagnosing people with signs of dementia (e.g., memory changes in cognitive processes). They used fNIR as the imaging sensor in the ZPLAY system to combine cognitive and physical behavioural data. ZPLAY helps gather and record valuable information, engages the subject in a fun activity, and provides the person with an increased level of independence that improves the quality of life of the individual suffering from the dementia-related disease (Makedon et al., 2010).

b) neurofeedback

In 2021 Nouchi et al. developed a portable cognitive training with neurofeedback (CT-NF) system using 2ch NIRS (near-infrared spectroscopy)) and conducted a randomized control trial (RCT) to compare CT-NF with cognitive training (CT) alone and active control (ACT) on cognitive functions. Based on the results of this study, CT-NF increase DLPFC activity after four weeks of training. Additionally, the CT-NF improved episodic memory, working memory, and attention more than any other training group. As a result of enhancing left DLPFC activity, all cognitive functions improved. Based on these findings, CT-NF enhances cognitive performance and the activities engaged in during training contribute to its beneficial effects (Nouchi et al., 2021).

c) virtual HIE-based intervention

In another study Bruin et al. (2020) specifically, they sought to demonstrate physical, cognitive, and neural benefits from a virtual high-intensity intermittent aerobic exercise (HIE), HIE-based intervention (vHIE) performed exclusively by the virtual body. As a result from the experiments and using fNIRS, it was discovered that physiology of the body could be directly affected by the virtual full-body illusion that was produced by an immerse virtual reality (IVR), they also confirmed that cognitive component and their neural

substrates can benefit from virtual training with certain characteristics and it can be used by people suffering from cardiopathy or neurological conditions that affect their motor skills (Burin et al., 2020).

d) Dual-task

In another study (Sugihara et al., 2021), the Activation of the Prefrontal Cortex and Improvement of Cognitive Performance with Standing on One Leg was investigated. The study indicates that postural task-dependent changes in PFC activity affect cognitive performance. The prefrontal cortex (PFC) is involved in executive functions, attention, and motor planning. Additionally, dual-task paradigms have been shown to impact both postural control and cognitive performance, suggesting that these two functions utilize the same brain areas. The N-back task is a continuous performance task and is used as a functional test of the PFC (Kirchner, 1958). CBF was measured with a portable device (HOT-2000, NeU, Tokyo, Japan) and a data acquisition system (HOT Measure Ver 3.0, NeU). As a result, the cognitive performance was affected by postural task-dependent changes in PFC activity and HbT increase in the PFC under challenging motor regulation would lead to concentration on cognition ability (Sugihara et al., 2021).

2.4 Research Question

As demonstrated in the previous sections, with an aging population, it is essential to provide solutions that can improve the quality of life for elderly people. With age, elderly people often develop cognitive disabilities, weaker memory, disorganized behaviour, and even physical limitations that are closely related to MCI, in addition to fear and anxiety regarding interacting with technological systems. There are several methods for preventing MCI through cognitive training sessions. However, there are some research gaps in terms of a protocol for regular unsupervised cognitive training sessions for elderly people to prevent MCI that is going to be addressed in this master's thesis. As previously stated, many types of research have been conducted to improve cognitive ability through some cognitive sessions, but none of them specifically worked on the unsupervised cognitive sessions of elderly people whose brain ability has been declining for years. In other words, in the previous research user did not participate in the design of the protocol using relative devices and applications (ABC and Ex_Brain device). The target population

of this study is elderly people aged +65. In this study, an application will be investigated and developed in collaboration with French psychological experts, Japanese software developers and data analysts, Japanese XB-01 device researchers, and elderly people in Germany, France, Japan, and Italy by bringing together their real needs and demands. Psychologists' opinions will be included to estimate real possibilities, then during cognition workshop sessions, elderly interaction in the setup will be evaluated to estimate their real needs, and by consulting with app developers in Japan application potential and setup possibilities will be learned. Because Stakeholders (such as older adults and caregivers and relatives), in particular, could have very different expectations of what a brain cognition training application is than researchers. It is intended to use the findings of this investigation to create an artifact that will address the identified problems by enhancing practitioners' ability to create new or changed practices. Furthermore, to create context-specific artifacts that allow users of such artifacts to further develop their practices; the grounded design approach refers to such endeavours as "design case studies" (Wulf et al. 2011, 2015). This emphasizes the importance of considering the needs of seniors, psychologists, and neurologists, and incorporating their own feedback along the way. New method using participatory methods will be designed. User experience and usability should be the focus of the evaluation. The analysis will consist of qualitative data such as interviews, as well as quantitative usage metrics (Lazar, Jonathan, Jinjuan Heidi Feng. 2017). The development of cognition training rarely takes into account the target group. Based on these reflections, the following research questions are derived:

1) **Which are motivational factors and challenges of older adults of doing cognitive training sessions at home by using neurofeedback non-invasion methods?**
2) **How should features of an application be designed to encourage older adults to do the cognitive training sessions regularly and unsupervised?**

3 Methodology

3.1 Design Case Study

Understanding the experiences and private environment of elderly people is important for having meaningful interactions with them. Therefore, a design case study is selected as a research strategy, it involved an in-depth investigation of an unsupervised cognitive training session for elderly people at home. The design of case study has three parts (Figure 5), empirical pre-study, design and evaluation and appropriation. In this study, a repetitive cycle of pre-study, design and appropriation was done. Elderly active but retired people are selected as the target group and the setting of the event is living lab in the Siegen University. Information is collected via expert review, interviews, observations, usability tests, instant collaborative co-design and self-report of a user. With the aim of discovering users' needs and abilities as precise as possible users were actively engaged throughout the research and design process, including co-design activities in participatory design. The aim was to discover An in-depth study of the interactions between IT design and appropriation activities is limited (Wulf et al., 2011; Wulf et al., 2015) and therefore, more studies are required.

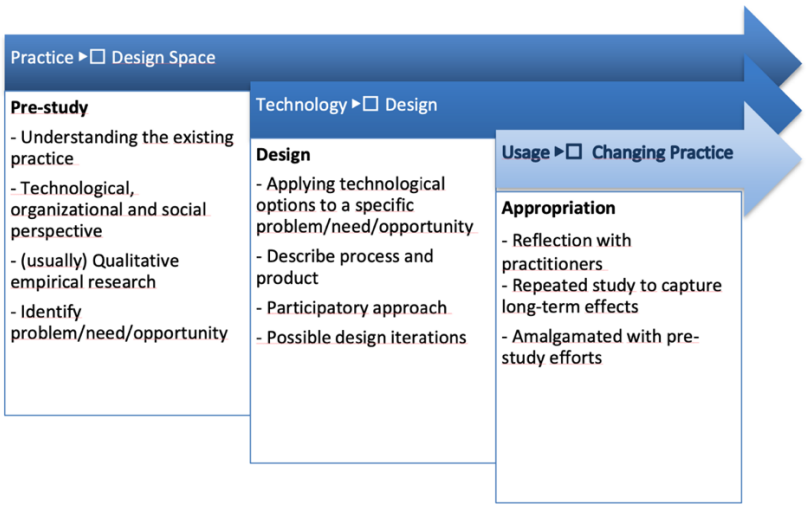


Figure 5 Design Case Study structure (Wulf et al., 2015)

HCI research is driven by direct feedback from interested individuals. There were some main questions to be asked from the user: What does a new tool need to do? Does the

design meet their needs? Are there any areas that need to be adjusted or revised?. There are different methods which each one has their own pros and cons, for example doing a survey method as a qualitative method could be very useful if a large number of people can be reached easily. Most surveys, however, are rather limiting: respondents only respond to the questions asked, so open-ended questions that invite lengthy responses are unlikely to be answered. This causes surveys to be broad, but not comprehensive. In this study, a better approach is to be specific, but not broad. By speaking with fewer participants, surveys cannot provide valuable data. Instead having conversations and interacting with the right people could be both a tremendous source of insight and a considerable challenge (Lazar, 2017). Other options could be interviews and/or focus groups. Interviews with individuals and focus groups that involve multiple users at the same time are the two most common ways of direct discussion with participants. Using focus groups or interviews to gather information has different strengths and challenges: deciding which strategy to be used is perhaps the first question that was answered. Structure and timing are also important considerations. It was decided to conduct a semi-formal, semi-structured interview rather than a fully structured interview. Between focus group and interview, the interview method was selected for several reasons. First, target group was selected based on their age and being retired, but they could be very different regarding digital literacy. This different background could make a few of them dominate the focus group session, then some of the inexperienced ones lose the opportunity to declare themselves, and consequently, their opinions and needs remains unrecognized. The other reason was that because of COVID-19 health protocol made gathering elderly people in a closed room difficult. They were the most fragile group against COVID-19 due to their age range. The other method is observation, observing users in natural settings in an undetectable manner might provide us with valuable insights into how the specific ICT is used in everyday life, but it can be extremely time-consuming. It is possible to wait for hours to discover while none of the individuals being observed have used the functions in which the researchers are most interested (Lazar, 2017). Even if it is possible to recruit a large number of users in a short period of time with the survey approach, the participants might misunderstand the questions. Further the data collected may not be reliable enough, and might be biased. The questions could be clarified during interviews and

explore participant responses in-depth with follow-up questions. In comparison to surveys, interviews are expensive. Usability tests could be helpful to identify key usability problems in an interface or application quickly and at a reasonable cost, but they do not guarantee that all critical design problems can be detected (Lazar, 2017). During the study, some of the users were eager to collaborate more actively, therefore, it was possible to engage them into the design platform, and do a collaborative instant co-design, that the P3 could see what and how his own ideas were applied to the prototype. Moreover, he did regular tests by his own at home, produced report documents based on his tests and send it. Finally, interview, observation, expert review and usability tests, collaborative instant co-design and self-report were chosen as the qualitative method for this study and the design and appropriation phases were repeated in several cycles.

The Bo system as a whole is a platform that includes cognitive training applications (ABC app and BrainHQ app), relaxation practices, and three types of training session feedback. Even though the Bo app prototype was designed for this study, the ABC app and BrainHQ app were used as system components. In other words, neither the ABC app nor the BrainHQ app was developed specifically for this study; they have already been ready to use. But the result of this study is reflected to the origin ABC developer, and some of them were applied in later interviews. Several useful points about the Bo app were discovered during the tests, reflected to the original designer (only for the ABC app in the Japanese partner of the E-VITA project), and some changes were made to the ABC app based on the reflected comments and feedback from the chain of interviews conducted in this study.

3.2 EVITA research project

This thesis is part of e-VITA project (Homepage of E-VITA, 2022). The e-VITA project is a European and Japanese collaboration to develop a virtual coaching service for older adults that uses ICT technology. With e-VITA (2021-2023), older adults in Europe and Japan will be able to actively age in place, function independently, and have a reduced risk of social exclusion. The multidisciplinary consortium behind this project will develop an innovative ICT-based virtual coaching system to detect gradual changes in older adults' physical, cognitive, psychological, and social well-being. In a smart home environment, e-VITA can provide personalized recommendations and interventions for sustainable wellness. This project is using device such as SanTo, NeU(ExBrain), Android, NAO, Gatebox, DarumaTO, which my thesis focus is on NeU(ExBrain). In the e-VITA platform, three thematically distinct domains are nevertheless interdependent and overlap in certain areas. Use of the e-VITA platform to facilitate meaningful moments in daily life (e.g., spirituality). Support, coaching, and interaction for the enhancement of subjective wellbeing. Health Coaching: to provide educational content, guidance, and coaching for health-related topics such as physical fitness, nutrition, cognition, and mental health. Utilization of the e-VITA platform to facilitate the social connectivity of older adults includes new forms of social participation and family involvement. For the elderly, the main reason for using an e-VITA coach is the satisfaction of their fundamental psychological needs. E-VITA coach interventions should be tailored to users' needs in order to provide a good user experience and increase subjective well-being as well as increase subjective well-being. All psychological needs should be considered. For example, with the E-VITA coach, older adults can get cognitive or physical training that is both stimulating and entertaining. Consequently, key domains relevant to the e-VITA coach were identified, and examples of successful practices that are supportive of subjective wellbeing were offered. Based on one of the reports from E-VITA project, Report on everyday practices that support subjective wellbeing, a partial list of relevant domains is as follows:

- Family life
- Romantic relationships

- Peers and friends
- Applying skills (e.g., knitting)
- Sharing knowledge (e.g., mentoring)
- Performance (e.g., singing, acting)
- Local communities of practice
- Care for pets
- Care for nature
- Relaxation
- Nature
- Daily Structure
- Sport and exercise
- Body care (e.g., hairstyle)
- Entertaining
- Games and Puzzles
- General knowledge (e.g., news)
- Self-knowledge
- Loss and death
- Spirituality
- Reminiscence

Care for nature, Relaxation, Entertaining, Games and Puzzles and Spirituality are directly covered in the designed protocol in this thesis and some of the other domains, based on the individual tastes, are able to be indirectly coverable because users can choose also the other activities in the Bo app activity dashboard (more info in Methodology chapter).

NeU System (Ex-Brain) and ABC

NeU (Neu-Brains, 2022) is a Brain Science company that was founded in 2017 by fusing the “cognitive brain science discoveries” with the “portable brain measurement technologies”. Ultra-compact Ex-Brain sensor, weighing 30 grams (battery included) and measuring 80 X 40 X 13 mm with a “butterfly-style” shape that bends in the middle to conform easily to any individual’s forehead it could be used in conjunction with a band-

type holder and stick to the forehead of users. This device uses optical topography devices (NIRS) to measure brain activity through cerebral blood flow volume changes in real-time.



Figure NeU device (XB-01)

The Active Brain CLUB (ABC) is a service that uses the XB-01 to improve cognitive health that uses NeU's XB-01 to measure brain activity during cognitive function training sessions and is intended for senior citizens in their who want to maintain their brain functions. The service combines the XB-01 and a visualization application for conducting cognitive function training sessions. For example, an individual's focus and memory ability can improve, leading to an improvement in brain performance. Since continuing training is vital, the application also includes several added programs such as different video games to improve cognitive function. The XB_01 device, in conjunction with the ABC application, will use fNIR to measure the cognitive load of the user's brain and provide feedback on their brain abilities in three different areas: Mental Speed, Concentration, and Memory via weekly self-check and real-time reports. Cerebral blood flow (CBL) will be measured using the XB_01 device.

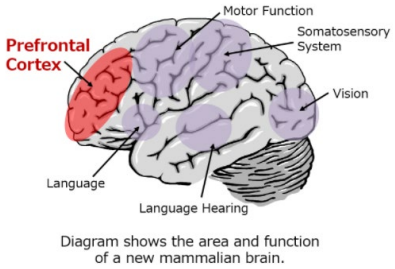


Figure 6 The prefrontal cortex, which is part of the frontal lobe in the cerebral cortex, controls cognitive functions such as memory, thinking, attention and inhibition. The decline of the prefrontal cortex blood flow leads to a decline in cognitive function which could be measured by Ex_Brain (NeU) sensor. The functions of the prefrontal cortex of the brain are: Short Term Memory, Attention, Thoughts, Decision Making and Suppression of Behavior

3.3 Participants

For the study, a total of five senior citizens (+65) were recruited in a series of 12 co-design sessions. They were recruited from previous parts of E-VITA project and mostly from a Szene-Café where old people were taught about ICTs voluntarily and they were learning from each other. The interviews, observations and co-design sessions were done in the living lap in Siegen University, as well as in their private homes and online. The list of the participants is shown below (Table 1).

#	ID	Age	Sex	Living situation	Job status	Technical competency	Took part in co-creation session
1	P1	65	M	Living alone	Retired	Relatively less experience	1st
2	P2	64	F	With husband	Retired	Good experience	5th,3rd
3	P3	71	M	With wife	Retired, active in industry, finance and personnel	Relatively less experience	6th, 7th
4	P4	80	M	With wife	Electro engineer, retired	Has very good experience	4th,2nd, 8th, 10th, 11th
5	P5	71	M	With family in one house	Retiered,	Has experience	9th

Table 1 Demographic information of participants.

3.4 Materials and methods

The study consisted of iterative sessions, which older adults shared their ideas for the design. Furthermore, two HCI experts participated in the online expert review and cognitive walk-through meetings. To get fresh perspectives and unbiased opinions, new

participants were brought in during the different phases of the study. Some participants withdrew their participation after a certain point because they had lost interest in co-designing of this technology. In the course of our 11 co-design sessions, the commercially available BrainHQ app and the ABC app together with its NeU brain measurement device, as well as the overall expectations for cognitive training were explored. The overall goal was to evaluate the ABC app and BrainHQ and develop an overarching application that provides a better user experience. As such, the Bo-App integrates interventions from the ABC App and BrainHQ, compensates shortcomings (e.g., lack of user onboarding and setup instructions) and adds additional features (e.g., motivating feedback and progression indicators). In the following, an overview over the content each session is briefly presented:

1st, 2nd and 3rd and 4th Session | Participant: P1, P2, P3, P4 | Bo-App Prototype Version: - | Phase: Pre-study

Material: Impression of NeU Device without ABC App, BrainHQ was used

Activity: Discovering their familiarities with ICTs, general opinion about cognitive training and games to improve cognitive ability, if a NeU sensor device (XB-01) is acceptable for them, what kind of activity is mostly their favorites to work with as a brain training activity.

5th Session | Participant: P3 | Bo-App Prototype Version: V1.0 | Phase: appropriation

Material: First prototype version of the Bo-App, the impression of NeU Device without ABC App, BrainHQ was used

Activity: Usability test to evaluate Bo-App, observing and monitoring reactions and body language. Asking opinion about using such a system to assess brain performance, the way of feedback presentation (feedback visualized in the shape of a growing plant or tree), and their interactions with the NeU device setup observed were explored during the interview and usability test.

6th Session | Participant: P2 | Bo-App Prototype Version: V1.0 | Phase: appropriation

Material: First prototype version of Bo-App, ABC App, NeU Device and BrainHQ were used during the test

Activity: Usability test to evaluate two different applications (ABC app and BrainHQ) as well as a first prototype of the Bo-App. P2's opinion about using such a system to assess brain performance, the way of feedback presentation (growing plant or tree), and her interaction with the NeU device setup instruction were observed and explored during the interview and usability test.

7th Session | Participant: P3 | Bo-App Prototype Version: V2.0 | Phase: mid-appropriation

Material: second prototype version of Bo-App, No ABC App, impression of the NeU Device

Activity: Usability test to evaluate Bo-App, interview to discover his opinion about the V2.0 and Observation of his reaction

8th Session | Participant: P4 | Bo-App Prototype Version: V2.0 | Phase: mid-appropriation

Material: second prototype version of Bo-App, No ABC App (due to missing android Device), impression of the NeU Device

Activity: Collaborative Instant co-design (by sharing Figma design page on his screen), interview and observation

9th Session | Participant: P4 | Bo-App Prototype Version: V2.0 | Phase: mid-appropriation

Material: ABC App limited functionality (bugs due to ongoing development), second prototype version of the Bo-App, impression of NeU Device, BrainHQ

Activity: Usability test and interview to evaluate new version of Bo-App, and testing ABC app (no game but brain meter section)

10th Session | Participant: P5 | Bo-App Prototype Version: V3.0 | Phase: Final-appropriation

Material: ABC with limited functionality (bugs due to ongoing development), third prototype version of the Bo-App and BrainHQ was used.

Activity: Interview and usability test for three apps. Self-report for Bo app and ABC app. Observing P5's interaction with Bo-App and exploring his opinion about the feedback method. Usability test about ABC app doing a walk-through test in the ABC app (game part) together but the brain meter was not working.

11th Session | Participant: P3 | Bo-App Prototype Version: V3.0 | Phase: Final-appropriation

Material: ABC App (both parts: real-time neurofeedback and games), NeU device on his own beforehand, limited functionality during the interview (bugs due to ongoing development), third prototype version of the Bo App.

Activity: Interview to evaluate what P3 experienced with the ABC app (both part gaming part and real-time neurofeedback part), co-design and improving verbal part of the Bo-App prototype. But during the interview due to using the last version of the ABC app, games were out of access.

12th Session | Participant: P3 | Bo-App Prototype Version: V4.0 | Phase: Final-appropriation

Material: Fourth prototype version of the Bo-App

Activity: Collaborative Co-Design in Figma, labelling and text

3.5 Pre-study methodology

3.5.1 Interviews

The participants were interviewed by a researcher and an accompanying German speaker (due to language barriers). The conversations were held according to a semi-structured interview guideline. The interviews took about 1-2 hours and took place partly online and partly in person (i.e., according to contemporary pandemic restrictions). For the on-line sessions one NeU device was sent to the P4 and all the interviews with him were conducted via Zoom. While the German speaker conducted the conversations, the researcher observed and guided the sessions.

After warming up the introduction presentation was done using PowerPoint to present the general idea and the aim of the project (in German, French, Italian and Japanese languages, because the prototype supposed to be tested in these partner countries of e-VITA project). There were some introductory questions about their life style such as state of health, social network, marital status, employment status and their job, then they were asked to describe their daily routine. They were asked about their experiences with technology artifacts, computer games, meditation, and physical exercise.

Questions and tasks for usability test were included in interview guideline. Participants worked through the predefined tasks and the evaluator (moderator) takes notes on what has been said. Participants also were asked if there are some cognitive issues in their daily routine tasks; or preferred activity in doing a cognitive training session to boost brain ability or prevent MCI-related diseases. Later, the content of the prototype was developed using the gathered information from user's test sessions. Furthermore, expert opinions are included in two areas, psychological area to satisfy psychological and technical aspects.

All the interviews (except one that was done in English), were conducted in the German language, Interviews were transcribed by Meetgeek (Meetgeek Virtual Meeting Assistant | AIVISION, 2022) platform. Later on, translated into English language by Deepl Translator (DeepL Translate, 2022). The transcribed interviews were coded by MAXQDA software. Based on the questions in the guide, three categories were developed.

3.5.2 Observation

In the exploratory phase of the design process, a casual or semi-structured observation (Nasar, 2007) was used to gather baseline information. There was a set of guiding questions, and observations were full consciously conducted, so deviations from the original plan were permitted in response to unexpected events (like when an unexpected error was happening). Despite the fact that ethnographic observations were informal, they were systematic and well documented with field notes, sketches, photographs, and raw video footage. Semi-structured observations were typically synthesized to guide design inspiration. Because the sample size was not big enough the use of the prototype and product was simply checked. During the observation to extract more realistic inference of the user's behaviour more questions were asked ("Universal Methods of Design," 2012).

3.6 Design

3.6.1 Basic Consideration of prototyping stage

Based on the empirical analysis of the pre-study the first version of the prototype was designed in Figma, the web-based design platform. Several items were considered during

the design process. Visibility of system status was considered during the prototyping. For example, several short descriptions were used in the design to keep users informed about status of every page. To make the design more realistic it is tried to use a language which is as close as possible to the users' language. For example, by using the words, phrases, and concepts familiar to the user, rather than internal terminology. Terms, concepts, icons, and images were selected based on the specific target group. By following real-world conventions and corresponding to desired outcomes (called natural mapping), possibly, it is easier for users to learn and remember how the interface works. This helps users to build an experience that makes them feeling intuitive. In the Bo app also stimulation-response compatibility (Paul Fitts and Charles Seeger in the 1950s (Expert Review in the Usability.de Glossary, 2022)) was considered. For example, in the "taking tour" guide users should drag the page to the left and right to forward and backward the pages.

Consistency and Standards

Consistency and standards of the words, scenarios, or actions are other points that were considered. When users are forced to learn something new as a result of not maintaining consistency, their cognition can be enhanced. Therefore, using new terms were avoided and instead the familiar ones were used. Some of standards (i.e., font size) (Experience, 2022) had to be dropped from the design as elderly target group had some limitations such as eye sight, physical movement impairment and so on. Therefore, some parts are structured according to the target group-specific needs. For example, in this study the font size is selected not based on the usability standards but based on the user's need.

Learnability and Efficiency

There are two basic usability metrics of the design: learnability (the user's ability to use a new design they have never seen before) and efficiency (the speed at which they complete tasks after learning the interface) are also other considerations. Five quality components of usability are considered (10 Usability Heuristics for User Interface Design, 2022) here includes Learnability, Efficiency, Memorability, Errors, Satisfaction. Learnability is about how easy it is for users to perform a task on their first encounter with the interface, as well as how many repetitions it takes them to become efficient (how quickly users can perform a task once they have learned it). Expert users prefer to spend

time on a task as less as possible and learn about the system quickly. So, designer tried design the app in a way that is efficient for expert users as well as new user. Balance between efficiency and learnability by considering shortcuts were made.

Cognitive Load

Users' ability to find content easily and complete tasks depends on the level of cognitive load on the app, therefore, Minimizing Cognitive Load to Maximize Usability(Experience, 2022) has been considered as well. Performance suffers when the amount of information coming in exceeds our ability to handle it. When users are unfamiliar with information, they may overlook important details or even get overwhelmed and give up. In user experience, “the cognitive load imposed by a user interface is the number of mental resources that are required to operate the system.“ (Experience, 2022) In psychology, cognitive load is defined as the effort required to learn new information (Experience, 2022). Users have to learn how to use an app's navigation, layout, and transactional forms. Although the app is fairly familiar for them, users have to still carry around the information that is relevant to their goal. Cognitive load cannot be eliminated entirely - even if it was possible, it would be unwise to do so (Experience, 2022). In the end, the app is used to train the brain and receive feedback. Intrinsic cognitive load is the effort of absorbing that new information and of keeping track of their own goals. However, it tried to wipe, or at least minimize, extraneous cognitive load in this study, for instance, mental resources are taken up by information that is not essential, such as font styles that do not communicate anything unique. Attention is a precious resource that should be prioritized accordingly. A number of usability guidelines are geared toward minimizing cognitive load, including chunking content and optimizing response times. To reduce cognitive load even further, three more tips were considered: visual clutter, redundant links, irrelevant images, and meaningless typography were avoided to prevent slowing users down. Users based on their previous experience with other applications, already have mental models of how applications work. To reduce the amount of learning, labels and layouts that they have encountered on other applications were used. Offloading of tasks was done by identifying tasks which required users to read text, recall information, or make a decision, and then replacing them with alternatives such as showing a picture or redisplaying

previously entered information. Consequently, by eliminating some tasks, more mental resources were available for the most important decisions.

Recognition rather than recall

Users were prevented from recalling information by adding field notes and menu items. They only need to understand what they can see to perceive the interface. A sufficient amount of information is visible to them. For experienced users, shortcuts are considered, then the user can customize frequent actions based on their preferences. Users can skip the tour on the first page and go directly to the login page in this study.

Aesthetic and minimalist design

It is tried in this study to eliminate irrelevant and rarely needed information, to prevent diminishing their relative visibility in comparison with relevant elements. To keep the content and visual design focused on the essential information to guarantee that visual elements are supporting the user's primary goals.

Error tolerance

Traditional error messages such as a yellow rectangle with an exclamation mark were designed, and a short but complete description were included about what is the problem and what they should do to correct it. For example, when the NeU device is running out of the charge the error page come up and shows the solution. Furthermore, it is common for users to make mistakes in their actions, therefore, an emergency exit should be provided. To do so, they can leave the unwanted action without having to go through a long process. To foster a sense of freedom and confidence for users, "Back" button in the main menu under all the pages were provided. This allows them to remain in control of the system and avoid feeling frustrated.

3.6.2 Instant collaborative instant co-design with the interviewee

An online zoom collaborative co-design was held on with P3. He was looking into the shared screen of the Figma design environment and actively expressing his opinion and some of them were instantly implied by the designer.

3.7 Appropriation and Mid-appropriation phase

Semi-structured interviews and observation methods were conducted in the mid appropriation phase. For observation videos from the user's interaction with the system were recorded, the screen of the android tablet (the tablet which was used as the main android device) were recorded, as well as recording the voice of the interview. During an online interview with P4, an instant co-design was accomplished, and the designing page of the Figma prototype was shared with him. He was watching the design page and engaged himself by expressing his ideas, which were immediately applied by the researcher, and the participant was determining whether or not it worked in his opinion.

HCI experts conducted two online cognitive walkthrough reviews to conduct the expert review method. HCI experts participated by using the Figma platform that was shared and leaving comments on the Figma prototype. Concurrent usability tests were conducted with multiple interviewees and three different applications, Bo app, BrainHQ, and ABC were tested.

3.7.1 Usability test - Thinking aloud

Researchers look for ways to improve the usability of an interface based on the tasks people perform in usability testing. (Gould & Lewis, 1985). The Think-Aloud Protocol technique is typically used for usability testing. The number of participants plays a direct role in problem detecting (Virzi, 1992), and therefore when there are more participants more problems could be detected (Jacobsen et al., 1998). In addition to experiment validity, empiricism, and bias avoidance, the usability test observation session was the only opportunity to see and weigh in on the prototype. The participation of a second observer was therefore necessary.

In this study, the purpose of usability testing was to observe how the user interacts with a digital application as they walk through the steps of given tasks. The method enabled the researcher to determine which parts of an interface confuse users and eventually these problems can be solved. Testing was based on tasks and scenarios that are representative of end-user goals. Specific, concrete and measurable tasks were assigned. The participants were not be pressured to solve a problem in a certain way, nor should the tasks.

3.8 Final appropriation

A set of semi-structured interviews with a usability test were done in this phase to check if the designed prototype fits into their individual needs and preferences. During the usability test, end users were asked to think aloud, this way is will be possible to recognize what is the exact problem and why it occurred. Meanwhile, some questions were mentioned in order to remind the users to think aloud cause during the test they were focused on doing tasks and sometimes could forget to tell what they are thinking and doing in real-time. They were told that: “We are not testing you and your ability, but we are testing the system.” Researchers had to interfere when they were stuck, therefore the process could go further and test the next pages of the app. Users went through the app freely and explored it. Afterwards, they did predefined tasks presented in a printed form, or on the laptop. Meanwhile, they loudly expressed their thoughts and sometimes they were reminded again to speak out their opinion loud by researchers.

- “What are you thinking right now?”
- “What do you think does this notification mean?”
- “What do you think will happen?”
- “What reactions did you expect?”

Also, some notes about what is happening in the meeting were taken by the researcher. In addition, an online cognitive walkthrough was done by a team of two HCI experts, who were engaged in the Figma platform and Skype, they left comments on the Figma prototype.

3.8.1 Expert review

Design review (Expert Review in the Usability.de Glossary, 2022) is a method of usability inspection to identify usability problems. One type of design review is expert review. In the expert review, UX experts inspect the system to identify usability problems. The expert interview (Expert Review in the Usability.de Glossary, 2022) is called an expert analysis as well, a heuristic review or an evaluation. In the expert interviews, the UX expert examines a design in order to identify usability issues and strengths. By expert review method, in this study, a set of written documents were produced. Experts presented their

conclusions besides mentioning reasons behind of them in the meetings. Usability findings were ranked based on severity or frequency. Having an expert review done by a UX expert then leads to the question of who is a UX expert. The examiners were chosen in a way that they not only have in-depth knowledge of usability best practices, and many years of experience conducting usability research, but also not have been involved in creating the design to be reviewed. Therefore, the outsiders could be more likely to give candid feedback because they are not emotionally invested in a design, are not affected by internal team politics, and can easily identify glaring issues that may go unnoticed by someone who has been staring at the same design for a long time. There were some minor issues that could be difficult to observe or measure in a qualitative study, according to expert reviews: It was possible, however, to detect issues that an expert may have missed, for instance when the user has very specific knowledge or requirements.

Cognitive walkthroughs

The prototype was analyzed independently by two experts to identify the majority of its usability problems. Often, heuristic evaluation and cognitive walkthrough (Cognitive Walkthrough in the Usability.de Glossary, 2022) are used in conjunction. In the early 1990s, Peter Polson, Clayton Lewis, John Reiman, and Cathleen Wharton introduced the cognitive walkthrough method from the University of Colorado's Institute of Cognitive Science. (Polson & Lewis, 1990). Polson developed the theory of exploratory learning, which is an approach to determining whether the cues and prompts in a system reflect how people think about tasks and anticipate their next steps (Polson & Lewis, 1990). Cognitive walkthroughs are used to determine whether a system is relatively easy to use in situations where preparatory instruction, coaching, or training are unlikely to take place. Each step of the interaction with a system can be seen as a step that either moves the individual closer to or further from his goal when an individual must actively engage with the interface in order to know what to do next, rather than relying on pre-existing knowledge of the system. The cognitive walkthrough is a technique usually used in expert reviews to identify usability problems without involving actual users. To be successful, it is important to identify beforehand who the users are and what tasks they perform with the product. Experts take on the role of users to complete their typical tasks based on this

knowledge. Every time the expert encounters difficulties or even is incapable of completing a task, they identify the cause and, thus, uncovers existing usability problems. In this study, three different online expert interviews were conducted with two HCI experts and one UI/UX designer. The link of the prototype that is designed in Figma was shared with experts and then they walked through the prototype, they left comments in the Figma and meanwhile there was an interview with questions like how does the prototype look like to you in terms of UI and UX. During the expert reviews, a set of questions were answered by the experts. The most important points are as followed:

1. List of usability strengths
2. List of usability problems
3. Severity rating for each usability problem (degree of severity was 1 or 2 or 3 and 1 means less severity and 3 is showing a high level of importance)
4. Recommendations

3.8.2 Self-report

In this method P4 was highly motivated about engaging into the study, and by his own suggestion he started to prepare some reports about the Bo app and ABC app. Consequently, he submitted some documents about his opinion and ideas to improve the apps.

4 Concept and ideation

A mind map (Figure 7) was used to generate ideas and explain concept. This mind map clarified the connections between various pieces of information in the early stages of the research. It is an effective mnemonic tool that promotes recall and understanding of a problem space due to its visual and diagrammatic nature. The steps are as follows: A focus sentence (central theme) is Independently Cognition Brain Training using dual task at home for elderly people considering different gender and culture (A design Case Study). The sub-branches were neurofeedback, constraints and condition, dual task, motivation, cultural variation, key questions. The final step was to strengthen concepts and their interconnections before declaring it complete in order to create new knowledge and understanding. The map is drawn via Miro (*Miro | Online Whiteboard for Visual Collaboration*, 2022) tool – an online web-based drawing tool to map the ideas, events and objects into visual materials.

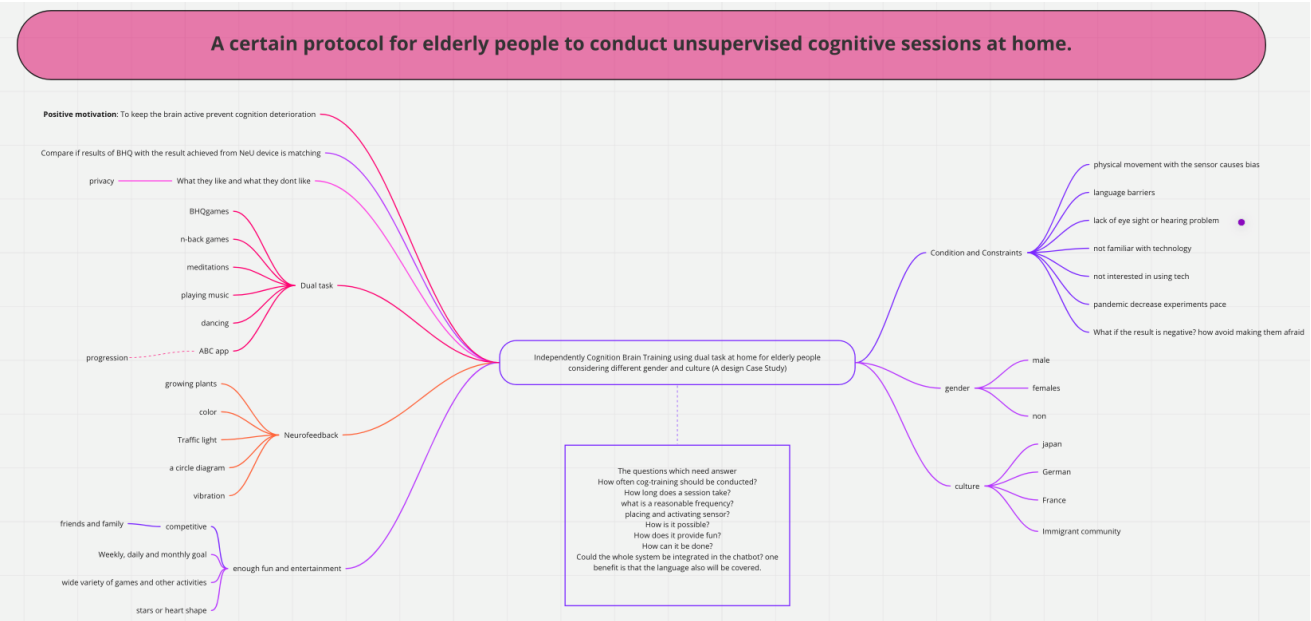


Figure 7 The mind map for analyzing early stages ideas based on the core idea

A concept map (figure 8) ("Universal Methods of Design," 2012) was used to incorporate new ideas into an existing understanding of the project elements. Individual concepts (NeU device, Bo app, BrainHQ and ABC app and E-VITA project) are linked by linking verbs in concept maps. When two or more concepts are linked by linking words, a

proposition is formed that results in a meaningful statement (Bo app. analyzes acquired data and provides users with feedback, NeU device is used by the users and so on. Using this method, the relationship between different elements in the project and the roles that they are playing was clarified easily.

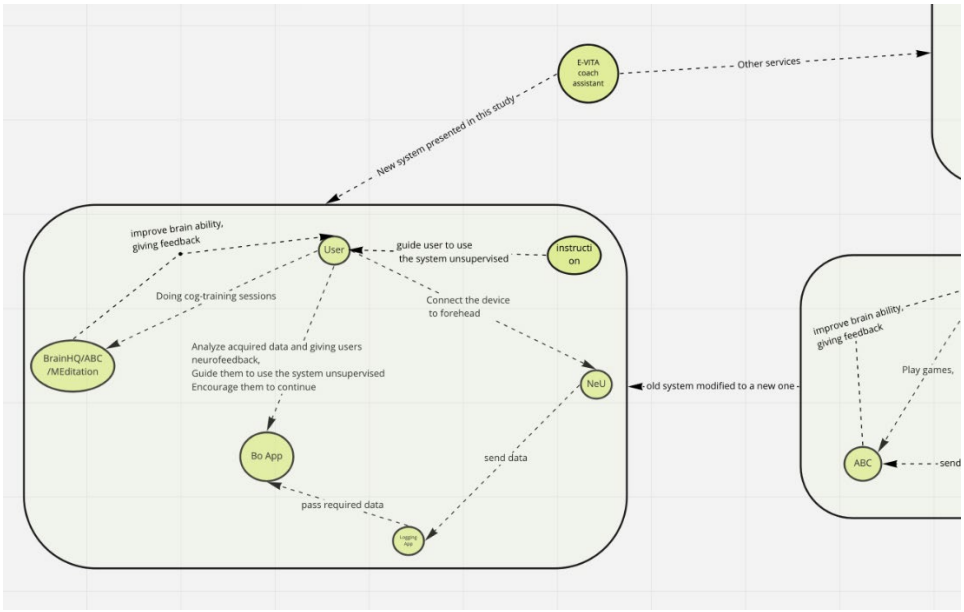


Figure 8 The concept map to find the relations between the project elements

A wireframe (Figure 9) was drawn to clarify the pages of the Bo App.

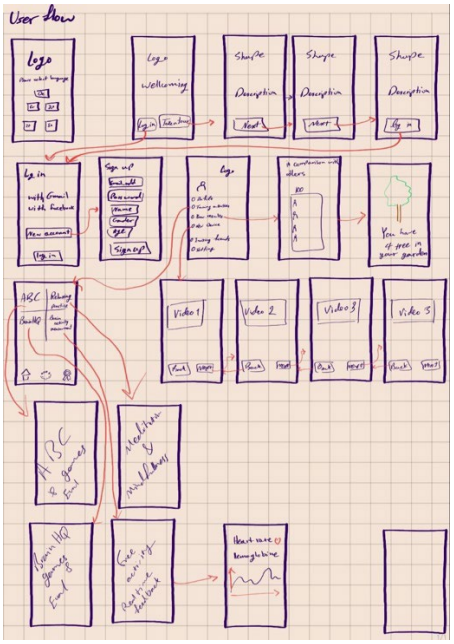


Figure 9 Wireframe of the Bo App

5 Description of the Bo System

The Bo system as a whole is a platform that includes various elements such as cognitive training applications (ABC app and BrainHQ app), relaxation practices, and four types of training session feedback that are shown in figure 11. Even though the Bo app prototype was designed for this study, the ABC app and BrainHQ app were used as components of Bo system. In other words, neither the ABC app nor the BrainHQ app was developed specifically for this study. Some useful points (issues with UI and UX designs for example lack of self-description) about the ABC app were discovered during the tests, sent to the original designers (only for the ABC app in the Japanese partner in the E-VITA project). Further, some changes were made to the ABC app based on the reflected comments and feedback from the chain of interviews conducted in this study.

5.1 ABC App and Cloud-Based System (STRAPI)

STRAPI

The STRAPI cloud-based data set was used for the system. ABC app is connected to the STRAPI and store data for every user for several aims, the end users' authentication and controlling their permission about having access to the different games and training feedback as well as increasing and decreasing the hardness of the games for every user based-on their last scores is possible via STRAPI remotely. First, a group for the institute was created, then a training plan with the date of the start and end of the project, an account for each user, adding different games to their account and therefore a QR code was produced for every user and they were able to have access to their predefined set of games via their personal account, and monitoring their results was possible through the STRAPI data base platform by researcher.

Active Brain CLUB (ABC)

The Active Brain CLUB (ABC) is a service that uses the XB-01 to improve cognitive health that uses NeU's XB-01 to measure brain activity during cognitive function training sessions and is intended for senior citizens in their who want to maintain their brain functions. The service combines the XB-01 and a visualization application for conducting

cognitive function training sessions. For example, an individual's focus and memory ability can improve, leading to an improvement in brain performance. Since continuing training is vital, the application also includes several added programs such as different video games to improve cognitive function. The XB_01 device, in conjunction with the ABC application, will use fNIR to measure the cognitive load of the user's brain and provide feedback on their brain abilities in three different areas: Mental Speed, Concentration, and Memory via weekly self-check and real-time reports. Cerebral blood flow (CBL) will be measured using the XB_01 device.

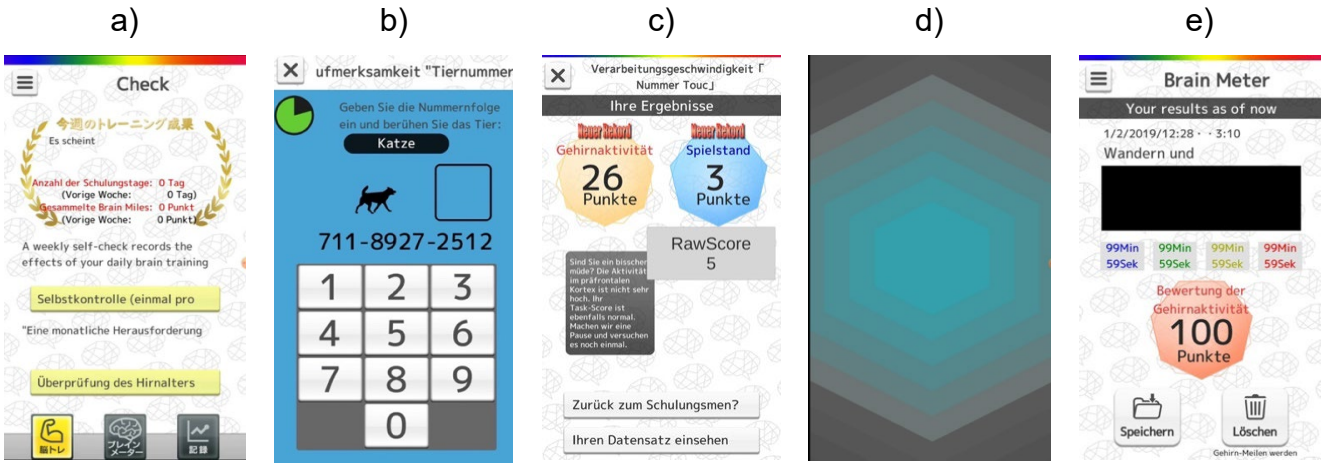


Figure 10 Different pages of the ABC app. a) Home page, b) A game sample, c) The feedback for the game shown in the (b), d) The real time feedback for an arbitrary activity (Brain Meter) e) The final feedback for the test shown in the (d) or Brain Meter result.

5.2 BrainHQ

Different neurobiologists provide a brain-training application with the goal of improving users' cognitive functionalities through entertaining computer games (BrainHQ) (*Brain Exercises, Brain Training, Brain Health – BrainHQ from Posit Science*, 2022). The app adjusts the difficulty of computer games based on the user's performance in previous games. For example, if the score is high, the level of complexity rises; if the score is low, the level of complexity falls. With each BrainHQ exercise, the brain is forced to process information accurately and quickly. Intelligent algorithms select the right challenge for each exercise, ensuring that it is not too easy or too challenging to induce brain change. Through BrainHQ exercises, the brain is rewired and better activation, timing, and

connectivity between brain regions are improved. The brain changes as a result of learning, experience, and training-a process known as brain plasticity. Improving brain function requires making these physical changes to the brain. Each BrainHQ exercise tests the brain's ability to process information quickly and accurately. Every exercise is designed to keep the challenge at a level just right for promoting brain change, not too easy, but not too hard. By rewiring the brain, BrainHQ exercises improve brain activation, timing, and connectivity between different parts of the brain. In addition to functional magnetic resonance imaging (fMRI) and electroencephalography (EEG), various advanced brain imaging techniques can be used to measure the brain's response to sounds, sights, or actions. The BrainHQ exercises are proven to activate specific parts of the brain related to cognition. A fMRI study (Scalf et al., 2007) shows training-related activation in visual attention areas. A further technique is to measure how quickly information travels through the brain using advanced techniques. Faster processing of information can result in better memory, quicker thinking, and faster reaction times. EEG findings (Anderson et al., 2013) indicate that BrainHQ training led to faster neural timing and quicker neural responses to speech-helping people comprehend faster speech. Each of the brain's 100 billion neurons has around 1,000 connections. Communication between neurons and among brain regions is therefore important. Brain imaging techniques enable researchers to study these connections. Study results (Strenziok et al., 2014) show that white matter insulation between brain regions specific to visual and attention processing has improved, indicating better connectivity between these regions. BrainHQ training focuses on speeding up users' brain processing. This is because the accurate, rapid processing of information that comes into your head from your ears, eyes, and other senses affects the user’s "higher" cognitive functions, such as memory and problem-solving. BrainHQ exercises have been shown in numerous studies to speed up the brain. Based on the IMPACT Study (*The IMPACT Study - BrainHQ from Posit Science*, 2022), using BrainHQ exercises resulted in an average increase of 135% in auditory processing speed. Three months after the training, this benefit was largely retained with no further training. The games work on different areas, like:

- Attentión Exercises
- Brain Speed Exercises

- Memory Exercises
- People Skills Exercises
- Intelligence Exercises
- Navigation Exercises
- Drive sharp Exercises

The brainHQ application also helps people to stay healthier and happier and feel more control in their life (Wolinsky et al., 2006; Wolinsky et al., 2009; Wolinsky et al., 2010). Different game categories of BrainHQ are shown in the homepage and the knowledge behind every game is described by selecting them.

5.3 NeU sensor and fNIRS

NeU is a Brain Science company that was founded in 2017 by fusing the “cognitive brain science discoveries” with the “portable brain measurement technologies”. Ultra-compact, weighing only 30 grams (battery included) and measuring 80 X 40 X 13 mm with a unique “butterfly-style” design that bends in the middle to conform easily to any individual’s forehead, Figure, it could be used in conjunction with a band-type holder and stick to the forehead of users. This device uses optical topography devices (NIRS) to measure brain activity through cerebral blood flow volume changes in real-time. Cerebral blood flow (CBL) will be measured using a NeU device, which uses fNIRS to measure blood flow in the user's prefrontal cortex. The non-invasive, low-cost functional Near-Infrared Spectroscopy (fNIRS) technique measures cortical blood flow (Fishburn, Frank Anthony, et al. 2014). fNIRS has been shown to be sensitive to both cognitive load and state, indicating that it can be used to answer a wide range of neuroimaging research questions and provide an alternative to fMRI (Fishburn, Frank Anthony, et al. 2014). At NeU, they utilize an ultra-compact brain activity sensor that can be used in any environment to provide products and services based on scientific evidence. Therefore, the brain's cognitive load will be evaluated and feedback will be given to the user during cognition sessions. So, the state of the brain could be checked independently and brain training will be done accordingly. It has been shown in chapter.2, that neuronal activity is linked to changes in blood oxygenation and deoxygenation caused by functional brain activities. This phenomenon is known as cerebral hemodynamics. In the near-infrared range,

oxyhemoglobin (HbO2) differs from deoxyhemoglobin (HbR). Thus, the concentration of each molecule can be determined independently. The wavelengths used in NIRS are 600 to 900 nm, which have very low levels of absorption in tissues, allowing deeper penetration into the brain (*Nais Lab*, 2022). Because, fNIRS measures relative values, so absolute value does not mean anything. The sensor just can detect changes from certain baseline values at baseline period. In general case, the "baseline period" is "resting period" which the participant does nothing.

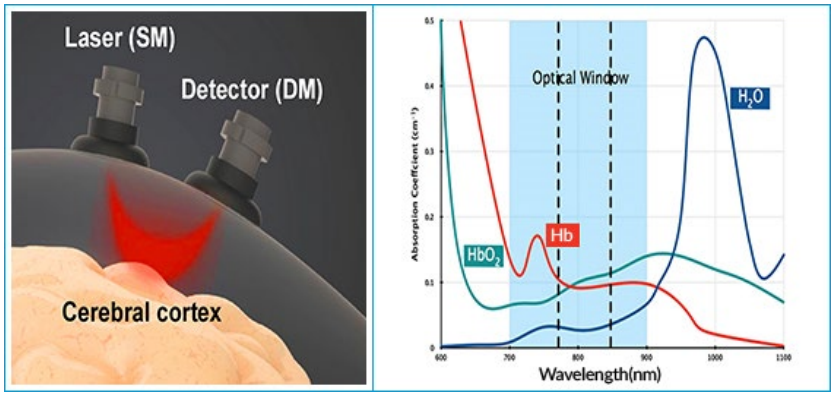


Figure 11 fNIRS sensor function. Blood volume increases as oxygen are sent to the brain and the concentration change of hemoglobin is measured with light. (Interviews with Leading Life Sciences Companies, 2022)

5.4 Describe design concept and development of Bo app

To build the prototype Figma (Figma, 2022)was selected due to its easy and practical features which enable the designer to edit vector graphics and design prototypes easily. In addition, in this study sharing the prototyping in different interview sessions is crucial and as the platform is web-based, it facilitates the interview process. It is possible to view and interact with Figma prototypes on mobile devices in real-time using the Figma mobile app for Android and iOS. In this study, a web-based prototype was designed to help the end-users to gain a comprehensive perception of the system.

The name of the application is “Bo”, which is another name for the Budhi tree in India which symbolizes growth and rebirth. According to Buddhism traditions, the tree of life refers to the Bodhi tree. The goal of this naming is to provide valuable life meaning to cognitive training sessions in order to encourage elderly people to continue training. The selected main colour is yellow. Understanding basic colour concepts and establishing a general language for colour are the first steps in determining colour in the design. It is, therefore, necessary to understand how colours are associated, how they affect individuals, and how they affect spatial moods. Active and dynamic colours and warm colours increase the focus on one area, stimulate and energize, provide a sense of mobility, vitality, and joy, and give user a mental image of fire, warmth, and sunshine. However, excessive use of these colours could be exhausting. Thus, it would be better if the warm dominant colour with a neutral one such as white or gray are combined. It represents wisdom, power, and reason in the same way that gold, sun, and light do. In addition to being used as a royal colour like purple, yellow stimulates the intellect and is an inspiring colour in bright and pure tones. It attracts attention to the space (Eren et al., 2022). The logo is a combination of a tree-shaped brain and a human head to present the main subject of the designed system.

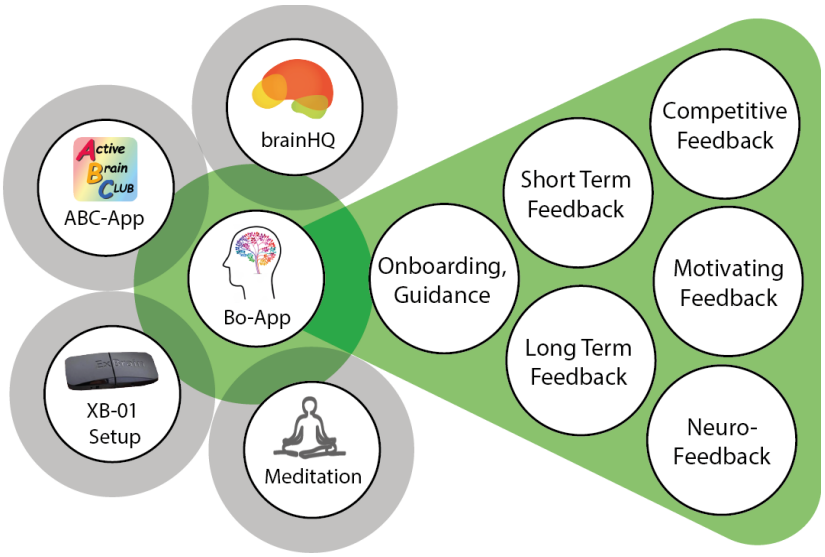


Figure 12 Bo-App infrastructure and design goals

Based on shared explorations the overarching Bo-App was co-created that integrates the ABC-App, provides instructions on how to use it and introduces additional forms of cognitive training (e.g., meditation). Over the period of co-design sessions, 4 successive prototype versions of the Bo-App have been created.

5.4.1 Development of the Bo-App

Based on shared explorations with older adults the overarching Bo-App is co-created that integrates the ABC-App, provides instructions on how to use it and introduces additional forms of cognitive training (e.g., meditation). Over the period of co-design sessions four successive prototype versions of the Bo-App have been created. Overall, the Bo-App provides an onboarding experience, user guidance, and various forms of motivating feedback and progression. An illustrative overview of the Bo-App can be found in the Annex.

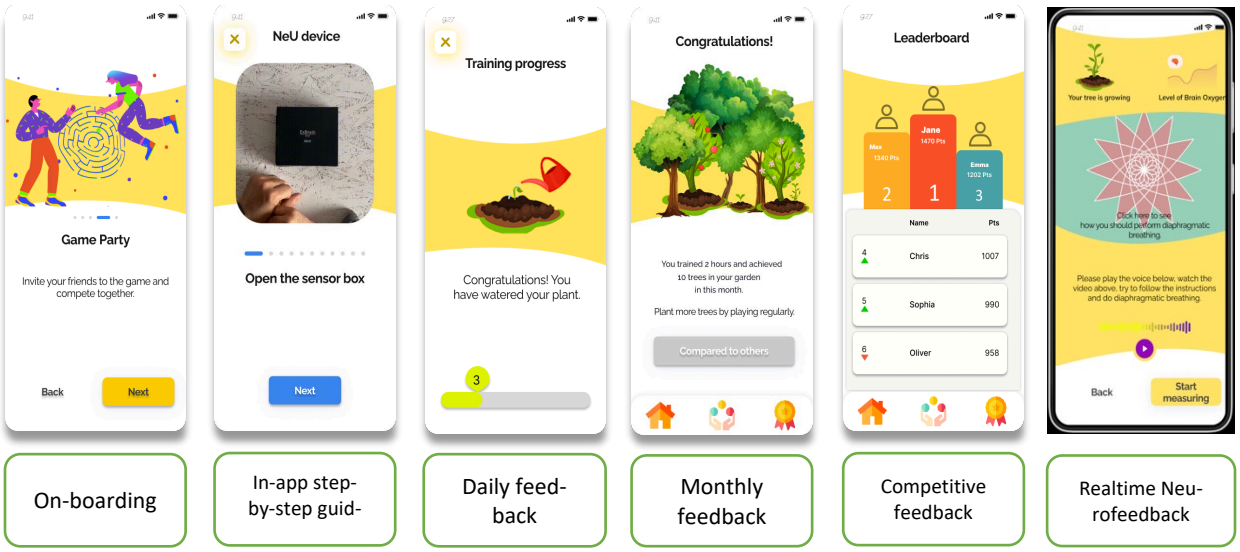


Figure 13 Features of the Bo-App

Over the course of our development process, features inside the Bo-App have been improved, altered and refined. A major focus was to offer a good indicator for the progression in cognitive training. For example, the screens that indicate the progression after each training session (i.e., short term feedback) have been iteratively improved. While the first version (i.e., V1.0) already featured a tree visualization to indicate progress there was no textual and numeric feedback. In V2.0 a motivating text and numerical rating has been included. In V3.0 the rating has been refined by using a visual slider with color grading.

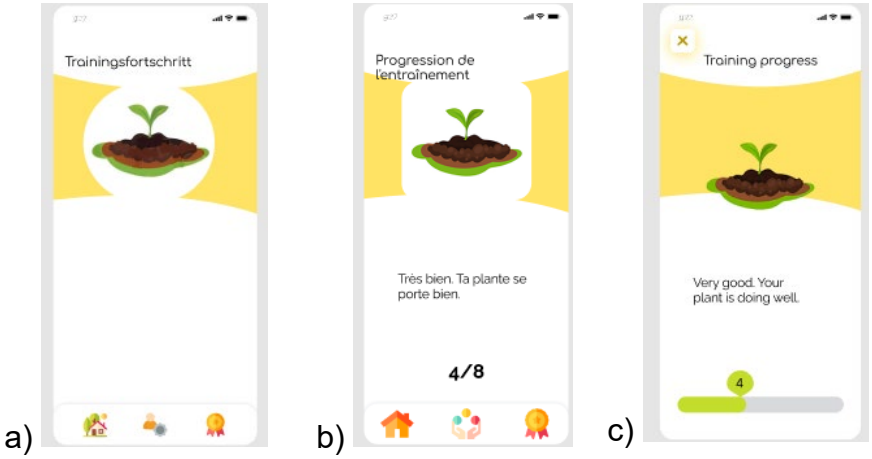


Figure 14 Iterative improvement of daily feedback. a) V1.0 b) V2.0 c) V3.0

A complete changelog with a list of all changes between the different version can be found in Table 2.

Version	Contents and Changelog
V1.0	• Growing tree visualization (without motivating text, and numeric clues) • A visual in-app integrated manual consisting of four pages of photos to learn about NeU visualization (four steps/pages)
V2.0	• The logo of the Bo-App was improved • changed labels on the home page of the Bo-App (e.g., setup of the NeU device) • added feedback numbers to the growing tree • added motivating textual feedback to the growing tree visualization • replaced photos with videos for the NeU device • added more instruction details to the visual in-app integrated manual (eight steps/pages) • added one page to warn the user about errors detected in the sensor signal due to the bad positioning of the NeU device • added a meditation song with the teaching concept of diaphragmatic breathing
V3.0	• changed progress bar with colour gradient from yellow to green • Rearranging the order of the step-by-step the visual in app integrated manual pages and adding more steps/pages (ten steps/pages) • Added one page as a low battery warning and show how to charge it again • French, Japanese, and Italian versions of the Bo-App were added to the prototype • Added a meditation song with the teaching concept of diaphragmatic breathing in French, Japanese, and Italian languages

V4.0	• Added two pages containing two real time neurofeedback with a thumbnail image of growing tree as well as numeric feedback in diagram form, and two guiding videos for teaching correct way of diaphragmatic breathing associated with verbal description. • Added a leaderboard to present the total result status of cognitive training course. • Added a total result page with the concept of the garden of planted trees which is made during the training session and based on what users has achieved before.

Table 2 Prototype Versions of the Bo-App

The goal of the Bo-App is to complement the existing ABC app for cognitive training. This App already features games for cognitive training and scoring. However, the feedback is not always understandable for users and could be more motivating.

The app is translated into German, French, Italian, Japanese and English language.

Bo app functionalities

On-Boarding Main Bo app functionalities is described here by meaningful pictures and texts to make users familiar with the features. In the app tour part, a brief overview of the app. Like what is Bo app, how it helps people, which sensor is using, how the neurofeedback is presenting their brain activity level and the possibility of inviting a playmate and doing a Game Party.

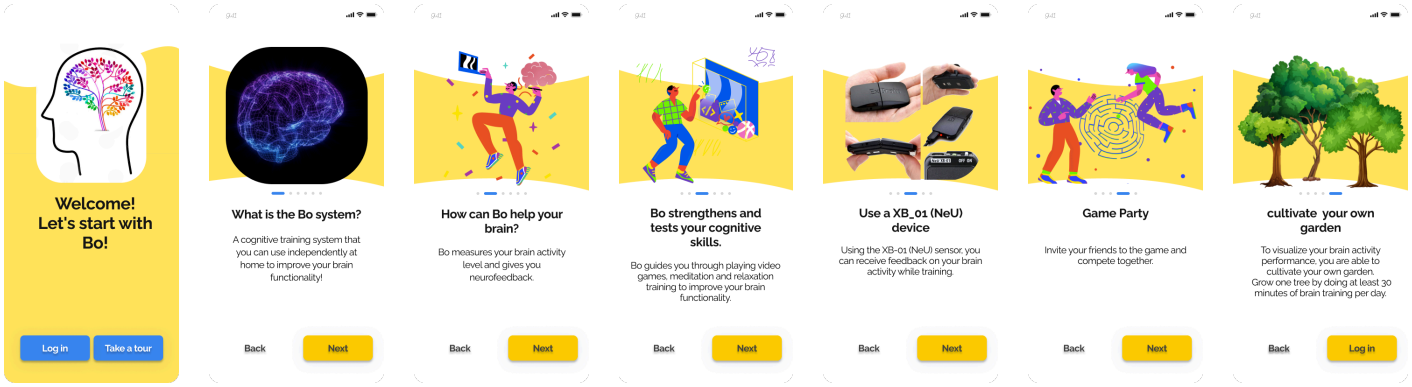


Figure 15 Application Tour

Goal Setting: On this page users are allowed to set their goals for their training sessions, later on, their monthly feedback will be calculated based on a combination of their own pre-defined goals, the number of the minutes they played and the sensor neurofeedback measures.

Reminder setting: Users determine that when they would like to receive reminder notifications on their smartphone. The notifications are periodic motivating messages, such as “Hey Max, your plant is getting dried if you don’t train your brain today.” Afterwards, they go to the home page. Motivating and reminder notifications pop up in the app once in a while based on their desires and goals. For example, “Congrats! Now you have water for your plant.”, “Cool! Now you have some fertilizer for your plant”, “Oh, now you have a sunny day, use the sunlight for your plant.”, or reminder messages such as, “If you do not play today your plant will be dried!”, “Your plant needs you, please come back and feed it with some more training!”

In-app visual manual: Step by step visual manual in micro levels (means the tutorial video is divided in short guiding videos (3 sec), and photos), which instructions, short videos are included. On the home page, they can select the Setup NeU device button to learn about the Neurofeedback sensor setup which is explained by several short video clips in gif format associated with a series of short verbal descriptions and Notifications. The pages are showing a step-by-step sequence of how the sensor should be turned on, charged, connected to the headband and finally connected to the Bluetooth. Every stage is easily returnable by integrated Back buttons and therefore end-user has enough controllability over the guiding processes, the user can navigate step by step and bullet progressions shows which state are they in and consequently, they have an overview of their status, Figure 16.

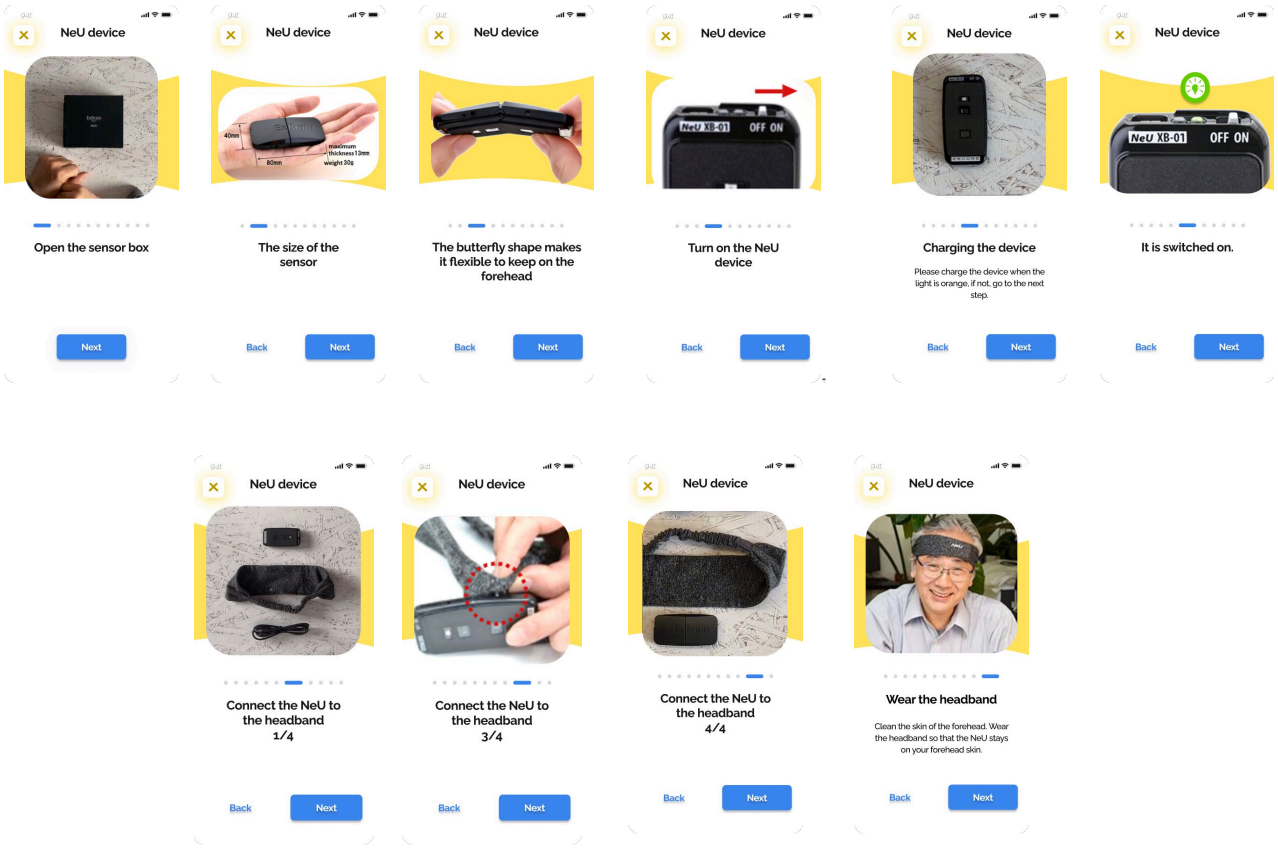


Figure 16 Inter app step-by-step manual. A combination of photos and videos are included.

There is a two button Back and Next below and one Close button up left direct the user to the previous and next step as well as main pages as soon as possible, when they are clicking wrongly, they can go back easily to correct it. Hence, the controllability, autonomy and error tolerance could be increased (Hassenzahl et al., 2013).

Activity page: included of four set of training activities such as ABC app, BrainHQ, Meditation and Mindfulness as well as a free activity. Users are free to choose the cognitive brain training activity such as BrainHQ, ABC game, A free activity and Meditation and relaxation section, in the activity page. On the activity page BrainHQ application, ABC application a series of relaxation and meditation practices and button for measuring the brain activity during any desired activity are considered. By choosing any of them the app will direct the user to a short relaxation practice, to boost their brain performance before any training. Because as it was shown in the chapter 2, having a short relaxation before any activity could improve brain performance during the training

sessions. Based on the research it was proven that meditation and mindfulness could boost the level of oxygen in the prefrontal cortex of the brain, so three minutes of diaphragmatic breathing instruction with a relaxing (flute and non-verbal music) were produced by native speakers of different countries, Japan, France, Italy and Germany and attached to the Figma prototype. In this guidance voice, a diaphragmatic breathing method was explained and used, as it was shown in the (Drigas & Mitsea, 2022) it is a positive effect on cognitive ability. In diaphragmatic breathing, the chest cavity is separated from the abdomen by the diaphragm. As a result of this type of breathing, all the parts of both lungs expand and contract in all directions, maintaining a maximum vital capacity. Through the modulation of the autonomic nervous system, diaphragmatic breathing affects the brain, cardiovascular, respiratory, and digestive systems. It is crucially considerable that the method could be harmful to some diseases such as dyspnea in patients with severe Chronic obstructive pulmonary disease (COPD) (Dhaniwala et al., 2020; Hamasaki, 2020).

The text for meditation practice:

Welcome to Bo.
We will practice proper belly breathing and help you concentrate and prepare for the game.
We will practice proper belly breathing.
Please take a relaxed position on a chair.
Close your eyes. place your feet on the floor and place your hands relaxed on your knees with palms down. empty your mind. try to breathe deeply, keeping your chest from moving because your abdomen is also lowering as the diaphragm contracts. The abdomen expands to fill the lungs with air.
Please keep in mind that any time you feel uncomfortable with this exercise, stop doing it and go directly to the next step.
Please repeat five times
Breathe in
Hold air
Breathe out

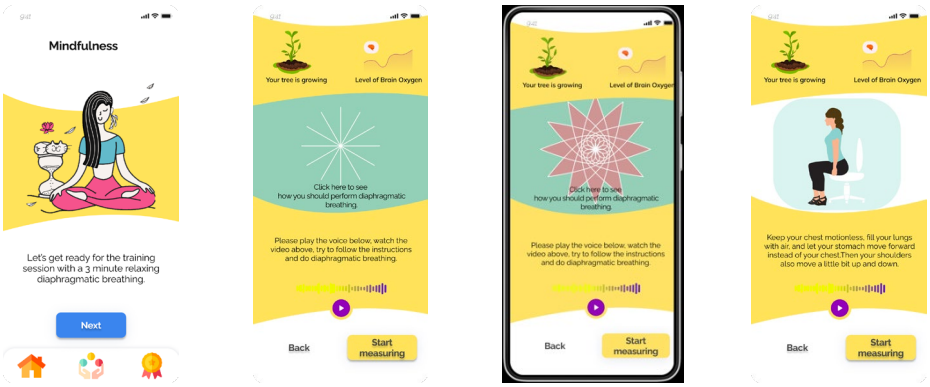


Figure 17 Diaphragmatic breathing guidance as a cognitive intervention

In second and third figure from left a geometric shape which is teaching diaphragmatic breathing by a short animation as well as text description is seen (Figure 17). The animation automatically encourages the users to inhale and exhale at the constant and appropriate speed. In addition, user can learn about diaphragmatic breathing by clicking on the geometric moving animation gif in the middle of the page, Figure 17. In the fourth page a short gif animation (a woman is doing diaphragmatic breathing) is shown to lead the users to practice deep breathing when they are listening to the meditation voice instruction (associated with three minutes of a soothing music). Meanwhile, if they press Start measuring button two different thumbnail feedback will be shown up on the page. One is the growing plant and the other one is a diagram neurofeedback to reflect their brain reflection.

Training activities: Mindfulness, ABC app, BrainHQ, Free activity

A three minutes diaphragmatic training before every session

Customizable feedback: Real-time Neurofeedback, Daily, Monthly and Competitive feedback

Real-time Neurofeedback: visualized in the Diagram shape motivate them to conduct the training session as well as helps them to regulate the hardness of the trainings

Daily feedback - Growing tree: The growing tree feedback is associated with a coloured numeric bar feedback, to help end-users perceive their status by providing auditory and visual stimuli such as a voice that speaks out the levelling up every time and it presented the performance after every session.

Monthly feedback: A garden that is visualized based-on the defined goals and training time duration. The resulted trees, bushes and plants that were grown up in daily practices

by users will be shown in this section all together, which participants liked it based on the interviews.

Competitive feedback - Leader board: Shows the result to the competition between users

Aesthetic, Customizable typography and Colors

Game Party and inviting others: Users are able to invite their friends or family members to do the training sessions together. Users can select to invite their friends or families to the application on this page (Figure 18). And the purpose behind this page is to invite the elderly to socialize with those around them and in this way, they will have more motivation to continue training due to the competition created. Consequently, they can have a leaderboard result based on the number of minutes that they trained.

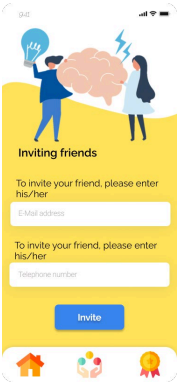


Figure 18 Inviting friends to the Bo app

Additionally, two pages are designed to warn the end-user about errors in Bluetooth connection and empty battery, for elderly people with eyesight and/or auditory impairment a tactile error sign via the vibration of the mobile phone will notify them.

6 Results of interviews, observation and usability tests

The study consisted of iterative sessions, during which older persons contributed their ideas for the design. To get fresh perspectives and unbiased opinions, new participants were bought in during the different phases of the study. Some participants withdrew their participation after a certain point because they had lost interest in co-designing this technology. Thus, not all participants participated in sessions until the study's end. In the course of our 9 co-design sessions, the commercially available BrainHQ app and the ABC app together with its NeU brain measurement device, as well as the overall expectations for cognitive training were explored in this study. The overall goal was to evaluate the ABC app and BrainHQ and develop an overarching application that provides a better user experience. As such, the Bo-App integrates interventions from the ABC App and BrainHQ, compensates shortcomings (e.g., lack of user onboarding and setup instructions) and adds additional features (e.g., motivating feedback and progression indicators).

6.1 Results of pre-study based on view of Sessions

In General, it was observed that elderly participants had difficulty expressing their opinions, perhaps out of a fear of saying something incorrect. As well as emphasizing that all ideas were valid, it was important to encourage their participation at all times. It was observed in the pre-study session that all the participants possess smartphones, and they are almost completely familiar with the smartphone functionalities. They all believed that they are mentally healthy, however most of them need to set reminder for their appointments on their smartphones.

1st, 2nd and 3rd and 4th Session | **Participant**: P1, P2, P3, P4 | Bo-App Prototype Version: - | **Methods:** Observation and Interview

This session was conducted as a pre-study with P1 to investigate overall interest in the technology. He did not have a positive point of view on this kind of technology. There was a trust issue:

"*Maybe waves can damage the brain*" (P1).

But he answered some of the questions, he generally, liked a technology which can measure brain activity and reflect brain performance and also, he liked the brainHQ games. He was eager to emphasize that he did not like technology, however when it came to playing games his opinion changed and he was entertained. He found them effective and entertaining at the first glance. But he was reluctant to put on the NeU device as he believed that technology and AI is going to take control of human being life in future. P2 showed initial interest in a system that is able to measure the brain activity level and give feedback about it. She was confident about using ICTs, and she was supporting other old people to get familiar with technology. She had some experience with yoga and meditation. Her rationale was to postpone MCI and dementia-related diseases. To do so, she was willing to try out the system for cognitive training. Her main motivation was to remain independent till old age. She reasoned:

"I prefer to be more independent. Especially in doing some personal tasks, taking shower, and toilet, I would not like to have help from others." (P2).

She was interested to do technical tasks with their mobile phone independently. Researcher asked her that which kind of manual is her preference. She was provided by a hardcopy preprepared version of manual and a video and she chose the video version.

"When I set up the phone for my husband, I watch a lot of videos because I want to fix it alone."

P3, who was familiar with ICTs, relaxation practices, and computer games. He was also comfortable with the measuring device. He liked to use reminders on his smart phone for taking medicines. P3 was interested in using such a device to measure regularly his brain activity, however, he said that when he goes to the doctor for a check blood pressure check-up, he feels stressed. When the participant was leaving it was observed that he forgot to remove the measuring device. This behaviour was interpreted as a sign that he was feeling very comfortable wearing it. Regarding the relaxation exercise before each session, P3 mentioned he regularly does some relaxation exercises every night, before going to bed and after he turned the TV off. He expressed that for a guiding manual his preference is a visual one mixed with a verbal one.

P4 likes to keep his brain active by studying scientific texts to prevent MCI and being dependent on others when he is older. He works with different ICTs easily and was interested to do it on his own and independently. He was comfortable wearing the NeU Device. He used to some mental practice to control his feet pain at night.

6.2 Results of appropriation phase based on view of Sessions

5th Session | Participant: P4 | Bo-App Prototype Version: V1.0, BrainHQ | **Methods**: Interview, usability test and observation

P4 pointed out about onboarding pages that these imageries would help him to understand his progression better. He wanted to have a clearer indicator of his progression (e.g., weak, middle, strong level of the brain ability), Figure 14 (a).

He could successfully set up the NeU device by interacting with the Bo-App, although firstly he connected the sensor to the headband inversely (e.g., inside out) he noticed immediately and corrected it, Figure 15.

To make an appropriate setup of the measurement device easier P4 suggested that the system should remind users whenever the sensor is not appropriately connected to the forehead and he prefers to be informed by an audio voice such as a beep ring rather than a visual sign because if he has the mobile phone in his pocket, he will not be able to hear it. Regarding the graphics, fonts and pictures he was mostly satisfied, about the logical sequence he had some preferences to make the steps clearer and also the German language was not clear to him. P4 liked the feedback in the shape of a growing tree.

Regarding the BrainHQ app, he needed more descriptions. He chose an eye training game, read the instructions and started the game. He asked for help and had to start the game again. Then he chose a memory game where he should click on the bird's position after they disappeared. His feedback shows that it was not clear to him when he should click on the display. Overall, he was interested in the games but struggled with system control tasks.

"*It is demanding. There is no explanation how to select this."* (P4).

6th Session | Participant: P2 | Bo-App Prototype Version: V1.0, ABC, BrainHQ | **Methods:** Observation and Interview and usability test

Bo-App and XB-01 Setup. Regarding the Bo system, when P2 was searching for the button that directs her to the user manual for the NeU Device, she had trouble finding it.

"*I don't know where to look*" (P2).



Figure 19 Screenshot of the homepage during P2' interaction with the app

She could successfully set up the NeU device by interacting with the Bo-App, although firstly she connected the sensor to the headband inversely (e.g., inside out) she noticed and corrected it.

She mentioned that she likes the growing tree neurofeedback "*It is beautiful*" (P2), but the icons are not clear in the Bo-app (i.e., home page icon in the main menu). She also suggested to add motivating sentences under the images.

Generally, she liked the Bo app, "It was easy to understand.", but she also said "German words are sometimes not easy to understand."

ABC App. After her initial interest, she tried out the existing system, which consisted of the ABC App and the NeU Device. She could not understand how the ABC app works and the process was described to her by researchers:

"ABC app supposed to match to the XB-01 device and they work together."

When she was confronted with the ABC app, she became a bit nervous.



Figure 20 Home page of ABC app

"I don't get along here" (P2).

She had a lot of doubts about selecting buttons. She was searching for the games.

"*Where should start the play? Where exactly?"* (P2).

She was roaming around to find the games and her frustration was indicated aloud.

"I don't see any game?" (P2).

Then researchers interfered a bit and helped her to go to the next step. Then she chose one the game from above list. Before the game started, the app asked her to countdown 15 seconds and breathe deeply. Then the game started.

"What should do here? Who should play this game?
Is it for stupid ones, it is not my thing?"

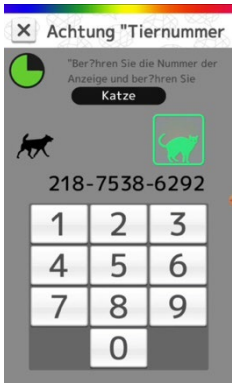


Figure 21 Screenshot from the game in ABC
when P2 was playing

In the textual descriptions in the app special characters were not displayed correctly which made reading hard. The scoring in the ABC app was not clear for her Figure 10.

"*The score is 30 and the brain measurement shows 30. Then maybe I am the stupid one."* (P2).

It was clear from researchers' observations, that when she removed the headband there was a red spot on her forehead because of the heat or the size of the headband. She was very relieved after removing the sensor. She was not enthusiastic about the cognitive exercises on the ABC app, which may have been due to usability issues (e.g., missing translations) and the lack of challenging content. After completing a particularly unchallenging and boring game (i.e., animals appear on the screen, the user has to tap on all cats) she exclaimed: "*This is designed for dumb people*!" (P2). Following this experience, she told us

"*This is nice technology but it is not my type. and I don't like to participate in the next interviews*" (P2).

Overall, her initial interest had faded after her discouraging experience with the ABC app and discomfort with wearing the measuring device.

BrainHQ. Regarding BrainHQ usability test, P2 played some of the games.

In the first try, she was confused although there was some description in the app, having difficulty executing cognitive tasks with facial expressions, finger motions, and hand motions. In the second trying she could do the game better.

"*It is very hard for me to understand that what should do".*

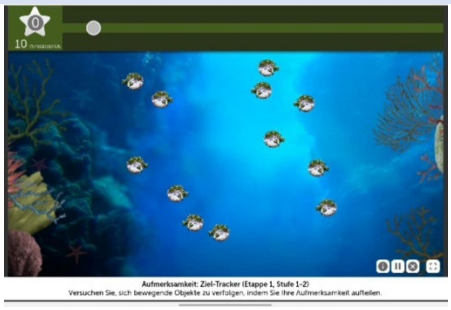


Figure 22 Screenshot from P2 playing with BrainHQ

She was reminded that the aim of this part is to make a comparison between BrainHQ and ABC app and she said it is not easy to learn them in a short time. However, she preferred BrainHQ over the ABC App.

"*I prefer BrainHQ it was a bit easier*." (P2).

6.3 Results of Mid-appropriation phase

7th Session | Participant: P4 | Bo-App Prototype Version: V2.0| **Methods:** Observation and Interview and usability test

Bo App. P4 liked the Bo App. Some parts were not clear for him. About the name and visualization elements of the app, he said:

"*I have not heard this before, very nice, very iconic. Great.*"

He further explained:

"*The beautiful scenery and the voice of nature is attractive for me; I like to sit and watch nature through the window. So, I like the growing tree feedback*."

He could easily use the instruction of the visual in-app integrated manual to learn how he should initially set up the NeU device.

"*It is good that this kind of result is used here, I like it because it is far from the technology world which remind me of beautiful nature",*

He also said:

"It is an interesting idea which encourages me to use the app"

It seems he was satisfied with the typography and visualization in the Bo app.

8th Session | Participant: P4 | Bo-App Prototype Version: V2.0 | **Methods:** Interview and usability test and co-design

P3 found the logo meaningful, Figure 23. It reminds him brain growing and strengthening.



Figure 23 Bo app logo

For the feedback in page Bo app, he suggested to incorporate various types of trees and bushes into the overall feedback, which he would design in the form of a garden. This idea was taken up later as a monthly feedback mechanism that visualizes regular training as nurturing a garden of trees. During the co-design, P3 suggested changing some of the words to make the aims of the Bo-App more familiar and more understandable.

The Activity page was still confusing for him, Figure 24.



Figure 24 Screenshot of activity page in the Bo app

The medical word “prefrontal cortex” in the feedback text was not easy to understand for him.

9th Session | Participant: P3 | Bo-App Prototype Version: V2.0, ABC | **Methods:** Observation and Interview and usability test

P3 thought the app is more understandable after some changes in the logical sequence of the app. He also mentioned:

“*I don’t like to use the NeU device every day because of self-determination*”.

It seems he does not like to measure a physiological parameter every day because as he mentioned in the first interview it can make him stressful.

ABC App. During the session P3 chose to test free activity, he started the ABC app with testing brain meter. During the test he was asked to type down the name of his intended activity to measure his brain activities level. Initially, he did not know what he should write. From the options hiking, journal reading and sleeping, he eventually selected sleeping. Then an animation was built on his brain activity level and presented by a changing colour geometric shape. The colors were not clear for him, in addition, the red color normally is associated with a negative result, however in the app it is indicating a high brain level activity and it made him feel insecure.

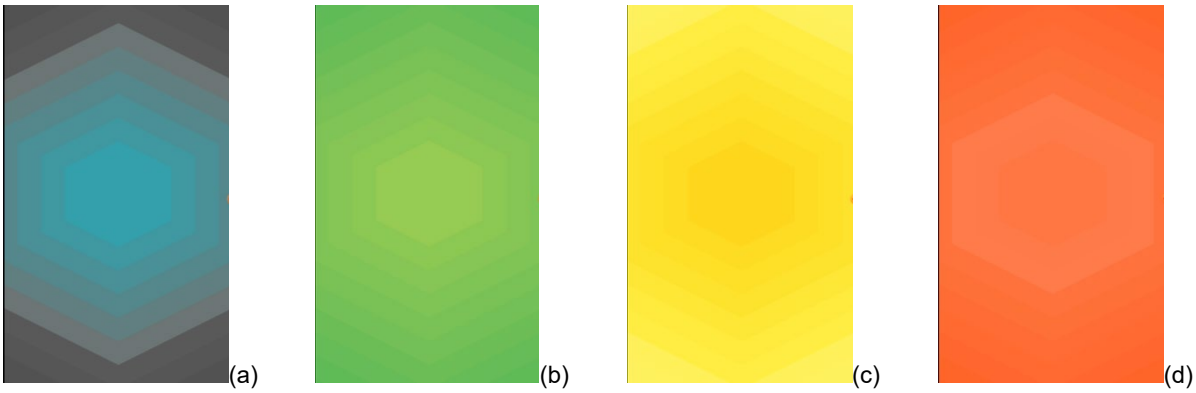


Figure 25 Different pages of the ABC app feedback when the real time brain meter test was doing by P4. a) Color gray is turning to blue means the lowest level of brain activity, b) A higher level of brain activity than what is shown in the (a), c) A higher level.

The brain meter visualization was accompanied by a sound and he said it is bothering him.

"*A smoother sound would be nice*." (P3).

He could not test the games due to some technical problems in the ABC App. For him, it was not clear what the ABC App measures. The displayed result was incomprehensible for him, Figure 10.

Researchers asked him about his preferred feedback and he suggested to have different options for users, then they can switch between the result with numbers or the iconic result of growing tree. In his opinion, navigation in BrainHQ app is easier than ABC app.

6.4 Results of Final-appropriation phase

10th Session | Participant: P5 | Bo-App Prototype Version: V3.0, ABC, BrainHQ | **Methods:** Interview and usability test

P5 found the visualization interesting, Figure 26. P5:

"*I like these simple pictures in your Software*."

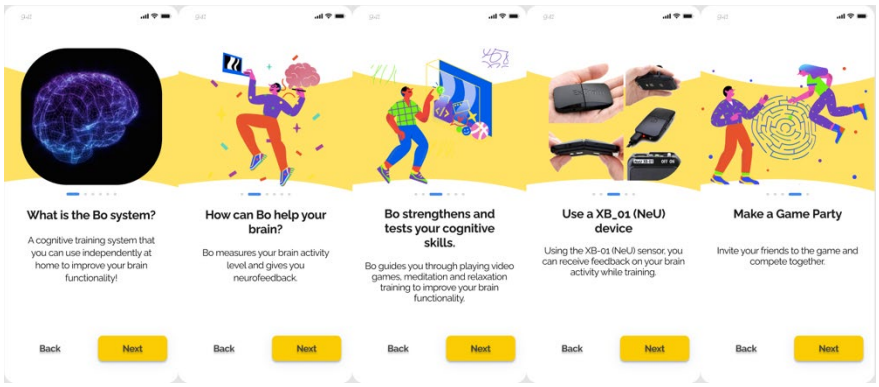


Figure 26 On-boarding

He could smoothly do the NeU device setting up with help of the step by step in-app video manual. P5 was unaccustomed to growing tree neurofeedback model and wished to see some numeric results too. During the session, the NeU device needed to be charged and P5 was searching for guidance on how to charge it. So far, it was not integrated any charging instructions.

ABC App. In the ABC app, P5 could not have access to Brain meter, due to some technical problems (i.e., Bluetooth connection did not work properly) but he could test the some of the games. He chose a game and encountered a page that was not translated yet into German (it was Japanese).

BrainHQ: He used to work with the BrainHQ, he mentioned that he is satisfied with it, but if it could be mixed with a sensor it will be more interesting for him.

11th Session | Participant: P4 | Bo-App Prototype Version: V3.0 | **Methods:** Interview and usability test | **Methods:** Interview, co-design and self-report

In our third session together, there was an on-line co-design meeting combined by interview and observation. So, P4 could see the result and what is changing or what is changeable and how. Afterwards, he could even be more aware of what kind of possibilities are in a design, therefore, he could be more creative in producing new ideas and insights. This method was effective for example, in terms of user's familiar terminology and at the end the participant was determining whether or not it worked in his opinion. P4 suggested to change some of the descriptions. For example, he suggested to change the grammatically gendered word for friend (i.e., Freund/Freundin) into the plural form (i.e., Freunde), as he found it more inclusive. So, the sentence on the playmate

invitation page became “To invite your friends please enter their email address or phone number ”.

He found the illustrative pictures useful for the onboarding process. Overall, he was very satisfied with the aesthetic of the app.

“*I like the harmonic usage of bright colours which encourage me to work with the app*” (P4).

He told the visualization in the app help him very good to feel the relaxation atmosphere of meditation, Figure 27.



Figure 27 Relaxation page shown before every activity

In the collaborative co-design section, he expressed:

“*I see the co-design method of collaboration as very positive and constructive and have looked forward to the next Zoom meeting with you every time. I found the experience that my ideas or suggestions found their way directly into the design of your app and that one has the result right in front of one's eyes very nice. This fact also motivated me to continue to work intensively on the functions of the app during the design process and also kept my attention high. The method also inspired me to come up with more ideas. When I see the app today, I think about the fact that I supported you during the development.* “(P4).

Moreover, P4 was motivated to do self-report and prepare reports based on his own tests. He was curious to spend enough time with the apps and explore them. Based-on his experience with the apps he prepared documents which included his opinion and ideas about improving the apps.

12th Session | Participant: P4 | Bo-App Prototype Version: V4.0 | **Methods:** Interview and co-design

Bo-App. In our fourth and final session together, P4 helped to refine the Figma prototype (i.e., the editable version was shared via a video conferencing tool). Together some

textual descriptions and labels were changed. This direct form of collaboration was very interesting for him, because he could see the instantaneous impact of his suggestions.

"*It is an interesting idea which encourages me to use the app*" (P4).

He found the descriptions useful and practical because they are guiding the users by short sentences and photos and videos. Further, he asked why only icons of older adults faces in the leaderboard were used. What about younger people who might also be interested to use this app? While the appeal of cognitive training for older adults was purposefully explored, it might be reinforced ageist stereotypes if the application is strictly targeting older populations. P4 suggested to use icons that show people of medium age. Overall, P4 found the visual and textual introduction clear. Yet, on some pages he suggested alternative words which made more sense for him. For example, he suggested a German word (i.e., *Darstellung*) to frame the tree garden as a representation of cognitive health. He also liked the term Game Party (i.e., Spieleparty), which competitive play was chosen. For him, this virtual opportunity of socializing was appealing, due to his mobility impairments. In terms of the visual in-app integrated setup guide, he liked that the user was guided step by step.

"*Small steps are easier to understand for older adults. I think it's much better than doing too many texts and too many expressions and descriptions in one picture. I can understand very easy how to follow up single steps.*"

In one instruction text where the user is required to perform a conditional step (i.e., charge the device when the light is orange), P4 suggested to highlight the conditional clause by using the word otherwise (i.e., *andernfalls*).

About the new instruction pages for relaxation and meditation P4 liked the way that video and gifs are guiding users.

"*I would not change anything in this page they are super super pictures and videos. I cannot imagine how it could be better"* (P4).

Regarding the leaderboard his idea was to change the old face icons of elderly people in the leaderboard. In the leaderboard the old face icons of target group were, but he mentioned that maybe the users are not only people. In the end, P4 thought that the app can motivate people to train or play. He believed it is complete because it is combining numeric(diagram) and visualized (growing plant) style of the feedback and it will cover a

wider range of users' preferences. He is interested in further cooperation in developing the prototype.

ABC App. P4 had enough time to work with the ABC app at home. He could use some of the features of the ABC app at home and he liked the experience of working with it, but he also mentioned some points, such as the help part of the app is not translated in German and it is in Japanese language. Also, he said that the descriptions are needed to be clearer and in more details. He had access to 5 games in his account which was prepared before by the researcher randomly and he said:

"I don't understand some of the games",

he briefly compared the Bo app and ABC app, and in his opinion the ABC app is too much technical, which is not intuitive for users, specially, for older adults, on the other hand the "Bo app design is much more understandable to handling to use the app". He expressed that the Brain Meter feature does not function in the last version, but with his previous experience he liked the colors, the sequence of gray, blue, green and red color, but not the sound. He mentioned that I like the soothing sounds but not this kind of sound,

"I am not friend with this kind of sounds. And I prefer the feedback from the Bo app because it is also much more understandable for the users."

Regarding the feedback he also mentioned that, Figure 10.

"I missed the explanation here, what is the relation between the colors and numbers, and what does this numbers mean?",

In addition, P4 suggested to use more explicit German words.

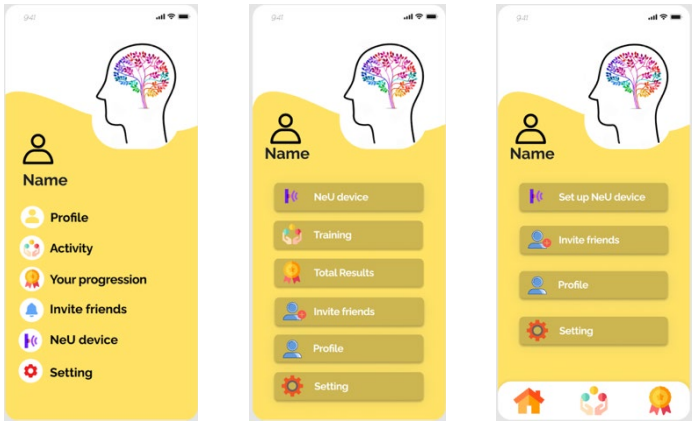
In self-report part he suggested to change some of the German words and terms. He pointed out that the Help section in the app is not translated in German language. For example, the image below shows a typical game result screen. It shows the incorrectly overlaid text "The level of "?" will be increased by one-point next time" and an OK button. He said maybe the field of this text is not probably formatted correctly. This report come across an attached photo of the android device that is showing related screen. The buttons need to be clearer and separated from the text appropriately in a way that user can recognize them easily.

6.5 Results of the expert interviews

Two expert interviews were carried on in this study. In the expert review, several insights were achieved. Consequently, the logical sequence of some pages were modified. Following Usability strengths were mentioned:

- The design is minimal, with a minimum number of elements.
- NeU device initialization is explained explicitly which is effective for doing it independently by elderly people and without help
- Various activities could be very helpful in encouraging users to do the practices regularly
- The Prototype is clearly designed in such a way that users can merely be confused

Furthur some Usability suggestions were suggested, for example it was expressed that “Here (Home page) all buttons are the same size and style it would be better that main buttons that use should click generally look like it so I know first time to click on NeU device or Training maybe if NeU device name changes to setup/connect device would be better”. Regarding the homepage sometimes it was not clear that the texts are clickable. Therefore, it developed as it is shown below, Figure 28.



(a) (b) (c)

Figure 28 (a) First version of Bo app, (b) Second version of Bo app, (c) Third version of Bo app

Also, It was sugeested to change the place of the leader board button to the total result page. He also suggested to remove main menu from the real-time feedback page to

prevent users touching it accidently, and instead provide a close button in the top corner for exiting the measuring process in case of cancelation need.

The other expert review was done by the second expert, she wrote her comments and answered the questions in a google doc. For example, on this page she wrote: “Is this the highest level? Use green instead of red, as red is associated with danger or error usually.” She also suggested that the colour of the Bluetooth icons needs to be red for alarming about an error, and green if the connection process is successful. She said: “Bo App can motivate people to meditate and have some cognitive exercise. It explains well what is happening with feedback and progress function. Navigation is clear and well structured. It guides users through the content in an appropriate way.”

In the first version of feedback only a growing tree was shown and in the second cycle of interviews it was figured out that they need something besides this to help in understand where exactly are their level, then numbers were added like 1 out of 8, which mean user pass one level. Based on the analysing the resulted data a gradient-coloured bar was added to show their exact state. The colour gradient which was staretd from yellow and ended to red is changed to yellow ro green, because red could be interpreted as a negative sign and users feels that their status is not good enough.

Summery Conclusion of the Results

The interviews of the pre-study phase with elderly adults indicated that their needs and expectations concerning a cognitive training coach system was almost exclusively medical. Their goal was to remain independent in old age, or at least to prolong their independence. Consequently, they might be interested in the progression towards this goal. For one of the users who suffers from physical movement limitation, and has limitation to connect to other people out of his house socializing through the application was a nice to have element. However, from an experience perspective one user emphasized that he liked the metaphorical connection to nature (i.e., through the tree visualization). For him, this made the technology less artificial and softened the logical reasoning, because his brain activity was not reduced to ones and zeros. Growing and nurturing a tree was associated with positive emotions. During the participatory process

it was observed that older adults liked to see the instant effect of their ideas and feedback. One participant who collaborated in creating the Figma prototype was very interested to see what happens “behind the scenes” when a designer creates a digital app. This form of collaboration allowed him to have direct impact on the end product and also see the limitations and possibilities more clearly. This made the participation particularly enjoyable for him, stirred up his creativity and created a stronger sense of ownership. The online collaboration in Figma, observing and seeing what happens when a designer creates an artifact, this form of direct impact and seeing what is possible and what, the instant effect of his ideas, enjoyed the co-creation, stirred up creativity, confidence and feelings of ownership.

On the usability side it became evident that a minimalist approach works well for the design. Reducing the complexity of the screens and guiding the users with step-by-step instructions was particularly fruitful. Video instructions were also considered helpful by the participants since it allowed them to see what exactly has to be done. Notably, often only small changes were needed to make the app more understandable for users. In our co-design process this became evident, when users suggested to alter or exchange specific words to improve the usability.

Overall, it became clear, that older adults understand the value of cognitive training, relaxation, meditation or yoga practices. The goals and motivations to engage in different behaviours was manifold. The wish to create either health-oriented or ritualized habits that are enjoyable, energizing and have a good balance of challenge and skill, was the biggest motivators for our participants.

A learnable system is not always efficient. However, a step-by-step explanation workflow is helpful for new users to learn the process as soon as possible because of a lot of explanation and instructions, this system could be highly learnable in that users could successfully complete the task as fast as possible even as they do it for the first time. But if draw the learning curve by plotting a user’s time on task over a set number of trials, the line would be found pretty flat, because the user has to go to the same screens over and over so they wouldn't be able to complete the task much faster, even as they become expert in the interface (Experience, 2022).

On the one hand, the goal of the Bo system is to train the brain and improve cognitive ability and working memory capacity by increasing cognitive load through challenging activities such as games that require deep focus and thoughts; on the other hand, based on interface design standards and in terms of HCI, the interfaces should be as simple as possible to minimize the required cognitive load and maximize usability; the take-home message could be complexity is needed. It is about autonomy and having control over the conditions that users are subjected to. When it is necessary to boost users' brain ability, it is necessary to do so in a controlled environment with appropriate training and profound knowledge behind it, but when working with an app such as learning to use a new sensor device, it is necessary to grasp the most important points and reach the desired point as quickly as possible with minimal cognitive load.

7 Disscustion and Lessons Learned

With the aging population, there are several challenges in digital artifact appropriation for elderly people. Digital artifact appropriation to enhance everyday life is an important area of HCI, and elderly people are one of the target groups. It was the objective of this study to assess the effectiveness of unsupervised cognitive training for older adults using digital artifacts. Through the pre-study, a bunch of problems were discovered that prevented elderly people from taking advantage of digital artifacts, moreover a series of their real needs and expectations when they face to a new device which they never have faced to before they were detected. After discovering the problems during the pre-study, the prototype was developed to address those issues. An evaluation of the first prototype was conducted through interviews, observations, etc. Following the evaluation, the prototype was redesigned based on the collected and analyzed data. Next, these steps were repeated, followed by a final reflection. Doing cognitive training sessions with an unsupervised approach was the novelty of this thesis. The designed application which called Bo app combined with a non-invasion measuring device (XB-01 NeU sensor to evaluate the brain activity level via blood flow in the prefrontal cortex of the brain) and commercial apps like ABC app and brainHQ were used to age healthy. Based on the research questions introduced in chapter 2:

1) **Which are motivational factors and challenges of older adults of doing cognitive training sessions at home by using neurofeedback non-invasion methods?**

2) **How should features of an application be designed to encourage older adults to do the cognitive training sessions regularly and unsupervised?**

The learned lessons will be described below

7.1 Motivational factors and challenges

A number of problems were detected during the pre-study, including usability issues, learning paces that differed due to knowledge backgrounds, language barriers, different needs, a lack of trust, and being critical about using digital artifacts on a regular basis. Also, other researchers found out similar challenges. They stated that elderly people are initially selected based on their age and that their social class, education, and technical ability are highly heterogeneous, therefore, the design process must take into account stakeholders' varied skills (Cerna, Katerina et al., 2020). In some cases, behaviour differences may be primarily explained by cognitive decline, but others may be explained by cultural differences, values, and lifestyles (Gordon et al., 2019). Further, older adults are often critical of technology if they do not see the value of using it every day. However, their attitudes can be changed if they see how the technology can support their everyday lives (Bengs et al., 2018). Another challenge is that trial and error learning is hampered by various concerns and fears, including a lack of trust and concerns about unwanted costs (Müller et al., 2015a). Finally, there are barriers related to usability and an insufficient understanding of elderly people preferences and needs (Czaja & Lee, 2007). A solution is needed to prevent the users from giving up, thus, encouraging and motivating to do regular training sessions is essential. There are some features that can help in this process, such as good usability, being user-friendly, sociability functions (Meurer et al., 2018), timed and tailored feedback (Callari et al., 2012(Rienzo et al., 2021), challenging games (Rienzo et al., 2021), adoptable activities with users' preferences and needs (Vaziri et al., 2019). Mentioned features that were integrated in the Bo app were welcomed by users.

"*I like the harmonic usage of bright colours which encourage me to work with the app*" (P3).

Individual cognitive ability status awareness, preventing MCI in future, staying independent from others, entertaining individually or with a group of people and socializing are different interest and motivation for participants. P4 mentioned that he cannot physically contact others due to his physical movement problem, however he

found the Party Game feature (one of the Bo app features) a possible solution for his need of socializing. P3 was not interested in having everyday feedback of his performance because of the stressful situation that it can make but he was interested in preventing MCI by doing training sessions. P1 was entertained by doing the Cognitive Stimulation games but reluctant to use the NeU device because of trust issues. P2 was seeking for a safe method to strengthen her cognitive abilities as well as staying independent in future.

According to Rienzo and Callari studies, the key to motivate users to use a new app is to give tailored, timed, and adopted feedback during the training session (Callari et al. 2012; Rienzo et al., 2021; Callari et al., 2012). It was seen in the next versions that the Bo system is acceptable for users which motivated them to use the NeU device and Bo App. By providing different kinds of feedbacks (monthly, daily, competitive, and real-time), various preferences are purposful. Vaziri et al., (2019) considered a dashboard of related and transparent health data to prevent users from being frustrated and giving up due to the irrelated health data presentation as well as increase their awareness about health status. Participants found the Bo app to be very useful since it offered real-time feedback (a diagram and a growing plant which inspires them with growth and improvement of their brain functionalities in thumbnail size during the diaphragmatic breathing section). A combination of Neurofeedbacks that was visualized in diagram, score bar and growing tree form. These satisfied various preferences and personalities of Bo app that can lower the usage barriers by improving ease of use. Callari et al. (2012) found that a user-friendly device perception correlated directly with motivation to use the device. According to their results, elderly people were more motivated to use technological devices when they perceived them as user-friendly and when they believed that the technology will help them staying healthy and safe (Callari et al., 2012). Additionally, considering target group heterogeneous expectations is motivating for users (Vaziri et al., 2019). In the present study, supporting the users with different learning paces in using the NeU sensor device, via using an in-app step-by-step visualized manual which is also skippable is another motivating factor. With matching other studies with present data, it can be concluded that covering different needs of users with features such as being self-descriptive and

informative, socializing (Leaderboard and Game Party), adapted typography, icons and visualizations can motivate elderly people.

Providing intuitive and easy-to-use games that do not exceed the effort, and cause stress can help elderly people to avoid abandonment of the activity (Rienzo et al., 2021) and eventually stay motivated. Further, the games and training experiences are adaptable with individual training progress and motivational states (Kickmeier-Rust et al., 2012) that means reduce the frustration possibility. Similarly, both ABC and BrainHQ games are cable of being tuned based on the users' ability. If the level of the hardness is higher or lower than users' real ability, they could get frustrated. Therefore, in Bo app the level of the Cognitive Stimulation games could be adopted automatically based on the previous trying in BrainHQ, and in ABC app the level of the hardness is selectable by STARPI operator. P4 had enough time to spend with the ABC app and he expressed that the games need to be described in a more explicit language. P5 had previous experience with BrainHQ and he was happy about using BrainHQ. Finally, BrainHQ games were accepted as Cognitive Stimulation activity in the Bo system and it was seen that the Bo system is acceptable for users in terms of adaptability.

7.2 Design implications and methodological considerations

Cognitive training interventions cause slowing down MCI. Elderly people are the most fragile target group. MCI causes dependency, lack of self-confidence, depression and vice versa. These elements function as a destructive cycle of causes and effects. Different elements such as Diaphragmatic breathing, Cognitive Stimulation game and neurofeedback were used to boost cognitive ability. Cognitive ability is directly connected to the oxygen of the prefrontal cortex blood flow. Diaphragmatic breathing and focusing on the breathing can improve health and cognitive performance (Chelidoni et al., 2020). In this study, in Bo app a step-by-step visual guide is leading the users to do Diaphragmatic breathing, which was so satisfying for the participants. "I cannot think about a design better than this" (P4). Users can use this digital artifact to improve their performance. Makedon et al. (2010) studied a game-based intervention (ZPLAY) to treat people with AD, and monitor their symptoms via fNIRS sensor that measures blood

oxygen (Makedon et al., 2010). In addition to collecting and maintaining valuable information, ZPLAY engages the user in a fun activity and improves the quality of their life. Although this game enables them to be more independent, the complexity of available neuropsychological tests of this method was problematic and the users required extensive training. Hence, the ZPLAY system was not appropriated for elderly people to be used independently and unsupervised at home, this thesis tries to solve this problem. Another element was neurofeedback that is investigated in this thesis. Nouchi et al. (2021) developed a portable cognitive training with neurofeedback (CT-NF). According to these findings, CT-NF enhances cognitive performance and the training activities contribute to its positive effects (Nouchi et al., 2021). In this thesis, using a commercial version of fNIRS, so-called NeU (XB-01), the Bo app provided different neurofeedback presentation such as numerical, visualized, text messages and metaphorical growing tree, that are appropriated for different needs and expectations. P5 and P3 expressed that they also like to read the text and numbers besides seeing a visualization of growing tree, therefore, they liked the provided feedback of the last versions of the Bo app.

Researchers investigated the effectiveness of different cognitive interventions through using fNIRS and digital artifacts. In comparison with other methods to improve the cognitive ability, Bo app still possess several advantages. For example, in Z-Play studied only people with AD and the application is Web-based, it needs to be done in Labs with experts' supervisions, in addition for training sessions caregivers and family are involved, and the app is not designed with PD method. Regarding the virtual HIE-based intervention, Virtual body illusion can be replaced with real trainings and they used IVR technology, still no PD method is used in contrast with the Bo app. Finally, compared to the Bo app, the Dual-task intervention as well as doing the study without PD, the physical gesture such as standing on one leg and doing N-back task as intervention was used.

Elderly people have concern about cognitive ability reduction. Hence, accessing to acceptable, efficient and effective systems to aware them about their performance is crucial for their well-being. The combined Bo system which contains the neurofeedback portable sensor can be one of these systems. Based on the result of the interviews, participants prefer to live in their own homes rather than going to the care homes, as well they prefer to use the system without help due to their privacy and independency.

“I prefer to be more independent. Especially in doing some personal tasks, taking shower, and toilet, I would not like to have help from others.” (P2).

All the participants except P1 were interested to use NeU sensor with the aim of learning about their cognitive performance. P1 mentioned a trust issue as a reason of his reject. P3 was interested to use the NeU sensor for three days a week. Another point was lack of self-determination; P3 likes to use the device, but not every day. Because the idea of having a device measuring his brain activities give him a feeling of lack of self-determination. There is no PD study to compare with the result of this study to see how elderly people independently conduct cognitive training session and interact with a neurofeedback system, therefore more studies are required in this topic, however some studies were done to investigate the effectiveness of neurofeedback on cognitive training. Based on these findings it was revealed that CT-NF enhances cognitive performance and the activities engaged in during training contribute to its beneficial effects (Nouchi et al., 2021). In addition, some studies show that receiving tailored feedback is very important for participants (Callari et al., 2012, Rienzo et al., 2021).

One of the inventions of this thesis was to design a digital artifact that could be used independently and unsupervised by elderly people. The intended digital artifact could be a website or smartphone application. Both the researcher’s observation and conducted interviews show that most of the participants possess one smartphone or in some cases a tablet and using smartphones is common among this target group and they use them to do various tasks. The smartphone application is used during daily routine life therefore, researchers interpret that a smartphone application could easily be a part of the regular task during the week. Further, a smartphone facilitates accessibility and usability. For this purpose, a smartphone application with specific features such as being user-friendly, having an acceptable level of usability, efficiency and effectiveness was designed to support their independency. Based on the interviews to have such an application it is crucial to have a self-descriptive design that is also error-tolerant. To date, no study has been conducted on the design of an application with or without a PD approach for supporting elderly people to unsupervised use a sensor device for cognitive activities. From the deep interviews in pre-study, it was interpreted that they prefer to see the guidance messages and reading the text, therefore, an in-app visual tutorial associated

with short guidance texts were designed to help them during the NeU sensor setup. Using the integrated tutorial in the Bo app participants could easily setup the NeU device unsupervised.

"*Small steps are easier to understand for older adults. I think it's much better than doing too many texts and too many expressions and descriptions in one picture. You can understand very easy how to follow up single steps. (P4)*"

In the current thesis, older adults with the high possibility of having MCI in future were selected as target group. Selecting elderly people also was done in other studies such as (Ertl et al., 2021) which they tried to include the stakeholders who were excluded due to their differences. Considering target group's physical characteristics such as physical limitations insight, hearing, movement and their cognitive processes (Callari et al., 2012) is an important part of this thesis. One example of this consideration is tactile(vibration) feedback which could be used as an error alarm when the NeU sensor rans out of battery. Likewise, when the gathered data from sensor shows error (because of the wrong position of it on the forehead skin) user will be informed via a vibration alarm. Similarly tactile feedback was used in other studies (Gao & Zhou, 2020, page 316).

Regarding heterogenous expectations and needs, to provide better support for all older adults full spectrum of cognitive functions should be considered (Gordon et al., 2019).

Implementation of the sensor setup visual tutorial in Bo app that cover a wide spectrum of cognitive function was a consideration of heterogenous expectations. This issue was confirmed in the usability tests and observations. For instance, some users do the sensor setup in a short time (around 1 minutes) without mistake and some others spent more time as well as having one mistake like connecting the headband to the sensor (reversed inside out). However, immediately noticed and corrected the mistake in the next try. Imagery and speech should be customized according to the individual's needs and preferences to optimized interaction (Gasteiger et al., 2022). In this thesis, the font size and the used colors are selected based on the target group user's needs in mid-appropriation phase. In addition, Reminder feature in Bo app is customizable based on the user's preferences. For instance, some participants find everyday notification annoying, some like to set the reminder notification on weekly base. Also, the duration of each training sessions designed in a customizable way which users can choose between

three considered options (15, 20, 30 minutes). Covering various kind of brain functionalities (Reynolds et al., 2021) ABC and BrainHQ apps offer different cognitive games which could result a wider empowering practices for individual users. This training games could be memory, attention and executive function and visuospatial function (Reynolds et al., 2021) that observing the all aspects was not possible in this study due to the limitations.

Sociability is another factor in elderly people motivation to conduct cognitive training activities such as stimulating games. From other studies, it is concluded that the target group's willingness and motivation to engage them in the research sessions and in daily usage of the digital artifact. Their motivation could be significantly increased if some features and functions are integrated with a sociable atmosphere (Meurer et al., 2018)(Vaziri et al., 2019); (Cerna, Katerina et al., 2020). For example, elderly people in their studies were encouraged to discuss and share their experiences about technologies by developing a mobile demonstration kit comprised of a variety of devices (Cerna, Katerina et al., 2020) or during workshops or social events, users exchanged experiences with other participants or played the 'exergames' with their grandchildren (Meurer et al., 2018). In the current study, socializing in the Bo app could be seen in three stages, firstly in the co-design process which is described in 7.4. Secondly, in the Game Party that users are able to communicate with their friends and families; this way the training sessions will be an enjoyable socializing activity. One of the participants said he cannot communicate with people out of his house due to his sickness while this system could be a satisfying option and replacement for the interaction with other people. Thirdly, the Leaderboard feature for the participants who like to compete with others could be an option of communication in this thesis.

Gamification also is another factor which could play an effective role in users' motivation. Cognitive Stimulation games were found out effective to prevent mental decline (Kickmeier-Rust et al., 2012). Rienzo et. al (2022) recommended to adapt the software to user profiles; consider aspects such as age, familiarity with the use of technologies and educational level. This could be achieved by integrating a questionnaire about gender, age, knowledge background and calculating and anticipating users' characteristics. This functionality could be extended in future work by using AI and evaluating the app. In

addition, adapting the content of the training activities to daily life is effective. In Bo app, also adaptation was conducted. For example, measuring the level of brain activity is possible during any daily life activity (except the one with intense movement due to sensor technical limitations that could cause the error). Another gamification factor could be choosing different tasks and suggesting cognitive activity, based on prior evaluation. Similarly, designing different tasks that involve different cognitive domains in the software to avoid boring the user with the same activity (Rienzo et al., 2021) is an effective gamification factor. In this thesis, different tasks are integrated in the Cognitive Stimulation game applications (ABC and BrainHQ) which provide users with categorized cognitive games. Another factor in gamification to be considered is that the instructions of the activity must be clear, brief, striking and presented for a prolonged period of time, or maintained during the development of the activity. This is because some older adults mentioned that they had difficulty remembering the instructions (Rodríguez-Almendros et al., 2021). To consider these points, instructions in Bo app were redesigned brief and clear based on the users' feedback so that the instructions are reachable from different pages through main menu. Verbal messages as a kind of the instruction were modified in PD and direct user's participating.

Summary of Bo App, main outcomes and functionalities

- **On-Boarding** describes for users to get familiar with the Bo app functionalities.
- **Goal Setting (Reminder setting, Training duration setting:)** fulfill the psychological essential need: autonomy (Hassenzahl et al., 2013), because users can decide about both the evaluation method. Also, users can set their favorite periodic reminders and duration of the training sessions.
- **In-app visual manual:** This functionality helps users to feel confidence and fulfil their competency need (Hassenzahl et al., 2013) by guiding them to set up the NeU device independently from others and reduce the work load of the caregivers and relatives.
- **Training activities:** Mindfulness, ABC app, BrainHQ, Free activity

- **Mindfulness:** Which participants are familiar with this functionality and do some similar practices for relaxation. P2 prefer to do it in the gym and with a human trainer, P3 do it before going to bed when he turns the TV off, to prepare himself for a deep sleep. P4 do it in the bed to control the pain of his feet. They agree to do the three minutes relaxation before doing the training sessions or as the main cognitive training activity.
- **ABC app and BrainHQ:** Based on the users' feedback both could be useful if they are more self-descriptive and clearer.
- **Free activity:** Users can select different activities.

- **A three minutes diaphragmatic training before every session activity:**

Conducting a one-session FAM before a class or job can be much more helpful (Yamaya et al., 2021). Additionally, these findings may assist in establishing the impact of one-session FAM on meditation novices.

- **Customizable feedback, Real-time, Daily, Monthly and Competitive:**
- **Growing tree (Daily feedback):** Helps with the essential psychological needs such as competency because they feel that they are useful by helping the plant to grow (Hassenzahl et al., 2013). The growing tree which is adopted from brain empowering by meditation could be inspiring for users as it was observed from the PD sessions.
- **Real-time feedback visualized in both the diagram shape and the growing plant:** Motivate the users to conduct the training session as well as helps them to regulate the hardness and stimulants of the training.
- **Monthly feedback:** The resulted trees, bushes and plants that grown up in daily practices will be shown in this section all together, which participants liked it based on the interviews.
- **Competitive (Leaderboard):** This kind of feedback could satisfy their social needs when they conduct the sessions with their friends or family members. There is the possibility to compete with others based on their game results.
- **Game Party:** Helping them to socialize and address popularity and relatedness psychological needs (Hassenzahl et al., 2013). The need of relatedness is

somewhat fulfilled with the option of inviting friends and family to interact with the app.

- **Aesthetic:** Typography Aesthetic, Colors bright colors used in the Bo-app are attractive and bright, Photos that were used found out satisfying for the participants. Based what P4 expressed. This could be investigating in more details as a future study.

In order to guarantee a satisfying development in training sessions the involvement of the target group in design process is recommended (Rienzo et al., 2021). In this study, a prototype was designed based on the possible solutions figured out through Design Case Study (Wulf et al., 2011; Wulf et al., 2015). Then the designed prototype was tested by engaging participants in the evaluation sessions. As many of these sessions began, unlike predictions, co-design sessions were conducted more productive and detailed than what the researchers expected since participants could contribute well to these iterative sessions. In addition, to find the solution, using an optimized PD approach is so important. Previous research and experiences in the PD approach and application design were exploited to obtain the most beneficial results from this thesis. The design of digital artifacts that can be used in a real-world context is dependent on the diversity of stakeholders, their individual interests and motivations as well as personal skills and learning requirements. Associated strategies and decisions influence both the design of the project and the final digital artifacts product (Müller et al., 2015a).

Engaging the user in the design process increased the feeling of ownership which cause trust and innovative ideas production. In this thesis, the feeling of ownership through collaborative co-design encourages users to participate in the interviews and produce some self-reports. Also, other researchers similarly found out that by using PD, designers involve users in the process of amending and designing the artifacts by gaining a better understanding of their usage contexts (Muller & Kuhn, 1993).

In order to support the appropriation of digital tools in learning spaces, technology adoption and ownership of the learning process should be considered, as well as the possibility of influencing the learning setting. A learning process literacy must also be addressed besides digital literacy (Cerna, Katerina et al., 2020). P4 also mentioned that

he was inspired to engage in PD with the ownership that instant co-design creates. He was interested to conduct more PD sessions by suggesting the time of the next meeting in every session. He mentioned these sessions are interesting because of the researcher approach, and also, being fun the sessions. Therefore, one possible future work could be to make the PD sessions fun by warming up talking and make the atmosphere friendly for the participants.

In PD contexts, people can easily understand a technical tool if they feel comfortable using it (Joshi, 2016). In this research, the participants who were relaxed but not nervous could play an influential role in decision-making and understanding the Bo app. For example, from the usability test in Bo app, P2 was relaxed and confident till she confronted usability problems in the ABC app. Then, she was nervous and not able to continue the tasks on her own although she was told that all the answers are valid and the goal is not to test her ability.

To have a broad and holistic approach it is essential to conduct the researches with older adults in their real-world context (*Cerna, Katerina et al., 2020*; *Evans et al., 2018*; *Gerling et al., 2020*);. The Bo app usage is in the private home, therefore living lab could be an appropriate context to study their interaction with NeU sensor and Bo app. Usability test, observations, co-design and interviews were happened either in their own private home or in the Living Lab to simulate an environment similar to the real-world context and consequently achieving real results.

Based on PD and a living lab (Vaziri et al., 2019) study, building trust via considering spent enough time by the users and a clear and logical structure of the app is possible. Older adults need to try the app and spend some time (around one hour) to make sure that the app is useful over time to eliminate concerns about privacy, trust, etc. The clear and logical structure of the app was relatively successful in trust building. Similarly, Vaziri et al. (2022) found out that the trust could be gained by providing a structure, appearance, and color scheme that are clear and logical. Accordingly in the Bo app, participants showed their satisfaction with the logical structure of the Bo app, its appearance and colors. P4 showed his trust because he felt enough fun and friendly atmosphere and the meetings with him were in English language, however, P2's meeting was held on in German language which continued easily before she was confronted with ABC usability

issues. From the researcher's interpretation, because of a combination of ABC usability issues (which the researcher of the study does not have any control on it) and the language barrier problem, she could not be convinced and almost lost her trust.

7.3 Limitations and Future Work

A number of potential limitations that need to be considered in this thesis are listed here. First and foremost, because of the small number of participants and the use of a qualitative research method, the findings of this study cannot be generalized. Recruiting enough participants to provide a big sample size and represent the population well was not possible. Second, based on user feedback, the headband and sensor are not comfortable to be wore for extended periods of time, and the temperature of the sensor rises after 20 minutes, irritating the skin on the forehead. The size of the considered headband is not changeable and sometimes too tight as well. Third, performing physical exercises such as dance or aerobic trainings is significantly beneficial as a cognitive training activity and was also a user preference. However, the sensor's produced unreliable data during intense movements. Furthermore, due to COVID-19, it was risky to conduct the workshops and focus group in present format as the target group was sensitive into COVID-19. ABC app German version was not appropriately accessible in the early stages and later the translated version had some issues. BrainHQ has only one version and its development was not possible. As of now, the Bo-App only exists on a conceptual level. The click dummy created in Figma illustrates the potential for guiding the users in cognitive training. However, it has not been evaluated with a larger number of users, rather co-created based on the preferences of five older adults. Whether the outcome is appealing to a larger user base, needs to be further explored. The co-design of the Bo-App took place in Germany and the resulting prototype has not yet been tested in other countries (e.g., France, Japan, Italy). Potentially, older adults could ascribe culture-specific associations and meanings to cognitive training. The wording itself, and there with the onboarding and user guidance has only been tested in German.

A future work could be conducting research in other countries of the E-VITA project such as Japan, France and Italy to find out the difference between different cultures, as well as different genders. Because it could be interesting to compare the experience of

cognitive training across different countries. Another future work could be figuring out that how can measuring devices be designed within an app that supports the self-determination and autonomy needs (Hassenzahl et al., 2013) of users. Designing a sensor resistant to intense movements is recommended for future work. One possible future work could be to make the PD sessions fun by warming up talking and make the atmosphere friendly for the participants but considering biases that could be caused.

8 Conclusion and outlook

As a result of the project, it has been learned that designing minimalist apps and accompanying them with adequate details is important, as well as taking into consideration users' physical and mental characteristics, and what can fear or inspire older adults when designing for them. Developing a well-described in-app manual that is visual and easily controllable is essential. Additionally, extensive tests should be run before deploying it to older adults as they can be worried about their health conditions. From the result, it became clear that they like the combination of numerical scores (e.g., diagrams) with the inspiring metaphorical (e.g., growing tree). They like to see both monthly reports and real-time neurofeedback on their performance, however, some of them do not prefer to be informed about their real-time cognitive performance because they can feel stress. Therefore, the feedback could be customized based on their preferences and different personalities. As stated, notifications or reminders should be used sparingly and users should have the option to customize the interval. Gentle reminder (e.g., Your plants need you.), A high number of notifications can, however, annoy users. There are some studies related to meditation that observed some result that shows that meditation and diaphragmatic breathing helps improve brain cognitive ability, but they did not design an app for conducting training sessions by the users themselves. Based on the results it became clear that users feel calm after doing the integrated diaphragmatic breathing section, however, more research is needed to discover the long-lasting positive effects of this section. Researchers observed different results, the reason can be that they had enough time for different ages of people as well as testing the system for a longer time, however, to the best of our knowledge, there is no study regarding different feedback. Thus, more studies are required to get more insight into how the feedback is important from this Bo system and also related to cognitive training. Furthermore, regarding the PD it was discovered that building communication and trust with participants can highly motivate users to encourage them to engage more creatively and be productive. Furthermore, regarding the PD it was discovered that building trust between researcher and participant can highly motivate users to encourage them to engage more creatively and productively.

MCI prevention and cognitive empowering are essential criteria for the health of the elderly but with increasing age, the danger of depression, loneliness and dementia increases. A combined system as a feasible technology solution for older adults to tackle this problem is proposed and the resulted insights are encouraging. The PD allows users to benefit from the smartphone applications based on their preferences, expectations, and needs. To facilitate the interaction between users and the smartphone applications, the combined Bo system was developed to guide them through the process of providing their necessary data. Having said that, the evaluation of the Bo app provided many valuable insights about how older adults interact with a new sensor device that they never have faced before and our generally positive experiences indicate that such technology offers an exciting option. Thanks to our participatory design approach, the Bo app is considered to be a step forward compared to existing solutions.

Need fulfilment is the foundation of a positive experiences and wellbeing; and positive experience and wellbeing is crucial for preventing MCI and this paper provides a starting point or a framework towards designing.

In this study, on participatory studies that explore different approaches to coaching. In this study, explored the role of measuring and tracking progression with neurofeedback, as well as using gamification to improve motivation and adherence was explored for cognitive training.

During the study various weaknesses and usability issues of the ABC app have been identified. Accordingly, the researcher's insights can not only be used to develop the Bo-App, but also to enhance the ABC app. For the e-VITA platform this study shows the potential of using:

1. overarching systems to bring together content from different sub-systems (i.e., Bo-App)
2. dedicated apps that offer content in combination with dedicated devices (i.e., ABC App)
3. third party apps that offer out of the box content that cannot be altered (i.e., BrainHQ)

References

10 Usability Heuristics for User Interface Design. (n.d.). Retrieved August 17, 2022, from https://www.nngroup.com/articles/ten-usability-heuristics/

Ageing and health. (n.d.). Retrieved August 17, 2022, from https://www.who.int/news-room/fact-sheets/detail/ageing-and-health

Albert, M. S., DeKosky, S. T., Dickson, D., Dubois, B., Feldman, H. H., Fox, N. C., Gamst, A., Holtzman, D. M., Jagust, W. J., Petersen, R. C., Snyder, P. J., Carrillo, M. C., Thies, B., & Phelps, C. H. (2011). The diagnosis of mild cognitive impairment due to Alzheimer's disease: Recommendations from the National Institute on Aging-Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease. *Alzheimer's & Dementia*, *7*(3), 270–279. https://doi.org/10.1016/j.jalz.2011.03.008

Altmeyer, M., Lessel, P., & Krüger, A. (2018). Investigating Gamification for Seniors Aged 75+. *Proceedings of the 2018 Designing Interactive Systems Conference*, 453–458. https://doi.org/10.1145/3196709.3196799

Anderson, S., White-Schwoch, T., Parbery-Clark, A., & Kraus, N. (2013). Reversal of age-related neural timing delays with training. *Proceedings of the National Academy of Sciences*, *110*(11), 4357–4362. https://doi.org/10.1073/pnas.1213555110

Bengs, A., Hägglund, S., & Wiklund-Engblom, A. (2018). Applying Experience Design to Facilitate Wellbeing and Social Inclusion of Older Adults. *Interaction Design and Architecture(s)*, *36*, 11–30. https://doi.org/10.55612/s-5002-036-001

Brain Exercises, Brain Training, Brain Health – BrainHQ from Posit Science. (n.d.). Retrieved August 17, 2022, from https://v4.brainhq.com/?v4=true&fr=y

Burin, D., Liu, Y., Yamaya, N., & Kawashima, R. (2020). Virtual training leads to physical, cognitive and neural benefits in healthy adults. *NeuroImage*, *222*, 117297. https://doi.org/10.1016/j.neuroimage.2020.117297

Callari et al. - 2012—Elderly-technology interaction accessibility and .pdf. (n.d.).

Callari, T. C., Ciairano, S., & Re, A. (2012). Elderly-technology interaction: Accessibility and acceptability of technological devices promoting motor and cognitive training. *Work*, *41*, 362–369. https://doi.org/10.3233/WOR-2012-0183-362

Cerna, Katerina, Dickel, Martin, Müller, Claudia, Kärnä, Eija, Gallistl, Vera, Kolland, Franz, Reuter, Verena, Naegele, Gerhard, Bevilacqua, Roberta, Kaspar, Heidi, & Otto, Ulrich. (2020). *Learning for life: Designing for sustainability of tech-learning networks of older adults*. https://doi.org/10.18420/ECSCW2020_WS04

Cerna, Katerina et al. - 2020—Learning for life Designing for sustainability of.pdf. (n.d.).

Chelidoni, O., Plans, D., Ponzo, S., Morelli, D., & Cropley, M. (2020). Exploring the Effects of a Brief Biofeedback Breathing Session Delivered Through the BioBase App in Facilitating Employee Stress Recovery: Randomized Experimental Study. *JMIR MHealth and UHealth*, *8*(10), e19412. https://doi.org/10.2196/19412

Chiesa, A. (2009). Zen Meditation: An Integration of Current Evidence. *The Journal of Alternative and Complementary Medicine*, *15*(5), 585–592. https://doi.org/10.1089/acm.2008.0416

Cognitive Walkthrough in the usability.de glossary. (n.d.). Usability.De. Retrieved August 17, 2022, from https://www.usability.de/en/usability-user-experience/glossary/cognitive-walkthrough.html

Colzato, L. S., Sellaro, R., Samara, I., Baas, M., & Hommel, B. (2015). Meditation-induced states predict attentional control over time. *Consciousness and Cognition*, *37*, 57–62. https://doi.org/10.1016/j.concog.2015.08.006

Colzato, L. S., Van Der Wel, P., Sellaro, R., & Hommel, B. (2016). *A single bout of meditation biases cognitive control but not attentional focusing: Evidence from the global–local*

task—ScienceDirect. https://www.sciencedirect.com/science/article/abs/pii/S1053810015300532

Crabtree, A. (n.d.). *Ethnography in Participatory Design*. 14.

Crescentini, C., Fabbro, F., & Tomasino, B. (2017). Editorial Special Topic: Enhancing Brain and Cognition Through Meditation. *Journal of Cognitive Enhancement*, *1*(2), 81–83. https://doi.org/10.1007/s41465-017-0033-4

Czaja, S. J., & Lee, C. C. (2007). The impact of aging on access to technology. *Universal Access in the Information Society*, *5*(4), 341–349. https://doi.org/10.1007/s10209-006-0060-x

Deepeshwar, S., Vinchurkar, S. A., Visweswaraiah, N. K., & Nagendra, H. R. (2015). Hemodynamic responses on prefrontal cortex related to meditation and attentional task. *Frontiers in Systems Neuroscience*, *8*. https://doi.org/10.3389/fnsys.2014.00252

DeepL Translate: The world's most accurate translator. (n.d.). Retrieved August 17, 2022, from https://www.DeepL.com/translator

Dementia. (n.d.). Retrieved August 17, 2022, from https://www.who.int/news-room/fact-sheets/detail/dementia

Dhaniwala et al. - 2020—Pranayama and Breathing Exercises—Types and Its .pdf. (n.d.).

Drigas, A., & Mitsea, E. (2022). *Conscious Breathing: A Powerful Tool for Physical & Neuropsychological Regulation. The role of Mobile Apps*. *28*, 25.

Duncan, J. (2010). The multiple-demand (MD) system of the primate brain: Mental programs for intelligent behaviour. *Trends in Cognitive Sciences*, *14*(4), 172–179. https://doi.org/10.1016/j.tics.2010.01.004

Eren, E. T., Alpak, E. M., & Düzenli, T. (2022). *Color Associations in Landscape Design and Subscription Levels to These Associations* [Preprint]. In Review. https://doi.org/10.21203/rs.3.rs-1126391/v1

Ertl, T., Müller, C., Aal, K., Wulf, V., Tachtler, F., Scheepmaker, L., Fitzpatrick, G., Smith, N., & Schuler, D. (2021). Ethical Future Environments: Smart Thinking about Smart Cities means engaging with its Most Vulnerable. *C&T '21: Proceedings of the 10th International Conference on Communities & Technologies - Wicked Problems in the Age of Tech*, 340–345. https://doi.org/10.1145/3461564.3468165

Evans et al. - 2018—Active ageing in Denmark; shifting institutional l.pdf. (n.d.).

Experience, W. L. in R.-B. U. (n.d.-a). *Do Interface Standards Stifle Design Creativity?* Nielsen Norman Group. Retrieved August 17, 2022, from https://www.nngroup.com/articles/do-interface-standards-stifle-design-creativity/

Experience, W. L. in R.-B. U. (n.d.-b). *Minimize Cognitive Load to Maximize Usability*. Nielsen Norman Group. Retrieved August 17, 2022, from https://www.nngroup.com/articles/minimize-cognitive-load/

Expert Review in the usability.de glossary. (n.d.). Usability.De. Retrieved August 17, 2022, from https://www.usability.de/en/usability-user-experience/glossary/expert-review.html

Fang, R., Ye, S., Huangfu, J., & Calimag, D. P. (2017). Music therapy is a potential intervention for cognition of Alzheimer's Disease: A mini-review. *Translational Neurodegeneration*, *6*(1), 2. https://doi.org/10.1186/s40035-017-0073-9

Figma: L'outil de design à interface collaborative. (n.d.). Retrieved August 17, 2022, from https://www.figma.com/fr/?utm_source=google&utm_medium=cpc&utm_campaign=17448963726&utm_term=figma.com&utm_content=602950877991&gclid=CjwKCAjwo_KXBhAaEiwA2RZ8hKQps7wiERfL9-kadL0Z52kyHte1AwO7nkxaQlJ6IVpaMG1-Fk3YyRoC4o8QAvD_BwE

Følstad, A. (n.d.). LIVING LABS FOR INNOVATION AND DEVELOPMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY: A LITERATURE REVIEW. *Living Labs*, *10*, 33.

Fotuhi, M., Lubinski, B., Hausterman, N., Riloff, T., Hadadi, M., Raji, C. A., & Trullinger, M. (2016). A PERSONALIZED 12-WEEK “BRAIN FITNESS PROGRAM” FOR IMPROVING COGNITIVE FUNCTION AND INCREASING THE VOLUME OF HIPPOCAMPUS IN ELDERLY WITH MILD COGNITIVE IMPAIRMENT. *The Journal of Prevention of Alzheimer’s Disease*, 1–5. https://doi.org/10.14283/jpad.2016.92

Fratiglioni, L., & Wang, H.-X. (2007). Brain Reserve Hypothesis in Dementia. *Journal of Alzheimer’s Disease*, *12*(1), 11–22. https://doi.org/10.3233/JAD-2007-12103

Gao, Q., & Zhou, J. (Eds.). (2020). *Human Aspects of IT for the Aged Population. Technologies, Design and User Experience: 6th International Conference, ITAP 2020, Held as Part of the 22nd HCI International Conference, HCII 2020, Copenhagen, Denmark, July 19–24, 2020, Proceedings, Part I* (Vol. 12207). Springer International Publishing. https://doi.org/10.1007/978-3-030-50252-2

Gasteiger, N., Ahn, H. S., Lee, C., Lim, J., Macdonald, B. A., Kim, G. H., & Broadbent, E. (2022). Participatory Design, Development, and Testing of Assistive Health Robots with Older Adults: An International Four-year Project. *ACM Transactions on Human-Robot Interaction*, 3533726. https://doi.org/10.1145/3533726

Gates, N. J., Sachdev, P. S., Fiatarone Singh, M. A., & Valenzuela, M. (2011). Cognitive and memory training in adults at risk of dementia: A Systematic Review. *BMC Geriatrics*, *11*(1), 55. https://doi.org/10.1186/1471-2318-11-55

Gerling et al. - 2020—Critical Reflections on Technology to Support Phys.pdf. (n.d.).

Gordon, M. L., Gatys, L., Guestrin, C., Bigham, J. P., Trister, A., & Patel, K. (2019). App Usage Predicts Cognitive Ability in Older Adults. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–12. https://doi.org/10.1145/3290605.3300398

Gould, J. D., & Lewis, C. (1985). Designing for usability: Key principles and what designers think. *Communications of the ACM*, *28*(3), 300–311. https://doi.org/10.1145/3166.3170

Hamasaki—2020—Effects of Diaphragmatic Breathing on Health A Na.pdf. (n.d.).

Hassenzahl, M., Diefenbach, S., & Göritz, A. (2010). Needs, affect, and interactive products – Facets of user experience. *Interacting with Computers*, *22*(5), 353–362. https://doi.org/10.1016/j.intcom.2010.04.002

Hassenzahl, M., Eckoldt, K., & Diefenbach, S. (2013). *Designing Moments of Meaning and Pleasure. 7*(3), 12.

Herold, F., Wiegel, P., Scholkmann, F., & Müller, N. (2018). Applications of Functional Near-Infrared Spectroscopy (fNIRS) Neuroimaging in Exercise–Cognition Science: A Systematic, Methodology-Focused Review. *Journal of Clinical Medicine*, *7*(12), 466. https://doi.org/10.3390/jcm7120466

Homepage of e-VITA. (n.d.). E-VITA Virtual Coach. Retrieved August 17, 2022, from https://www.e-vita.coach/

Hoshi, Y. (2016). Hemodynamic signals in fNIRS. In *Progress in Brain Research* (Vol. 225, pp. 153–179). Elsevier. https://doi.org/10.1016/bs.pbr.2016.03.004

Https://neu-brains.site/en/. (n.d.). Retrieved August 17, 2022, from https://neu-brains.site/en/

Interviews with leading Life Sciences companies: NeU Corporation – Resource Center. (n.d.). Retrieved August 17, 2022, from https://resourcecenter.venturevaluation.com/2020/03/interviews-with-leading-life-sciences-companies-neu-corporation/

Iturria-Medina, Y., Sotero, R. C., Toussaint, P. J., Mateos-Pérez, J. M., & Evans, A. C. (2016). Early role of vascular dysregulation on late-onset Alzheimer's disease based on multifactorial data-driven analysis. *Nature Communications*, *7*(1), 11934. https://doi.org/10.1038/ncomms11934

Jacobsen, N. E., Hertzum, M., & John, B. E. (1998). The Evaluator Effect in Usability Studies: Problem Detection and Severity Judgments. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, *42*(19), 1336–1340. https://doi.org/10.1177/154193129804201902

Joshi, S. G. (2016). *Designing for Prolonged Mastery. On involving old people in Participatory Design*. *28*, 35.

Kane, R. L., Butler, M., Fink, H. A., Brasure, M., Davila, H., Desai, P., Jutkowitz, E., McCreedy, E., Nelson, V. A., McCarten, J. R., Calvert, C., Ratner, E., Hemmy, L. S., & Barclay, T. (2017). *Interventions to Prevent Age-Related Cognitive Decline, Mild Cognitive Impairment, and Clinical Alzheimer's-Type Dementia*. Agency for Healthcare Research and Quality (US). http://www.ncbi.nlm.nih.gov/books/NBK442425/

Kickmeier-Rust, M., Holzinger, A., & Albert, D. (n.d.). *Fighting Physical and Mental Decline of Elderly With Adaptive Serious Games*. 1.

Kirchner, W. K. (1958). Age differences in short-term retention of rapidly changing information. *Journal of Experimental Psychology*, *55*(4), 352–358. https://doi.org/10.1037/h0043688

Köbe, T., Witte, A. V., Schnelle, A., Lesemann, A., Fabian, S., Tesky, V. A., Pantel, J., & Flöel, A. (2016). Combined omega-3 fatty acids, aerobic exercise and cognitive stimulation prevents decline in gray matter volume of the frontal, parietal and cingulate cortex in patients with mild cognitive impairment. *NeuroImage*, *131*, 226–238. https://doi.org/10.1016/j.neuroimage.2015.09.050

Lazar, J. (2017). *Research methods in human computer interaction* (2nd edition). Elsevier.

Lin et al. - 2009—Applications of Near Infrared Spectroscopy and Ima.pdf. (n.d.).

Lippelt, D. P., Hommel, B., & Colzato, L. S. (2014). Focused attention, open monitoring and loving kindness meditation: Effects on attention, conflict monitoring, and creativity – A review. *Frontiers in Psychology*, *5*. https://www.frontiersin.org/article/10.3389/fpsyg.2014.01083

Lutz, A., Slagter, H. A., Dunne, J. D., & Davidson, R. J. (2008). Attention regulation and monitoring in meditation. *Trends in Cognitive Sciences*, *12*(4), 163–169. https://doi.org/10.1016/j.tics.2008.01.005

Makedon, F., Zhang, R., Alexandrakis, G., Owen, C. B., Huang, H., & Saykin, A. J. (2010). An interactive user interface system for Alzheimer's intervention. *Proceedings of the 3rd International Conference on PErvasive Technologies Related to Assistive Environments*, 1–5. https://doi.org/10.1145/1839294.1839336

Malinowski, P. (2013). Neural mechanisms of attentional control in mindfulness meditation. *Frontiers in Neuroscience*, *7*. https://doi.org/10.3389/fnins.2013.00008

Maruya, K., Arai, T., & Fujita, H. (2021). Brain Activity in the Prefrontal Cortex during Cognitive Tasks and Dual Tasks in Community-Dwelling Elderly People with Pre-Frailty: A Pilot Study for Early Detection of Cognitive Decline. *Healthcare*, *9*(10), 1250. https://doi.org/10.3390/healthcare9101250

Meetgeek Virtual Meeting Assistant | AIVISION. (n.d.). Retrieved August 17, 2022, from https://app.meetgeek.ai/setup

Meurer, J., Müller, C., Simone, C., Wagner, I., & Wulf, V. (2018). Designing for Sustainability: Key Issues of ICT Projects for Ageing at Home. *Computer Supported Cooperative Work (CSCW)*, *27*(3–6), 495–537. https://doi.org/10.1007/s10606-018-9317-1

Middleton, L. E., & Yaffe, K. (2009). Promising Strategies for the Prevention of Dementia. *Archives of Neurology*, *66*(10). https://doi.org/10.1001/archneurol.2009.201

Miro | *Online Whiteboard for Visual Collaboration*. (n.d.). Retrieved August 17, 2022, from https://miro.com/app/dashboard/

Miyoshi, T., Tanioka, K., Yamamoto, S., Yadohisa, H., Hiroyasu, T., & Hiwa, S. (2020). Revealing Changes in Brain Functional Networks Caused by Focused-Attention Meditation Using Tucker3 Clustering. *Frontiers in Human Neuroscience*, *13*. https://www.frontiersin.org/article/10.3389/fnhum.2019.00473

Morris, J. C., Storandt, M., Miller, J. P., McKeel, D. W., Price, J. L., Rubin, E. H., & Berg, L. (2001). Mild Cognitive Impairment Represents Early-Stage Alzheimer Disease. *Archives of Neurology*, *58*(3). https://doi.org/10.1001/archneur.58.3.397

Müller, C., Hornung, D., Hamm, T., & Wulf, V. (2015a). Measures and Tools for Supporting ICT Appropriation by Elderly and Non Tech-Savvy Persons in a Long-Term Perspective. In N. Boulus-Rødje, G. Ellingsen, T. Bratteteig, M. Aanestad, & P. Bjørn (Eds.), *ECSCW 2015: Proceedings of the 14th European Conference on Computer Supported Cooperative Work, 19-23 September 2015, Oslo, Norway* (pp. 263–281). Springer International Publishing. https://doi.org/10.1007/978-3-319-20499-4_14

Müller, C., Hornung, D., Hamm, T., & Wulf, V. (2015b). Practice-based Design of a Neighborhood Portal: Focusing on Elderly Tenants in a City Quarter Living Lab. *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, 2295–2304. https://doi.org/10.1145/2702123.2702449

Muller, M. J., & Kuhn, S. (1993). Participatory design. *Communications of the ACM*, *36*(6), 24–28. https://doi.org/10.1145/153571.255960

Nais lab. (n.d.). Retrieved August 17, 2022, from http://nais.kaist.ac.kr/research/fnirs.php

Nasar, J. L. (2007). [No title found]. *Journal of Environmental Psychology*, *27*(3), 252–253. https://doi.org/10.1016/j.jenvp.2007.05.001

Ng, T. K. S., Feng, L., Fam, J., Rawtaer, I., Kumar, A. P., Rane, G., Cheah, I. K.-M., Mahendran, R., Lee, Y. K., Tan, E. C., Goh, L. G., Kua, E. H., & Mahendran, R. (2021). Mindfulness Awareness Practice (MAP) to Prevent Dementia in Older Adults with Mild Cognitive Impairment: Protocol of a Randomized Controlled Trial and Implementation Outcomes. *International Journal of Environmental Research and Public Health*, *18*(19), 10205. https://doi.org/10.3390/ijerph181910205

Nouchi, R., Nouchi, H., Dinet, J., & Kawashima, R. (2021). Cognitive Training with Neurofeedback Using NIRS Improved Cognitive Functions in Young Adults: Evidence from a Randomized Controlled Trial. *Brain Sciences*, *12*(1), 5. https://doi.org/10.3390/brainsci12010005

Park, S.-H., & Han, K. S. (2017). Blood Pressure Response to Meditation and Yoga: A Systematic Review and Meta-Analysis. *The Journal of Alternative and Complementary Medicine*, *23*(9), 685–695. https://doi.org/10.1089/acm.2016.0234

Polson, P. G., & Lewis, C. H. (1990). Theory-Based Design for Easily Learned Interfaces. *Human–Computer Interaction*, *5*(2–3), 191–220. https://doi.org/10.1080/07370024.1990.9667154

Reynolds, G. O., Willment, K., & Gale, S. A. (2021). Mindfulness and Cognitive Training Interventions in Mild Cognitive Impairment: Impact on Cognition and Mood. *The American Journal of Medicine*, *134*(4), 444–455. https://doi.org/10.1016/j.amjmed.2020.10.041

Rienzo, A., Cubillos, C., & Soto, G. (2021). Gamification Elements in Digital Applications for the Evaluation and Cognitive Training of Older Adults. *2021 IEEE International Conference on Automation/XXIV Congress of the Chilean Association of Automatic Control (ICA-ACCA)*, 1–8. https://doi.org/10.1109/ICAACCA51523.2021.9465207

Rienzo et al. - 2021—Gamification Elements in Digital Applications for .pdf. (n.d.).

Rodríguez-Almendros, M. L., Rodríguez-Fórtiz, M. J., Hornos, M. J., Samos-Jiménez, J., Rodríguez-Domínguez, C., & Rute-Pérez, S. (2021). Design guide and usability questionnaire to develop and assess VIRTRAEL, a web-based cognitive training tool for the elderly. *Behaviour & Information Technology*, *40*(13), 1355–1374. https://doi.org/10.1080/0144929X.2020.1750702

Scalf, P. E., Colcombe, S. J., McCarley, J. S., Erickson, K. I., Alvarado, M., Kim, J. S., Wadhwa, R. P., & Kramer, A. F. (2007). The neural correlates of an expanded functional field of view. *The Journals of Gerontology. Series B, Psychological Sciences and Social Sciences*, *62 Spec No 1*, 32–44. https://doi.org/10.1093/geronb/62.special_issue_1.32

Shimada, H., Makizako, H., Doi, T., Lee, S., & Lee, S. (2017). Conversion and Reversion Rates in Japanese Older People With Mild Cognitive Impairment. *Journal of the American Medical Directors Association*, *18*(9), 808.e1-808.e6. https://doi.org/10.1016/j.jamda.2017.05.017

Silva, N. C. D. da, Valle, A. M. D., & Kawashita, I. M. S. (2019). Participatory design with the elderly in software development: A case study. *Proceedings of the 18th Brazilian Symposium on Human Factors in Computing Systems*, 1–10. https://doi.org/10.1145/3357155.3358486

Simons, D. J., Boot, W. R., Charness, N., Gathercole, S. E., Chabris, C. F., Hambrick, D. Z., & Stine-Morrow, E. A. L. (2016). Do "Brain-Training" Programs Work? *Psychological Science in the Public Interest*, *17*(3), 103–186. https://doi.org/10.1177/1529100616661983

Solutions. (n.d.). Retrieved August 17, 2022, from https://neu-brains.site/solutions/

Son and Yazici—2006—NEAR INFRARED IMAGING AND SPECTROSCOPY FOR BRAIN A.pdf. (n.d.).

Son, I.-Y., & Yazici, B. (2006). NEAR INFRARED IMAGING AND SPECTROSCOPY FOR BRAIN ACTIVITY MONITORING. In J. Byrnes & G. Ostheimer (Eds.), *Advances in*

Sensing with Security Applications (Vol. 2, pp. 341–372). Kluwer Academic Publishers. https://doi.org/10.1007/1-4020-4295-7_15

Strenziok, M., Parasuraman, R., Clarke, E., Cisler, D. S., Thompson, J. C., & Greenwood, P. M. (2014). Neurocognitive enhancement in older adults: Comparison of three cognitive training tasks to test a hypothesis of training transfer in brain connectivity. *NeuroImage*, *85*, 1027–1039. https://doi.org/10.1016/j.neuroimage.2013.07.069

Sugihara, Y., Matsuura, T., Kubo, Y., & Ohgomori, T. (2021). Activation of the Prefrontal Cortex and Improvement of Cognitive Performance with Standing on One Leg. *Neuroscience*, *477*, 50–62. https://doi.org/10.1016/j.neuroscience.2021.10.004

Tang, Y.-Y., Hölzel, B. K., & Posner, M. I. (2015). The neuroscience of mindfulness meditation. *Nature Reviews Neuroscience*, *16*(4), 213–225. https://doi.org/10.1038/nrn3916

Tang, Y.-Y., & Posner, M. I. (2014). Training brain networks and states. *Trends in Cognitive Sciences*, *18*(7), 345–350. https://doi.org/10.1016/j.tics.2014.04.002

The IMPACT Study—BrainHQ from Posit Science. (n.d.). Retrieved August 17, 2022, from https://www.brainhq.com/world-class-science/published-research/impact-study

Train the Brain Consortium. (2017). Randomized trial on the effects of a combined physical/cognitive training in aged MCI subjects: The Train the Brain study. *Scientific Reports*, *7*(1), 39471. https://doi.org/10.1038/srep39471

Universal methods of design: 100 ways to research complex problems, develop innovative ideas, and design effective solutions. (2012). *Choice Reviews Online*, *49*(10), 49-5403-49–5403. https://doi.org/10.5860/CHOICE.49-5403

Vaziri, D. D., Anslinger, M., Unbehaun, D., Wieching, R., Randall, D., Schreiber, D., & Wulf, V. (2019). *Mobile health platforms for active and healthy ageing support in older adults: Design ideas from a participatory design study*. 29.

Virzi, R. A. (1992). Refining the Test Phase of Usability Evaluation: How Many Subjects Is Enough? *Human Factors: The Journal of the Human Factors and Ergonomics Society*, *34*(4), 457–468. https://doi.org/10.1177/001872089203400407

Voilmy, D., & Duchêne, J. (2014). *Developing home-based social and healthcare solutions for the elderly using participatory design*. 7.

Wierenga, C. E., Hays, C. C., & Zlatar, Z. Z. (2014). Cerebral Blood Flow Measured by Arterial Spin Labeling MRI as a Preclinical Marker of Alzheimer's Disease. *Journal of Alzheimer's Disease*, *42*(s4), S411–S419. https://doi.org/10.3233/JAD-141467

Wilson, R. S., Boyle, P. A., Segawa, E., Yu, L., Begeny, C. T., Anagnos, S. E., & Bennett, D. A. (2013). The influence of cognitive decline on well-being in old age. *Psychology and Aging*, *28*(2), 304–313. https://doi.org/10.1037/a0031196

Wolinsky et al. - 2006—The Effects of the ACTIVE Cognitive Training Trial.pdf. (n.d.).

Wolinsky et al. - 2009—The Effect of Speed-of-Processing Training on Depr.pdf. (n.d.).

Wolinsky et al. - 2010—Does Cognitive Training Improve Internal Locus of .pdf. (n.d.).

Wulf et al. - 2011—Engaging with practices design case studies as a .pdf. (n.d.).

Wulf et al. - 2015—Practice-Based Computing Empirically Grounded Con.pdf. (n.d.).

Wulf, V., Müller, C., Pipek, V., Randall, D., Rohde, M., & Stevens, G. (2015). Practice-Based Computing: Empirically Grounded Conceptualizations Derived from Design Case Studies. In V. Wulf, K. Schmidt, & D. Randall (Eds.), *Designing Socially Embedded Technologies in the Real-World* (pp. 111–150). Springer London. https://doi.org/10.1007/978-1-4471-6720-4_7

Yamaya, N., Tsuchiya, K., Takizawa, I., Shimoda, K., Kitazawa, K., & Tozato, F. (2021). Effect of one-session focused attention meditation on the working memory capacity of meditation novices: A functional near-infrared spectroscopy study. *Brain and Behavior*, *11*(8). https://doi.org/10.1002/brb3.2288

Zhang, X., Lian, S., Zhang, Y., & Zhao, Q. (2022). Efficacy and safety of donepezil for mild cognitive impairment: A systematic review and meta-analysis. *Clinical Neurology and Neurosurgery*, *213*, 107134. https://doi.org/10.1016/j.clineuro.2022.107134

Appendix

XB-01 : Raw data file which you can download from the server for the cognitive training system

	Parameter	Definition	Unit
1	Tick	time	sec
2	heartRate rate	heart rate	rate (bpm) *unstable version
3	HbT change(SD1cm)	blood flow change in skin	(mMmm)
4	HbT change(SD3cm)	blood flow change in skin and brain	(mMmm)
5	saturation(SD1cm)	Light saturation count (#) *should be zero for stable measurement	(mMmm)
6	saturation(SD3cm)	Light saturation count (#) *should be zero for stable measurement	
7	Mark		

8	Reaction		
9	Color	calculated brain activity (a.u.) *Shown as color in the app	(a.u.)
10	accelX		
11	accelY		
12	accelZ		
13	gyroX		
14	gyroY		
15	gyroZ		
16	Stability		
	Parameter	Definition	Unit
17	Pulse Coeff1 L1		
18	Pulse Coeff2 L1		
19	Pulse Coeff3 L1		
20	Pulse Coeff1 L3		
21	Pulse Coeff2 L3		
22	Pulse Coeff3 L3		
23	AC SD1cm		
24	AC SD3cm		
25	accelX-SD		
26	accelY-SD		

27	accelZ-SD		
28	gyroX-SD		
29	gyroY-SD		
30	gyroZ-SD		
31	Battery Volt		
32	Estimated pulse rate	heart rate	(bpm) *stable version

		Specification (Draft)
Brand		NeU
Model		XB-01
NIRS Sensor	Measurement data	Forehead cerebral blood flow change (Any one point) Pulse rate
	Measurement CH	1 CH (Any place)
	Mobility	Manual adjustment
	Sampling	100 msec
Accelerometer	Type	6 axis
	Sampling	100 msec
	Wireless	Bluetooth 4.1LE
	Modulation	F1D
	Frequency range	2,402 MHz - 2,480 MHz
	Peak Output Power	4.0 dBm
	Antenna	P/N 2405AT18A100
	Antenna Type	Chip Antenna
	Antenna Gain	0.5 dBi
	Channel Number	40 Channels
USB		with Charge function with Communication function
Battery	Battery	Built-in Lithium Ion Battery 3.7 V, 370 mAh
	Operating time Charging time	3 hours 2 hours
Dimensions		80 x 40 x 13 mm (3.2 x 1.6 x 0.5 in)
Weight		30 g (Including Battery)
Recommended Operating Conditions		Temperature: 0 to 35 ℃ (0 to 95 ℉) Humidity: 20 to 80% Rh
Design Variation		Black

A flowchart was drawn in the early stages for the Bo app

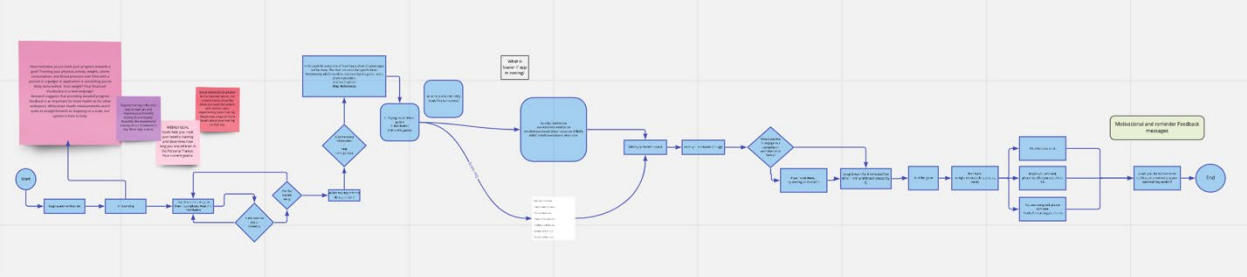


Figure 29 Flowchart of Bo app

Bo App

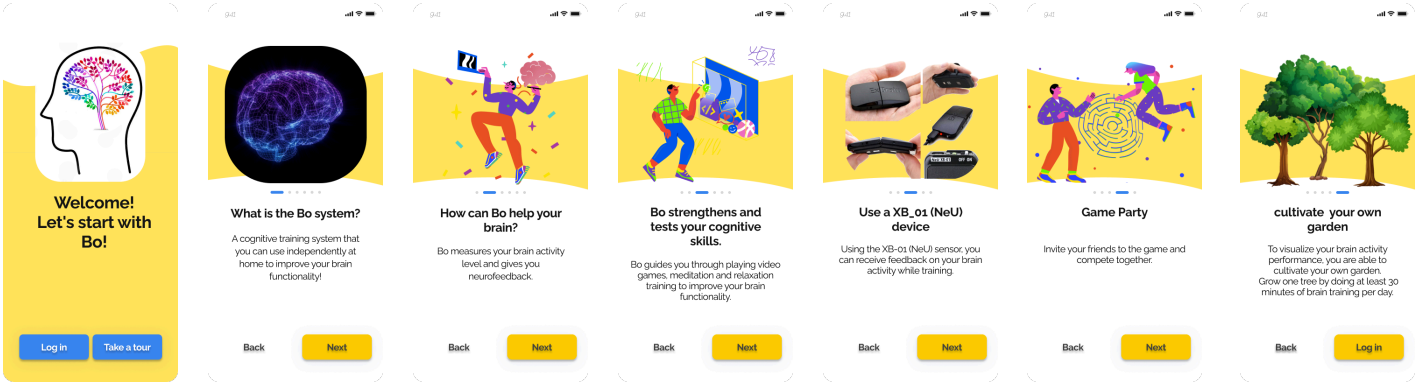
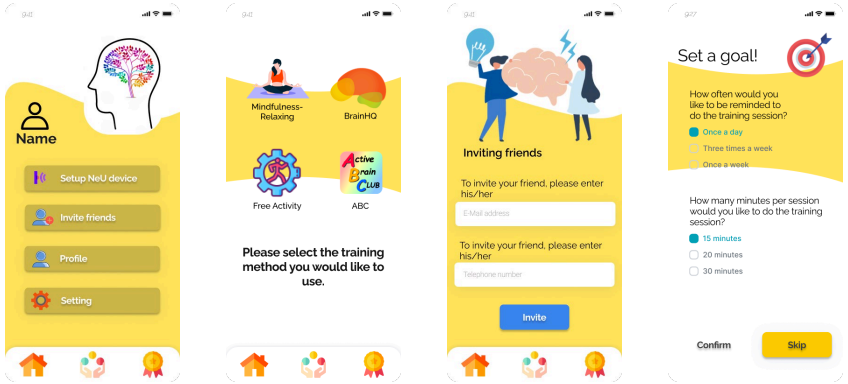


Figure 30 On-boarding pages



(a) (b) (c) (d)

Figure 31 (a) Home page, (b) Activity page, (c) Inviting friends, (d) Setting the training goals

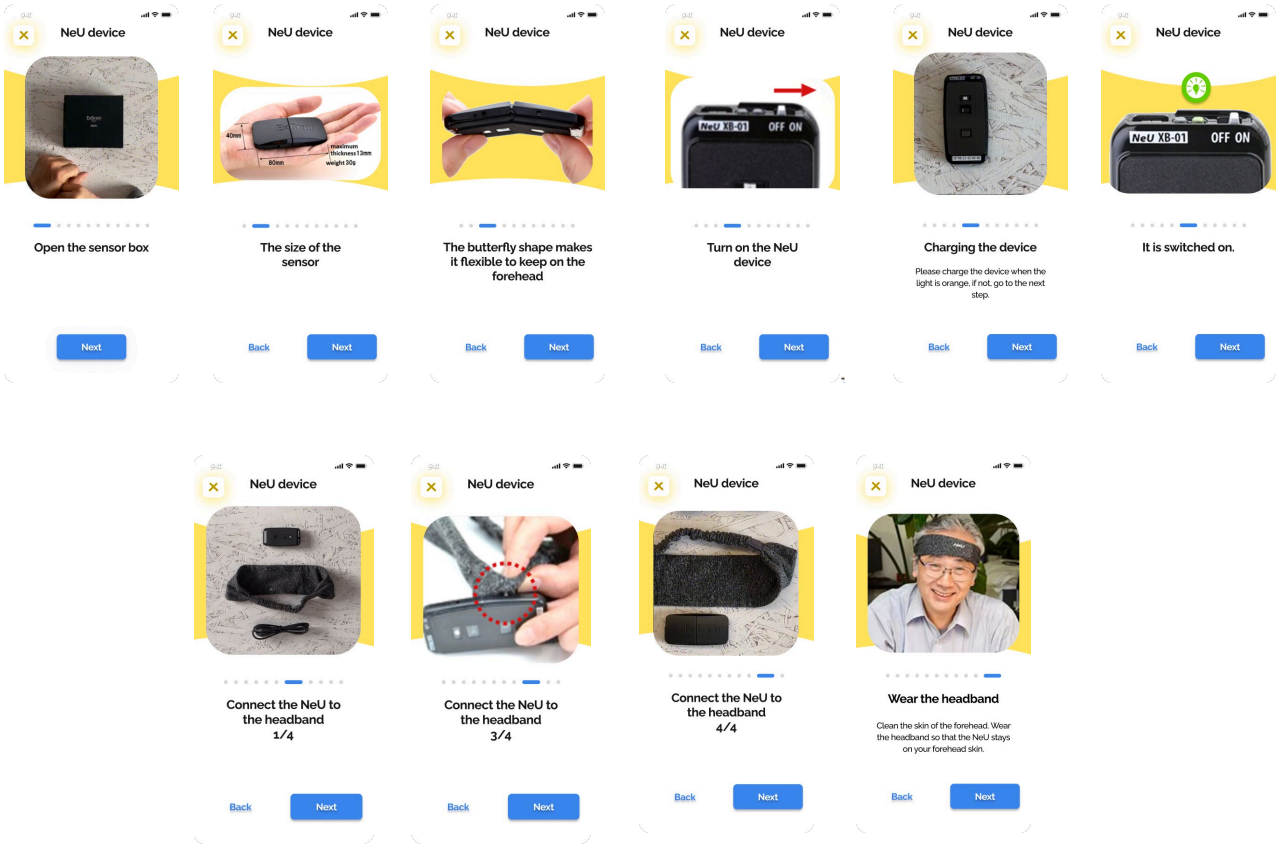


Figure 32 NeU device step-by-step tutorial

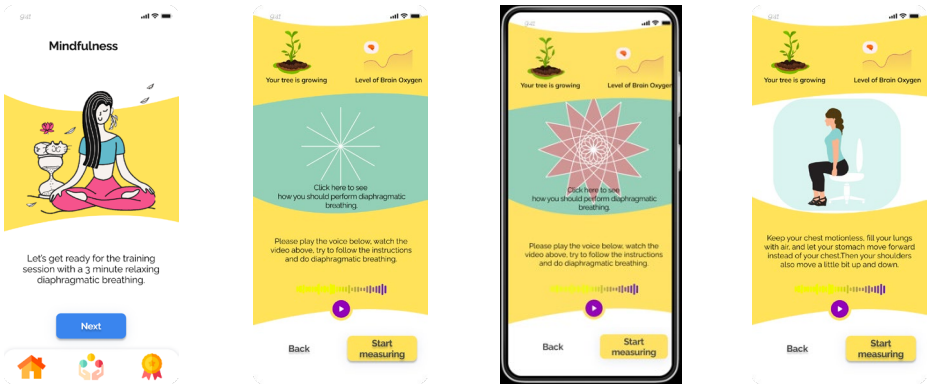


Figure 33 Diaphragmatic relaxation and mindfulness guidance

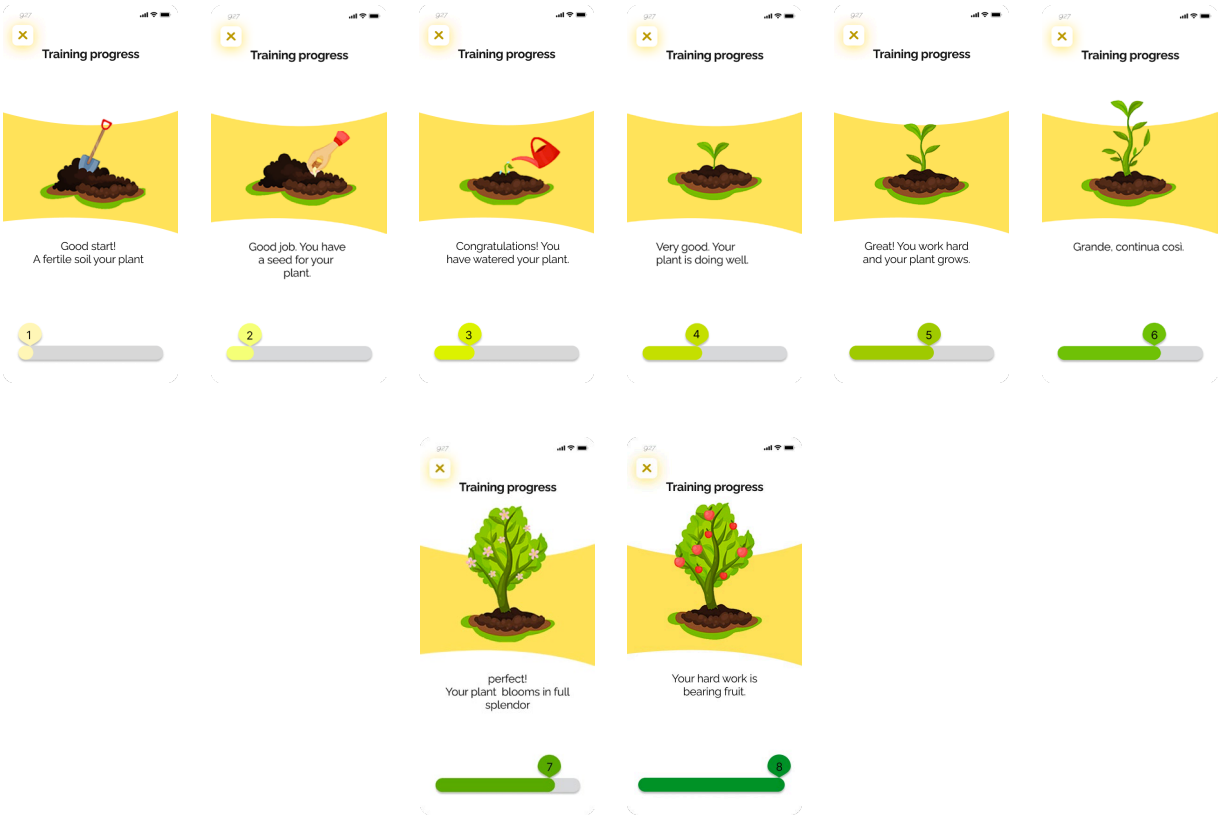


Figure 34 Daily feedback

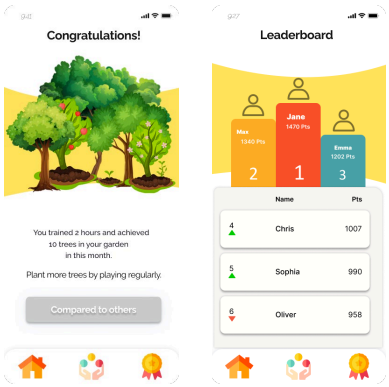


Figure 35 (a) Long-time feedback, (b) Competitive feedback

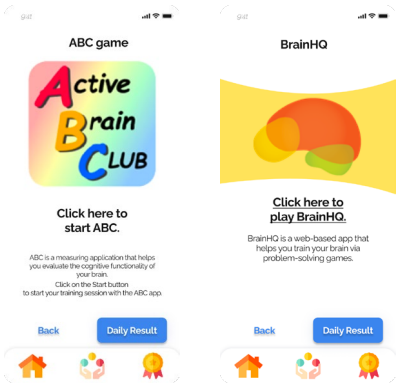


Figure 36 Cognitive training games

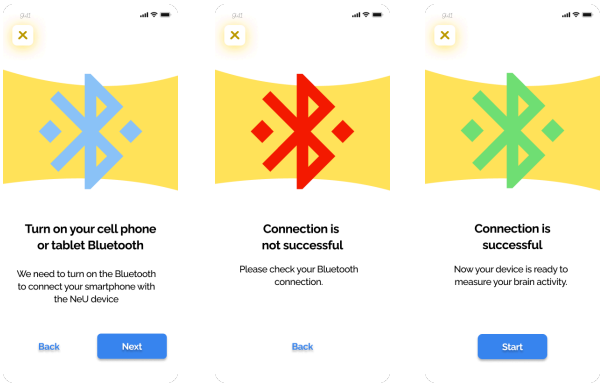


Figure 37 Bluetooth setup

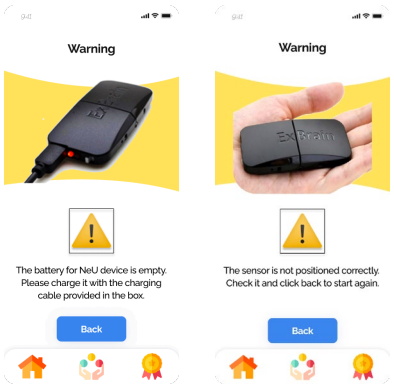
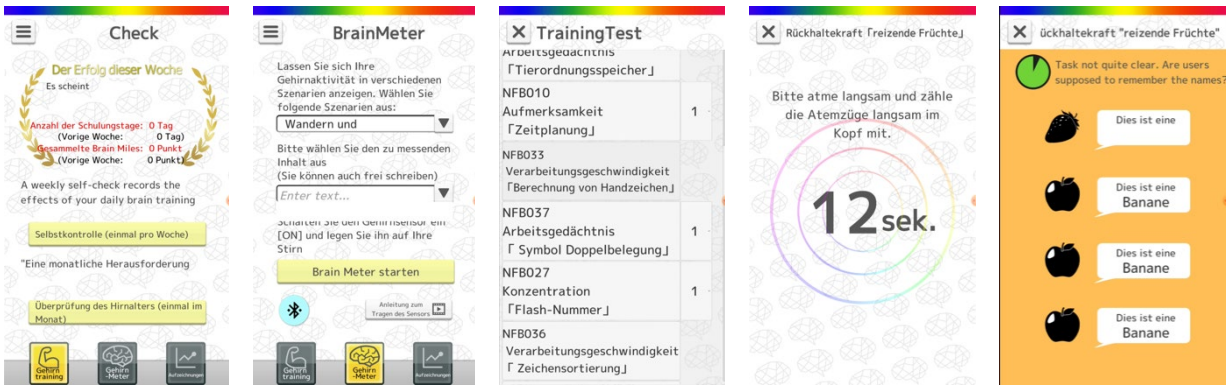


Figure 38 Error informing

ABC



(a) (b) (c) (d) (e)

Figure 39 (a)On-boarding page, (b)BrainMeter page, (c)Training game list, (d)Deep breathing, (e)A game sample

BrainHQ

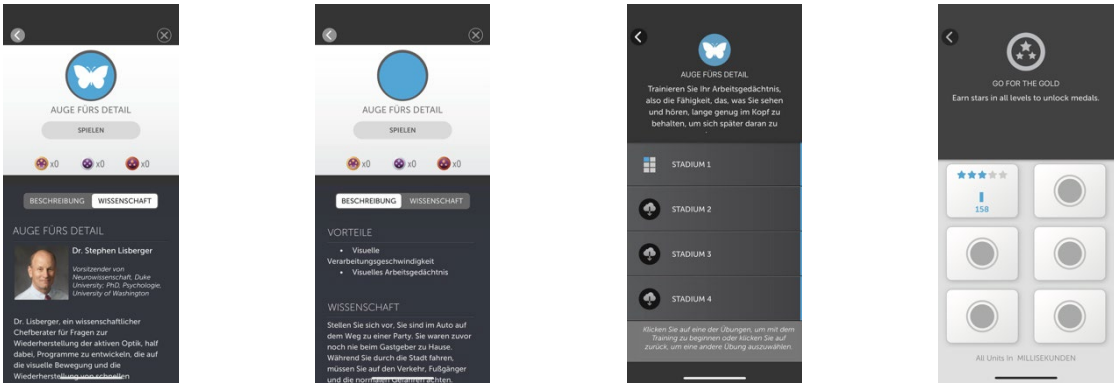


Figure 40 A game sample of BrainHQ with background knowledge

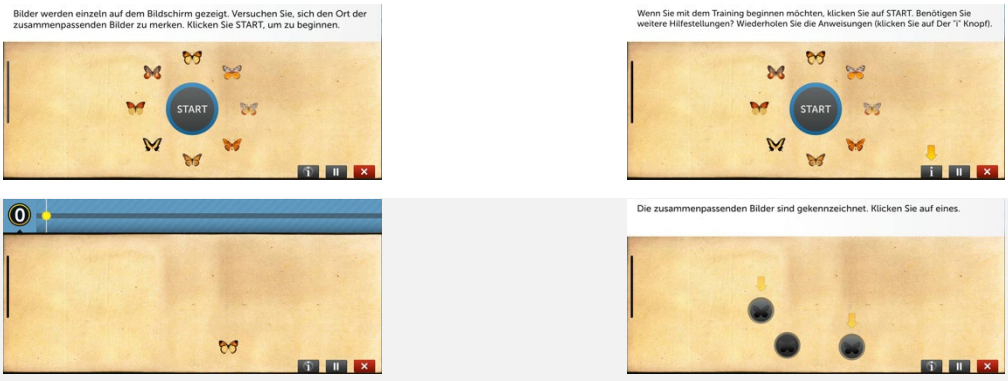


Figure 41 Initializing guidance of the game sample

Declaration

Hereby I confirm that I have composed the present thesis independently. I only have used the sources and means specified in this thesis. Especially from the internet, I only have used the denoted references. I have taken note of the section in the examination regulations concerning attempts to cheat.

I confirm that the electronic version of the thesis which I deliver is identical to the printed version with respect to the content. I agree that an electronic version of the thesis will be stored for purposes of inspection of plagiarism.

Roodabeh Seif – 1486533
18 August 2022